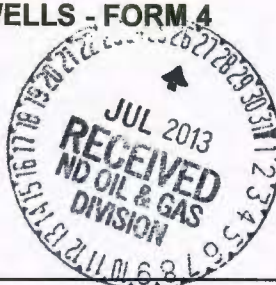


**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
222100-01

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 1, 2013
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	
Approximate Start Date	

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Central production facility-commingle prod

Well Name and Number (see details)					
Footages		Qtr-Qtr	Section	Township	Range
F	L	F	L	12	153 N
Field Baker		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests permission to add the following wells to CTB # 222100-01.

Well File #22740 Larry 5301 44-12B SESE 12-153-101 API 33-105-04981

Well File #22099 Yukon 5301 41-12T SWSW 12-153-101 API 33-053-03911

Well File #25571 Colville 5301 44-12T SESE 12-153-101 API 33-053-04981

Well File #22221 Innoko 5301 43-12T SWSE 12-153-101 API 33-053-03937

The following wells are currently being commingled in the subject CTB:

Well File #22100 Achilles 5301 41-12B SWSW 12-153-101 API 33-053-03912

Well File #22220 Jefferies 5301 43-12B SWSE 12-153-101 API 33-053-03936

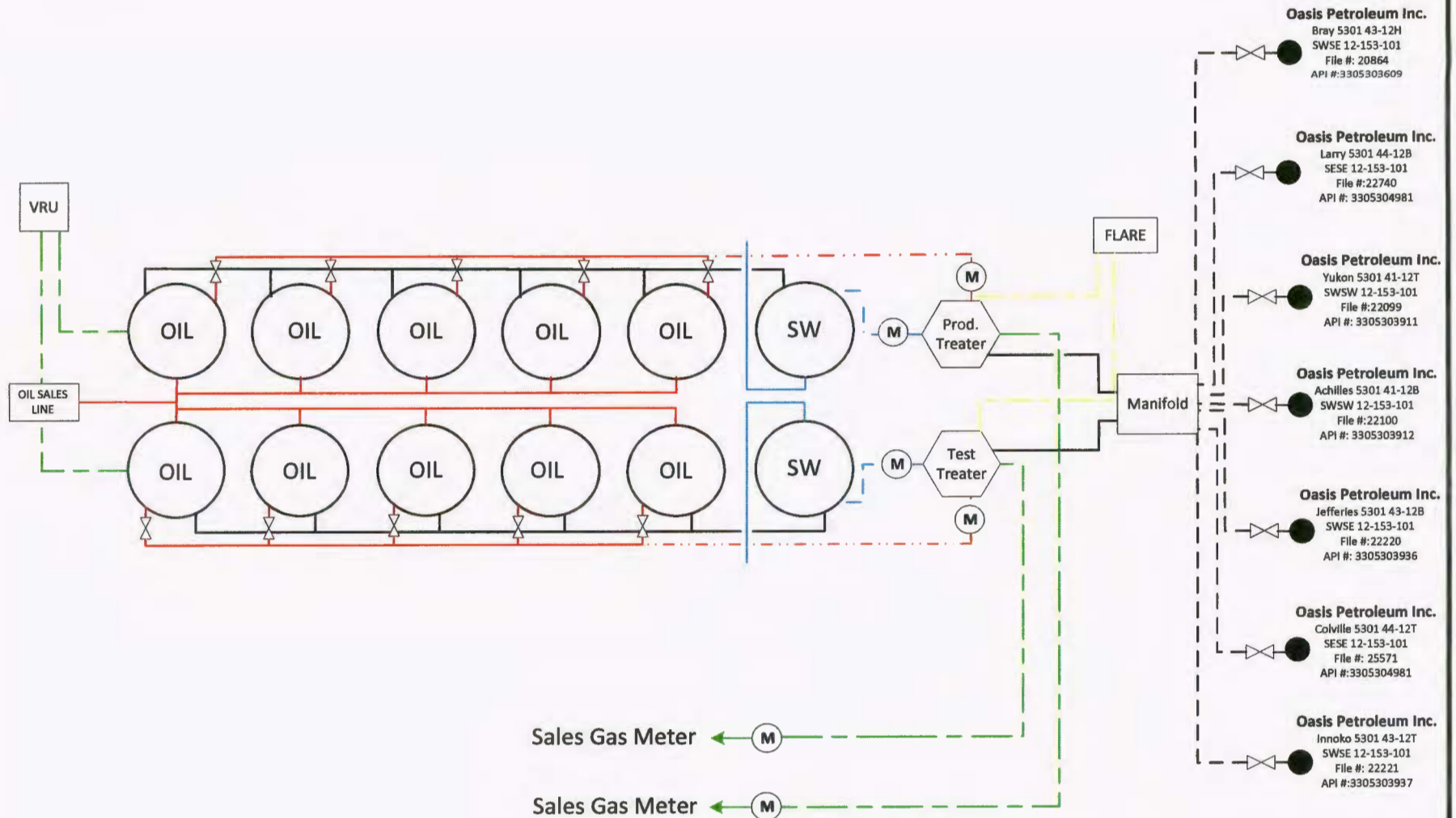
Well File #20864 Bray 5301 43-12H SWSE 12-153-101 API 33-053-03609

Well File #22740 Larry 5301 44-12B SESE 12-153-101 API 33-053-04071

Please find the following attachments: 1. A schematic drawing of the facility which diagrams the testing, treating, routing, and transferring of production. 2. A plat showing the location of the central facility 3. Affidavit of title indicating common ownership.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Brandi Terry	
Title Regulatory Specialist		Date July 24, 2013	
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 7-30-13	
By ORIGINAL SIGNED BY	
Title DARYL GRONFUR	
METER SPECIALIST	



5301 13-24 ACHILLES CENTRAL TANK BATTERY

DATE	REV.	BY	APPR.	SCALE
JULY 23, 2013	0	LEE		NA
LOCATION	FIELD			
NORTH DAKOTA	BAKER			

COMMINGLING AFFIDAVIT

STATE OF NORTH DAKOTA)
) ss.
COUNTY OF MCKENZIE)

Tom F. Hawkins, being duly sworn, states as follows:

1. I am the Vice President - Land and Contracts employed by Oasis Petroleum North America LLC with responsibilities in the State of North Dakota and I have personal knowledge of the matters set forth in this affidavit.
2. Sections 13 and 24, Township 153 North, Range 101 West, 5th P.M., McKenzie County, North Dakota constitute a spacing unit in accordance with the applicable orders of the North Dakota Industrial Commission for the Bakken pool.
3. Four wells have been drilled in the spacing unit, which are the Bray 5301 43-12H, Achilles 5301 41-12B, Jefferies 5301 43-12B, Larry 5301 44-12B; and three wells have been permitted in the spacing unit, which are the Colville 5301 44-12T, Innoko 5301 43-12T and Yukon 5301 41-12T.
4. By Declaration of Pooled Unit dated August 26, 2011, filed in McKenzie County, North Dakota, document number 422312, all oil and gas interests within the aforementioned spacing unit were pooled.
5. All Working Interests, Royalty Interests and Overriding Royalty Interests in the Bray 5301 43-12H, Achilles 5301 41-12B, Jefferies 5301 43-12B, Colville 5301 44-12T, Innoko 5301 43-12T and Yukon 5301 41-12 wells are common.

Dated this 9TH day of July, 2013.



Tom F. Hawkins
Vice President-Land and Contracts

STATE OF TEXAS)
) ss.
COUNTY OF HARRIS)

Subscribed to and sworn before me this 9TH day of July, 2013.

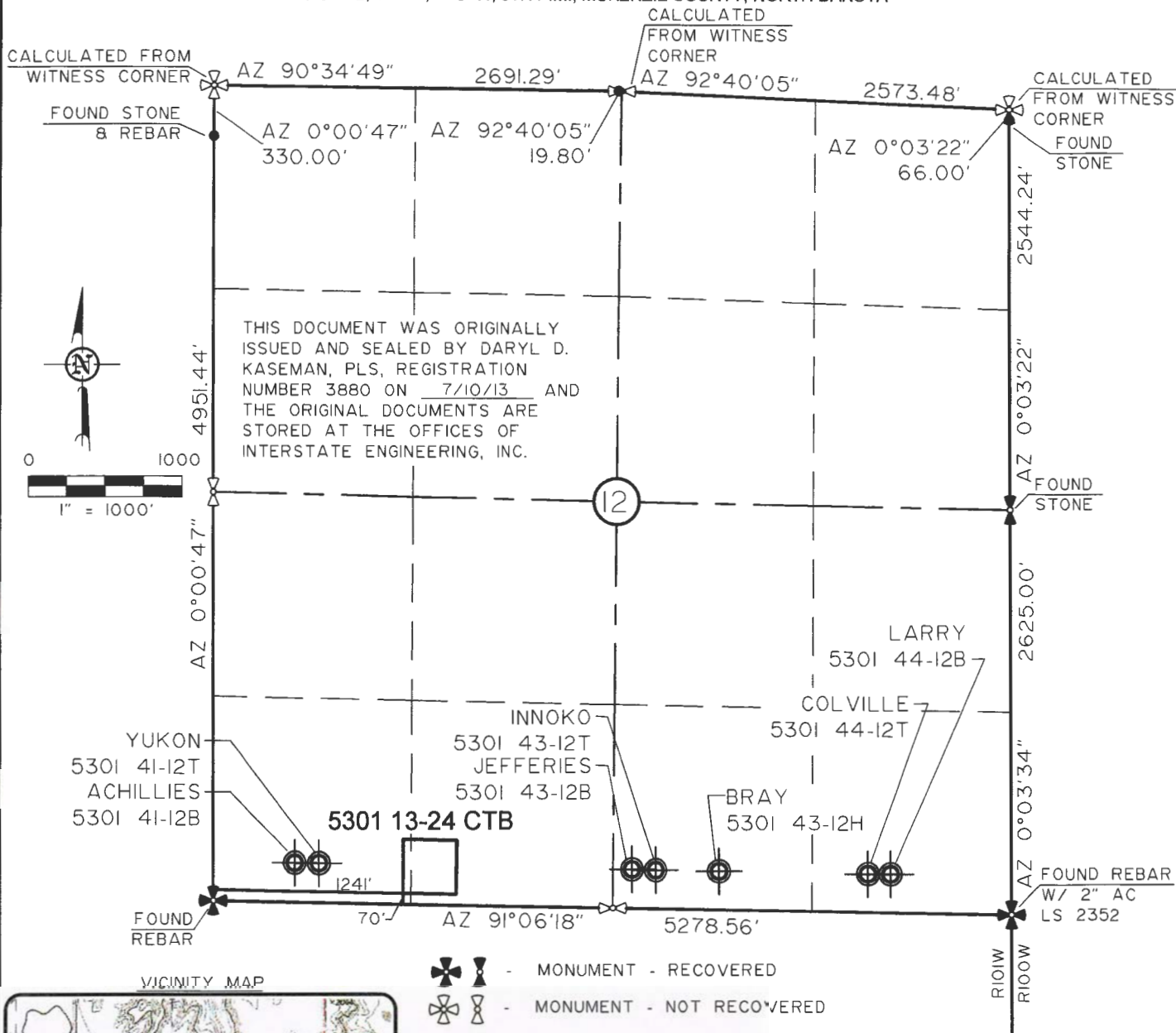


Notary Public
State of Texas
My Commission Expires: August 14, 2017

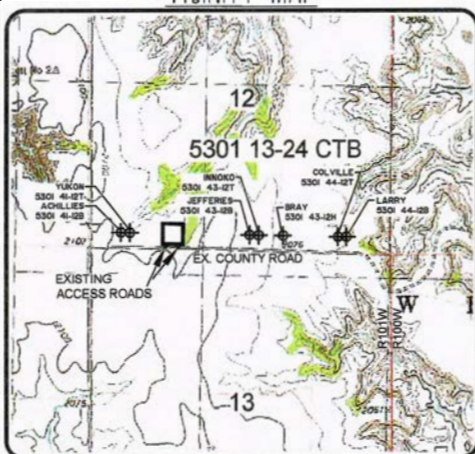


BATTERY LOCATION PLAT
OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 202 HOUSTON, TX 77002
 "5301 13-24 CTB"

SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



VICINITY MAP



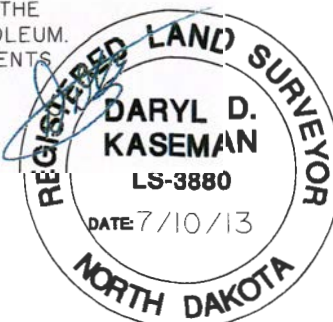
- MONUMENT - RECOVERED
- MONUMENT - NOT RECOVERED

STAKED ON 3/08/12
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 13 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE
 REQUEST OF FABIAN KJORSTAD OF OASIS PETROLEUM.
 I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
 WORK PERFORMED BY ME OR UNDER MY
 SUPERVISION AND IS TRUE AND CORRECT TO
 THE BEST OF MY KNOWLEDGE AND BELIEF.

[Signature]

DARYL D. KASEMAN LS-3880



© 2012, INTERSTATE ENGINEERING, INC.

1/5



Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.iengi.com

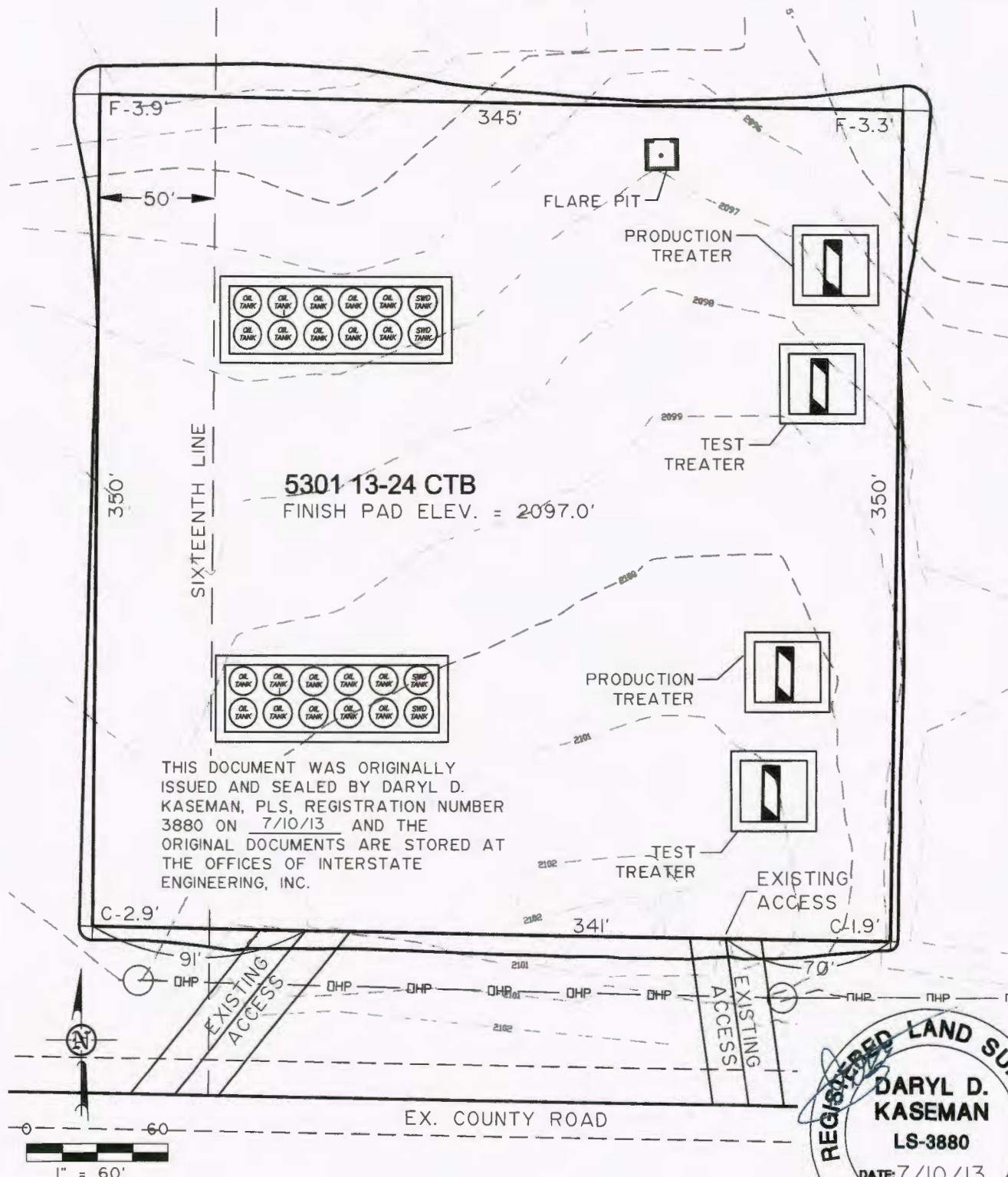
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 WELL LOCATION PLAT
 SECTION 12, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.D.M. Project No.: S12-09-249
 Checked By: DDK Date: SEPT. 2012

Revision No.	Date	By	Description
REV 1	7/10/13	JDM	ADDED WELLS

PAD LAYOUT
OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 202 HOUSTON, TX 77002
 "5301 13-24 CTB"
 SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

© 2012, INTERSTATE ENGINEERING, INC.

2/5

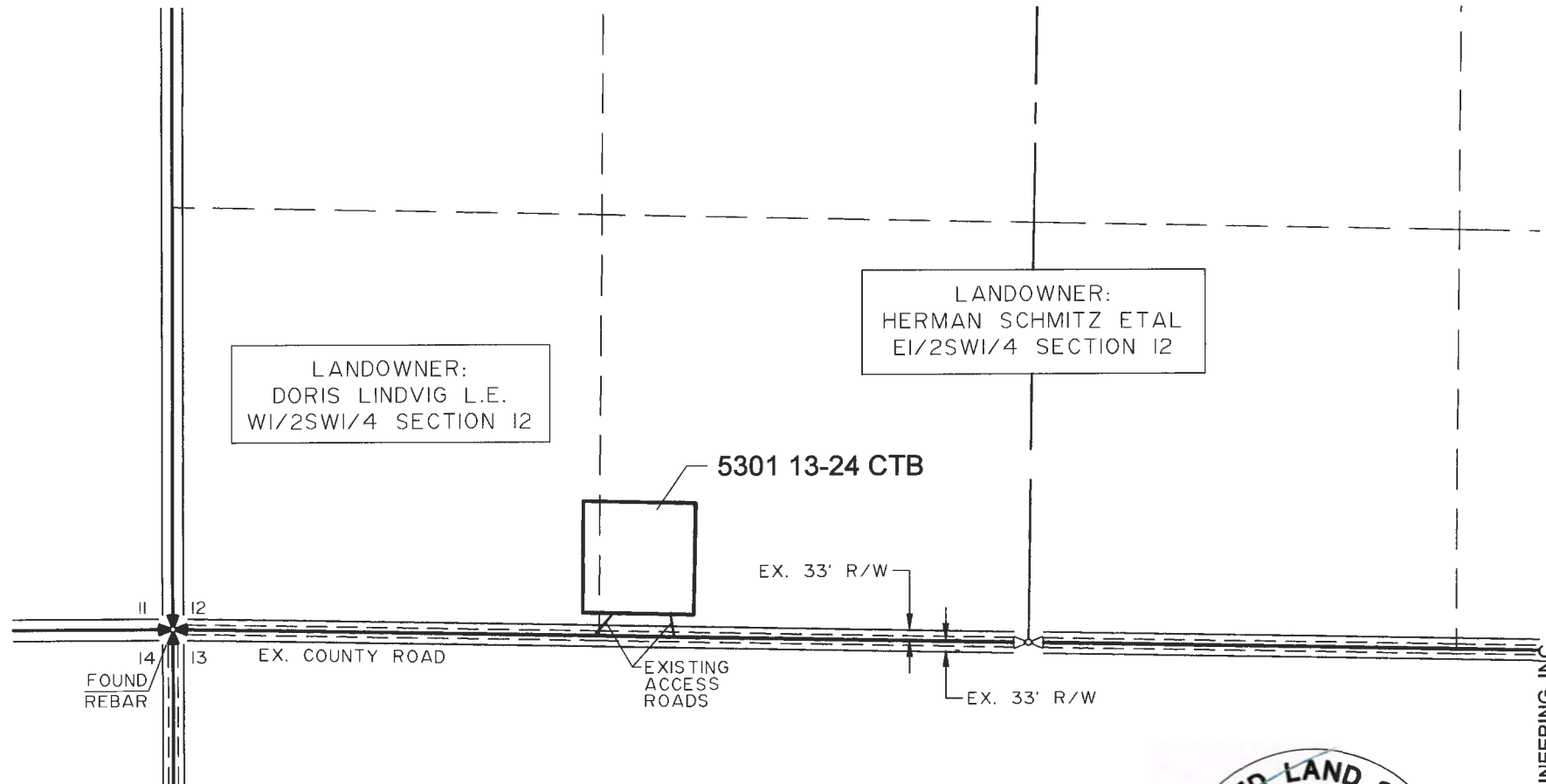


Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.iengi.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 PAD LAYOUT
 SECTION 12, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: J.D.M. Project No.: S12-9-249
 Checked By: D.O.K. Date: SEPT 2012

Revision No.	Date	By	Description
REV 1	7/10/13	JDM	ADDED WELLS

ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 202 HOUSTON, TX 77002
 "5301 13-24 CTB"
 SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY
 ISSUED AND SEALED BY DARYL D.
 KASEMAN, PLS, REGISTRATION NUMBER
 3880 ON 7/10/13 AND THE
 ORIGINAL DOCUMENTS ARE STORED AT
 THE OFFICES OF INTERSTATE
 ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete
 utilities location is recommended before construction.

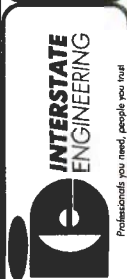


© 2012, INTERSTATE ENGINEERING, INC.

Revision No.	Date	By	Description
REV 1	7/10/13	DM	ADDED WELLS

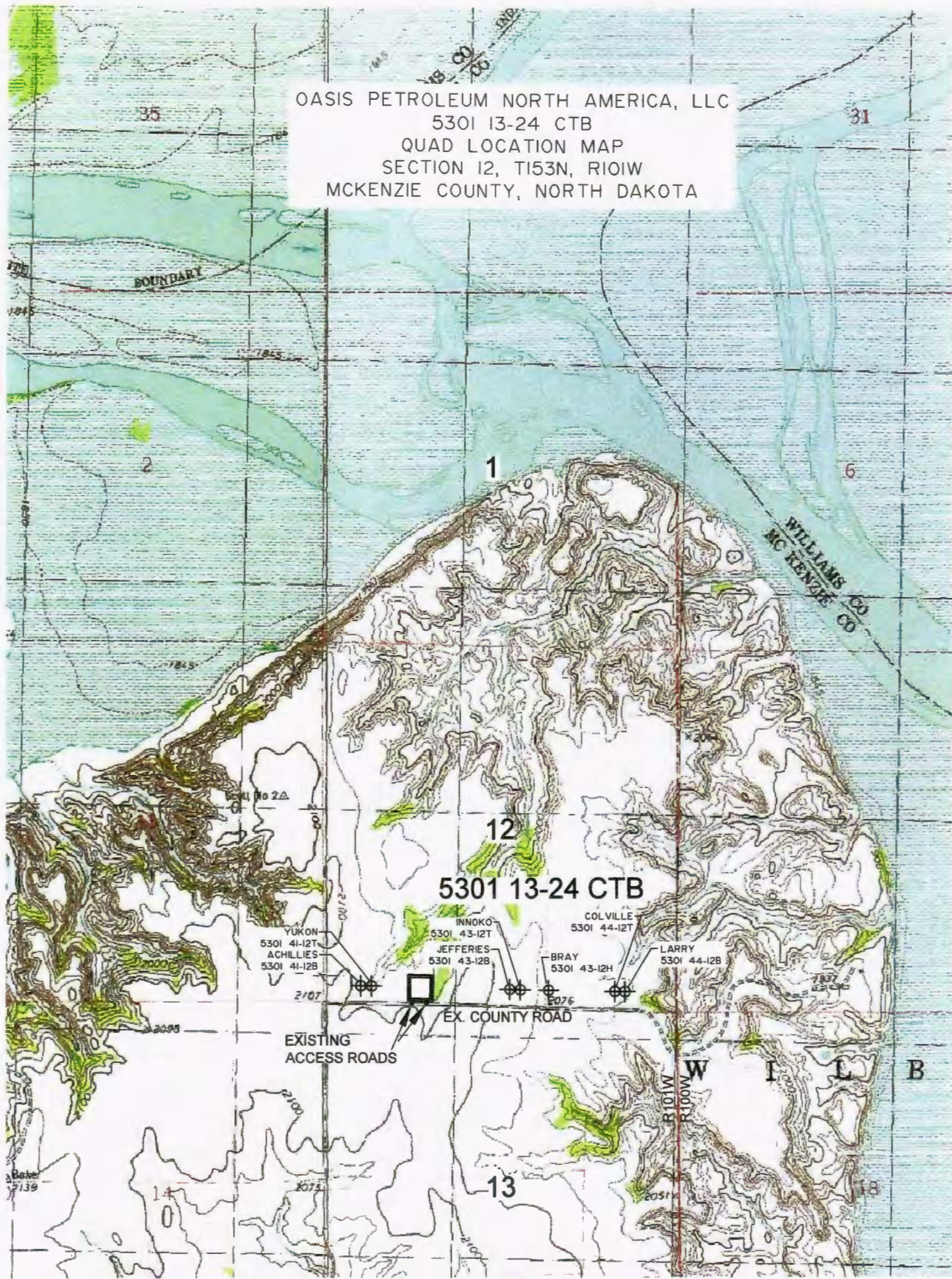
OASIS PETROLEUM NORTH AMERICA, LLC ACCESS APPROACH SECTION 12, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA	Project No.: 51209-249	Date: SEPT. 2012
Drawn By: J.D.M.	Checked By: D.K.K.	

Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, North Dakota 58270
 PH (408) 433-5817
 FAX (408) 433-5818
 Web: www.ie-inc.com
 Other offices in Minnesota, North Dakota and South Dakota



3/5
 SHEET NO.

OASIS PETROLEUM NORTH AMERICA, LLC
 5301 13-24 CTB
 QUAD LOCATION MAP
 SECTION 12, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA



© 2012, INTERSTATE ENGINEERING, INC.

4/5
 SHEET NO.



Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.iengi.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 12, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.D.M. Project No: S12-09-249
 Checked By: D.D.K. Date: SEPT. 2012

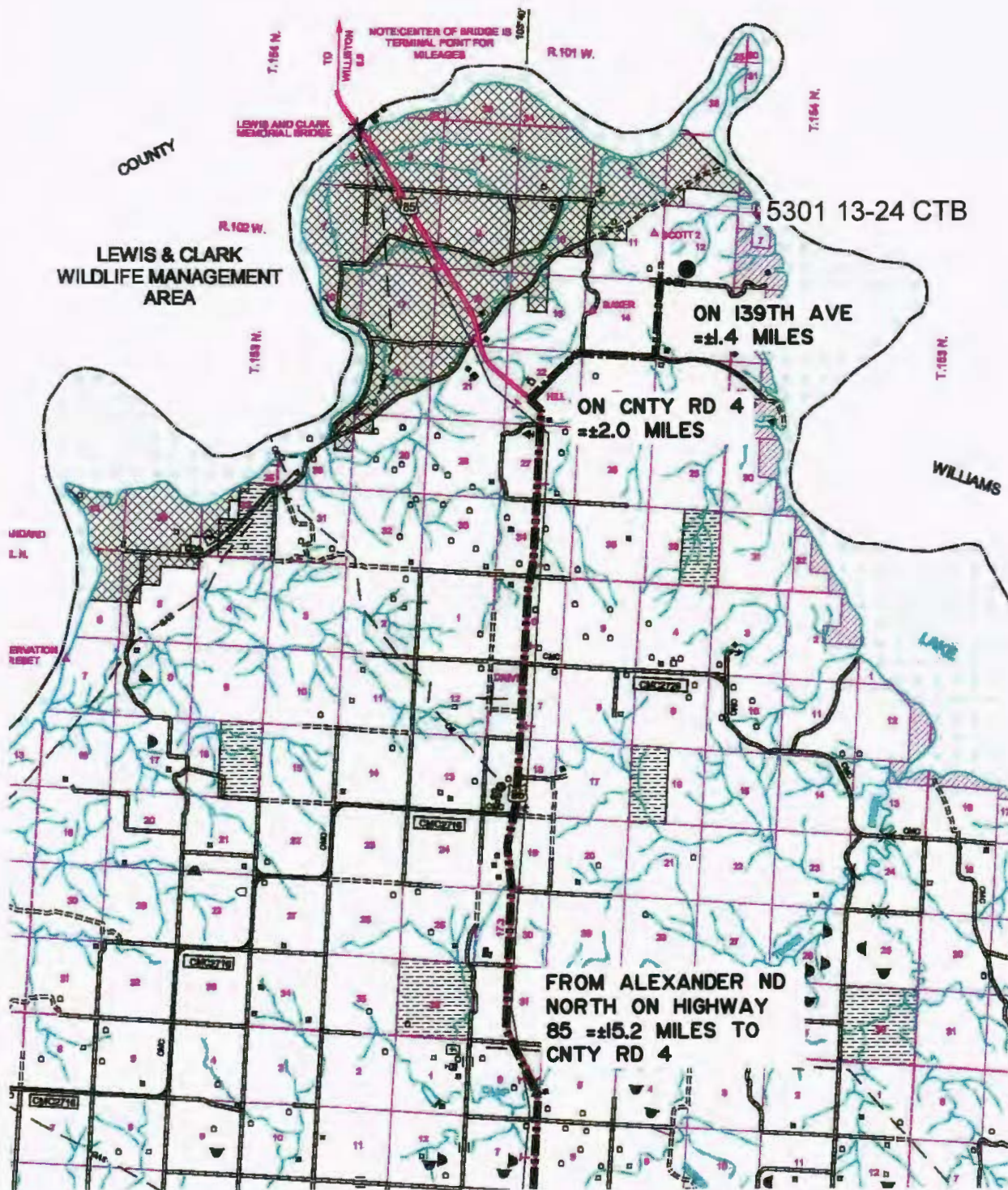
Revision No.	Date	By	Description
REV 1	7/10/13	JDM	ADDED WELLS

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 202 HOUSTON, TX 77002

"5301 13-24 CTB"

SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2012, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

5/5
SHEET NO.

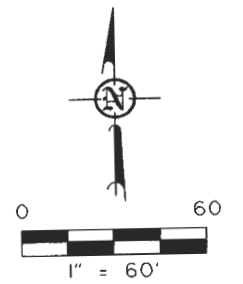
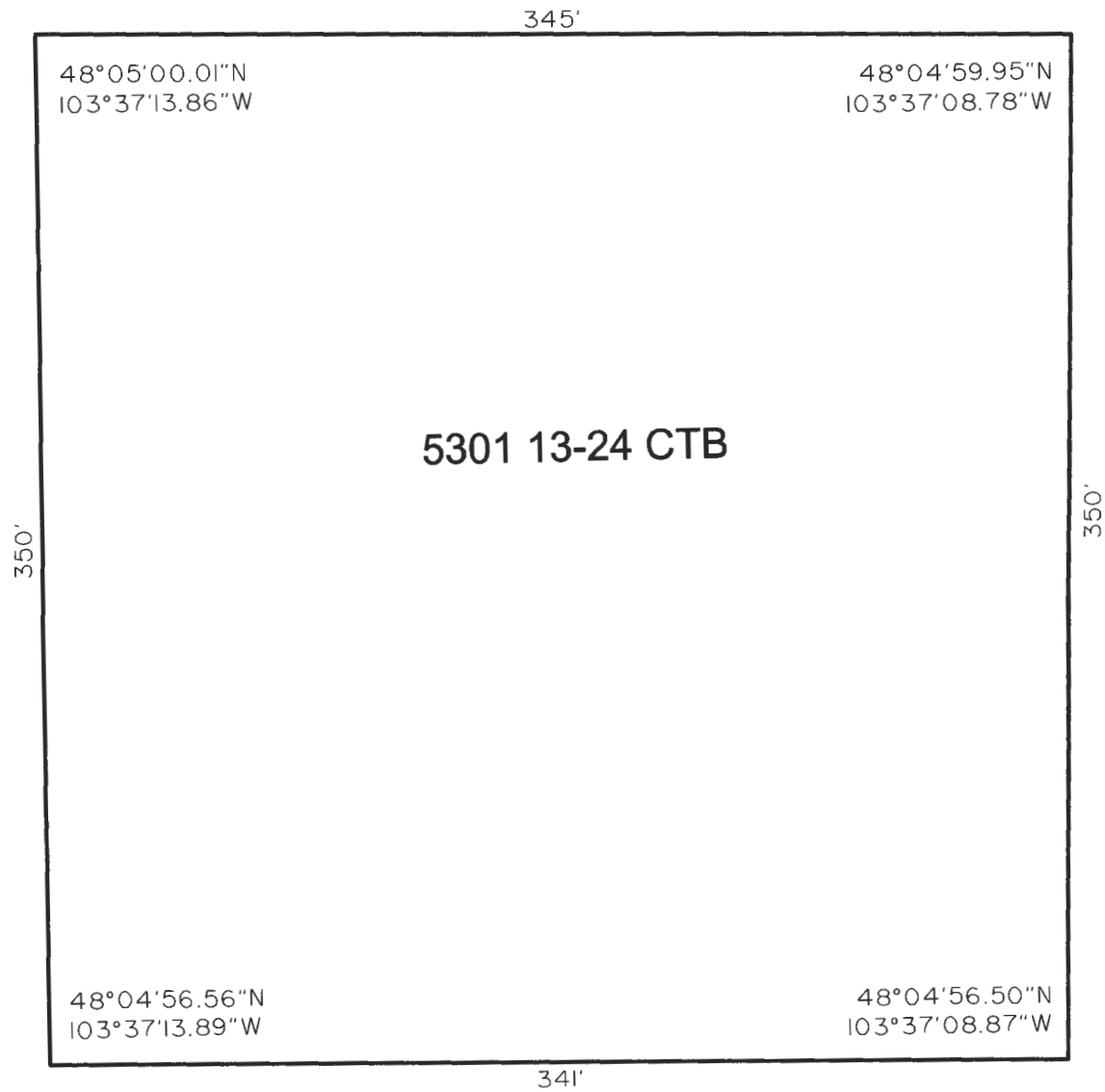


Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 58270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengi.com
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: J.D.M. Project No: S12-09-249
Checked By: D.D.K. Date: SEPT. 2012

Revision No.	Date	By	Description
REV 1	7/10/13	JDM	ADDED WELLS

LAT/LONG PAD CORNERS





SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA

OIL AND GAS DIVISION

600 EAST BOULEVARD DEPT 405

BISMARCK, ND 58505-0840

SFN 5749 (09-2006)

Well File No.

22220

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent

Approximate Start Date

☒ Report of Work Done

Date Work Completed

June 4, 2012

☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.

Approximate Start Date

☐ Drilling Prognosis

☐ Redrilling or Repair

☐ Casing or Liner

☐ Plug Well

☐ Supplemental History

☐ Temporarily Abandon

☐ Other

☐ Well Report

☐ Shooting

☐ Acidizing

☐ Fracture Treatment

☐ Change Production Method

☐ Reclamation

Well is now on pump

Well Name and Number

Jefferies 5301 43-12B

Footages

250 F S L

2510 F E L

Qtr-Qtr

SWSE

Section

12

Township

153 N

Range

101 W

Field

Baker

Pool

Bakken

County

McKenzie

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)

Address

City

State

Zip Code

DETAILS OF WORK

Effective June 4, 2012 the Jefferies 5301 43-12B is on pump.

Tubing: 2-7/8" L-80 tubing @ 10108

Pump: 2-1/2" x 2.0" x 24' insert pump @ 10086

Company	Telephone Number
Oasis Petroleum North America LLC	281 404-9491
Address	
1001 Fannin, Suite 1500	
City	State
Houston	TX
Signature	Zip Code
<i>Brandi Terry</i>	77002
Printed Name	Date
Brandi Terry	January 16, 2013
Title	
Regulatory Specialist	
Email Address	
bterry@oasispetroleum.com	

FOR STATE USE ONLY

☒ Received	☐ Approved
Date	
By	
Title	
PETROLEUM ENGINEER	



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - Form 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No. 22220
NDIC CTB No. 1-22220 222/100

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number JEFFERIES 5301 43-12B	Qtr-Qtr SWSE	Section 12	Township 153 N	Range 101 W	County McKenzie
--	------------------------	----------------------	--------------------------	-----------------------	---------------------------

Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9435	Field BAKER
--	---	-----------------------

Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9435	% Purchased 100%	Date Effective June 5, 2012
---	---	----------------------------	---------------------------------------

Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Field Address	City	State	Zip Code
---------------	------	-------	----------

Transporter Hiland Crude, LLC	Telephone Number (580) 616-2058	% Transported 75%	Date Effective June 5, 2012
---	---	-----------------------------	---------------------------------------

Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
---------------------------------	---------------------	--------------------	--------------------------

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other Transporters Transporting From This Lease	% Transported	Date Effective
---	---------------	----------------

Blackstone Crude Oil LLC	25%	June 5, 2012
---------------------------------	------------	---------------------

Other Transporters Transporting From This Lease	% Transported	Date Effective
---	---------------	----------------

Comments

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Date August 22, 2012
--	--------------------------------

Signature <i>Anette Terrell</i>	Printed Name Annette Terrell	Title Marketing Assistant
------------------------------------	--	-------------------------------------

Above Signature Witnessed By: Signature <i>Marian Hargis</i>	Printed Name Marian Hargis	Title Marketing Manager
--	--------------------------------------	-----------------------------------



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - Form 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No. 22220
NDIC CTB No. 22220

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number JEFFERIES 5301 43-12B	Qtr-Qtr SWSE	Section 12	Township 153 N	Range 101 W	County McKenzie
--	------------------------	----------------------	--------------------------	-----------------------	---------------------------

Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9435	Field BAKER
--	---	-----------------------

Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9435	% Purchased 100%	Date Effective April 1, 2012
---	---	----------------------------	--

Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Field Address	City	State	Zip Code
---------------	------	-------	----------

Transporter Banner Transportation Company LLC	Telephone Number (580) 616-2058	% Transported 100%	Date Effective April 1, 2012
---	---	------------------------------	--

Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
---------------------------------	---------------------	--------------------	--------------------------

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Comments		

ORIGINAL

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date June 8, 2012
Signature 	Printed Name Annette Terrell	Title Marketing Assistant

Above Signature Witnessed By: Signature 		Date JUN 14 2012
Signature 		Printed Name Dina Barron
		Title Mktg. Contracts Administrator

Oil & Gas Production Analyst

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)

Well File No.
22220



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Jefferies 5301 43-12B		Spacing Unit Description Sec. 13 & 24 T153N R101W	
Operator Oasis Petroleum North America LLC		Telephone Number 281-404-9500	Field Tyrone
Address 1001 Fannin, Suite 1500		Pool Bakken	
City Houston	State TX	Zip Code 77002	Permit Type ☐ Wildcat ☒ Development ☐ Extension

LOCATION OF WELL

At Surface 250 F S L	2510 F E L	Qtr-Qtr LOT2	Section 4	Township 155 N	Range 101 W	County Williams
Spud Date 12/13/2011	Date TD Reached 2/22/2012	Drilling Contractor and Rig Number Nabors 149		KB Elevation (Ft) 2118	Graded Elevation (Ft) 2093	
Type of Electric and Other Logs Run (See Instructions)						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8		2170	13 1/2	36			630	
Vertical Hole	Intermediate	7		11093	8 3/4	29 & 32			755	3296
Lateral1	Liner	4 1/2	10200	21220	6	11.6				

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD Ft)	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	21220	Perforations	11093 21220	10270		3/15/2012			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s). This Completion, Top and Bottom. (MD Ft) Lateral 1-11093'-21220'						Name of Zone (If Different from Pool Name)			
Date Well Completed (SEE INSTRUCTIONS) 3/22/2012		Producing Method Flowing		Pumping-Size & Type of Pump		Well Status (Producing or Shut-In) Producing			
Date of Test 4/7/2012	Hours Tested 24	Choke Size 52 /64	Production for Test	Oil (Bbls) 1962	Gas (MCF) 1798	Water (Bbls) 1793	Oil Gravity-API (Corr.) 42.0 °	Disposition of Gas Sold	
Flowing Tubing Pressure (PSI) 813		Flowing Casing Pressure (PSI)		Calculated 24-Hour Rate	Oil (Bbls) 1962	Gas (MCF) 1798	Water (Bbls) 1793	Gas-Oil Ratio 916	

orig.

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test:

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 3/15/2012	Stimulated Formation Bakken	Top (Ft) 11093	Bottom (Ft) 21220	Stimulation Stages 36	Volume 83063	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 4540494	Maximum Treatment Pressure (PSI) 8489		Maximum Treatment Rate (BBLS/Min) 46.0	
Details 20/40 ceramic-2,683,424 40/70 sand- 1,857,070						

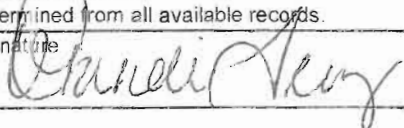
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address bterry@oasispetroleum.com	Date 5/18/2012
	Signature 	Printed Name Brandi Terry



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Monday, February 27, 2012

State of North Dakota

Subject: **Surveys**

Re: **Oasis**
Jefferies 5301 43-12B
McKenzie, ND

Enclosed, please find the original and one copy of the survey performed on the above-referenced well by Ryan Directional Services, Inc. Other information required by your office is as follows:

<i>Surveyor Name</i>	<i>Surveyor Title</i>	<i>Borehole Number</i>	<i>Start Depth</i>	<i>End Depth</i>	<i>Start Date</i>	<i>End Date</i>	<i>Type of Survey</i>	<i>TD Straight Line Projection</i>
Pitre, Jace	MWD Operator	O.H.	2170'	21206'	02/03/12	02/21/12	MWD	21250'

A certified plat on which the bottom hole location is oriented both to the surface location and to the lease lines (or unit lines in case of pooling) is attached to the survey report. If any other information is required please contact the undersigned at the letterhead address or phone number.

Douglas Hudson
Well Planner



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Monday, February 27, 2012

State of North Dakota

Subject: **Survey Certification Letter**

Re: **Oasis**
Jefferies 5301 43-12B
McKenzie, ND

I, Jace Pitre, certify that; I am employed by Ryan Directional Services, Inc.; that I did on the conduct or supervise the taking of the following MWD surveys:

on the day(s) of 2/3/2012 thru 2/21/2012 from a depth of 2170' MD to a depth of 21206' MD and Straight line projection to TD 21250' MD;

that the data is true, correct, complete, and within the limitations of the tool as set forth by Ryan Directional Services, Inc.; that I am authorized and qualified to make this report; that this survey was conducted at the request of Oasis for the Jefferies 5301 43-12B; in McKenzie, ND.

Jace Pitre

Jace Pitre

MWD Operator

Ryan Directional Services, Inc.

Report #: **1**
Date: **2-Jan-12**



Ryan Job #: **5279**
Kit #: **44**

SURVEY REPORT

Customer: **Oasis Petroleum**
Well Name: **Jefferies 5301 43-12B**
Block or Section: **24 153N 101W**
Rig #: **Nabors 149**
Calculation Method: **Minimun Curvature Calculation**

MWD Operator: **Jace Pitre**
Directional Drillers: **Jason Standelin**
Survey Corrected To: **True North**
Vertical Section Direction: **184.63**
Survey Correction: **8.56**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
Tie in to Gyro Surveys									
Tie In	2170	0.00	0.00		2170.00				
1	2183	1.40	176.10	50.00	2183.00	0.16	-0.16	0.01	10.77
2	2278	2.20	193.60	51.00	2277.95	3.11	-3.09	-0.34	1.01
3	2371	1.20	206.90	59.00	2370.91	5.77	-5.69	-1.20	1.15
4	2467	1.60	195.00	62.00	2466.88	8.02	-7.88	-2.00	0.51
5	2562	2.10	202.40	66.00	2561.83	10.98	-10.77	-3.01	0.58
6	2658	1.50	231.20	71.00	2657.79	13.52	-13.19	-4.66	1.11
7	2753	1.90	245.00	73.00	2752.74	15.15	-14.63	-7.05	0.60
8	2849	1.90	251.30	75.00	2848.69	16.57	-15.81	-10.00	0.22
9	2944	1.00	224.70	77.00	2943.66	17.83	-16.91	-12.08	1.16
10	3040	1.10	129.20	78.00	3039.65	18.99	-18.09	-11.95	1.62
11	3135	1.00	110.40	82.00	3134.64	19.74	-18.95	-10.47	0.38
12	3230	0.80	102.60	84.00	3229.62	20.05	-19.39	-9.05	0.25
13	3326	0.70	104.00	86.00	3325.62	20.24	-19.67	-7.82	0.11
14	3422	0.60	89.10	89.00	3421.61	20.29	-19.81	-6.75	0.20
15	3516	0.50	108.90	93.00	3515.60	20.34	-19.93	-5.87	0.23
16	3612	0.70	102.20	95.00	3611.60	20.52	-20.19	-4.90	0.22
17	3707	0.80	115.30	96.00	3706.59	20.83	-20.60	-3.73	0.21
18	3803	0.80	113.40	98.00	3802.58	21.28	-21.15	-2.51	0.03
19	3898	0.70	137.90	100.00	3897.57	21.90	-21.85	-1.52	0.35
20	3993	0.60	116.90	104.00	3992.57	22.48	-22.50	-0.68	0.27
21	4089	0.60	131.20	104.00	4088.56	22.97	-23.06	0.14	0.16
22	4184	0.90	255.10	105.00	4183.56	23.52	-23.58	-0.20	1.40
23	4280	1.00	263.50	107.00	4279.55	23.93	-23.87	-1.76	0.18
24	4375	0.90	265.40	109.00	4374.53	24.21	-24.02	-3.33	0.11
25	4469	0.70	258.70	111.00	4468.52	24.49	-24.19	-4.63	0.23
26	4564	0.70	269.90	113.00	4563.52	24.69	-24.31	-5.78	0.14
27	4660	0.30	278.60	113.00	4659.51	24.73	-24.27	-6.61	0.42
28	4755	0.30	293.10	114.00	4754.51	24.63	-24.14	-7.09	0.08
29	4850	0.10	100.50	116.00	4849.51	24.56	-24.05	-7.24	0.42
30	4946	0.30	114.80	116.00	4945.51	24.65	-24.17	-6.93	0.21
31	5041	0.30	230.30	118.00	5040.51	24.91	-24.44	-6.89	0.53
32	5137	0.60	272.00	120.00	5136.51	25.11	-24.58	-7.59	0.44
33	5232	0.80	316.50	122.00	5231.50	24.69	-24.08	-8.54	0.59
34	5328	0.70	265.20	122.00	5327.49	24.34	-23.65	-9.59	0.68
35	5423	0.70	248.70	123.00	5422.49	24.69	-23.90	-10.71	0.21
36	5519	1.10	248.00	125.00	5518.47	25.36	-24.46	-12.11	0.42
37	5614	1.30	239.30	125.00	5613.45	26.39	-25.35	-13.88	0.28
38	5710	1.30	231.80	127.00	5709.43	27.76	-26.58	-15.67	0.18
39	5806	1.50	223.20	131.00	5805.40	29.48	-28.17	-17.39	0.30
40	5901	1.90	219.50	131.00	5900.36	31.75	-30.29	-19.24	0.44
41	5996	1.70	222.10	132.00	5995.31	34.16	-32.56	-21.19	0.23
42	6092	1.70	227.90	132.00	6091.27	36.33	-34.57	-23.20	0.18
43	6187	1.30	274.10	132.00	6186.24	37.36	-35.43	-25.32	1.30
44	6283	2.10	312.00	133.00	6282.20	36.31	-34.18	-27.71	1.39
45	6378	2.60	335.00	136.00	6377.12	33.38	-31.06	-29.92	1.11
46	6474	1.60	346.80	138.00	6473.06	30.21	-27.78	-31.14	1.13
47	6569	1.90	358.10	138.00	6568.01	27.38	-24.92	-31.50	0.48
48	6664	1.60	6.60	140.00	6662.97	24.49	-22.03	-31.40	0.42
49	6760	0.70	348.60	140.00	6758.95	22.59	-20.12	-31.36	1.00
50	6855	1.00	14.40	143.00	6853.94	21.21	-18.75	-31.27	0.50
51	6951	0.50	164.90	143.00	6949.93	20.78	-18.34	-30.95	1.52
52	7045	0.80	180.00	147.00	7043.93	21.82	-19.39	-30.84	0.36
53	7141	0.40	180.00	149.00	7139.92	22.82	-20.40	-30.84	0.42
54	7236	0.80	195.90	150.00	7234.92	23.80	-21.37	-31.02	0.45
55	7331	0.80	187.10	152.00	7329.91	25.12	-22.66	-31.29	0.13
56	7427	0.60	175.00	154.00	7425.90	26.28	-23.83	-31.33	0.26
57	7522	0.60	143.40	156.00	7520.89	27.15	-24.73	-30.99	0.34
58	7617	0.50	119.40	158.00	7615.89	27.69	-25.33	-30.33	0.26
59	7713	0.80	125.40	156.00	7711.88	28.21	-25.92	-29.42	0.32
60	7808	0.80	137.90	159.00	7806.88	29.01	-26.80	-28.43	0.18
61	7903	0.50	99.10	159.00	7901.87	29.49	-27.36	-27.58	0.54
62	7999	0.40	78.30	160.00	7997.87	29.43	-27.35	-26.84	0.20
63	8094	0.30	81.70	160.00	8092.86	29.28	-27.25	-26.27	0.11
64	8189	0.30	70.50	160.00	8187.86	29.13	-27.13	-25.79	0.06
65	8285	0.40	32.30	167.00	8283.86	28.73	-26.77	-25.37	0.26

Report #: **1**
Date: **2-Jan-12**



Ryan Job # **5279**
Kit # **44**

SURVEY REPORT

Customer: **Oasis Petroleum**
Well Name: **Jefferies 5301 43-12B**
Block or Section: **24 153N 101W**
Rig #: **Nabors 149**
Calculation Method: **Minimun Curvature Calculation**

MWD Operator: **Jace Pitre**
Directional Drillers: **Jason Standelin**
Survey Corrected To: **True North**
Vertical Section Direction: **184.63**
Survey Correction: **8.56**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
66	8380	0.40	51.90	165.00	8378.86	28.21	-26.28	-24.93	0.14
67	8476	0.50	38.40	167.00	8474.86	27.63	-25.75	-24.41	0.15
68	8571	0.60	19.30	168.00	8569.85	26.81	-24.95	-23.99	0.22
69	8666	0.80	16.60	168.00	8664.84	25.68	-23.85	-23.63	0.21
70	8761	0.60	12.90	172.00	8759.84	24.53	-22.73	-23.33	0.22
71	8857	0.60	5.30	172.00	8855.83	23.53	-21.73	-23.17	0.08
72	8952	0.30	60.20	181.00	8950.83	22.90	-21.12	-22.91	0.52
73	9047	0.30	86.90	174.00	9045.83	22.72	-20.98	-22.45	0.15
74	9143	0.40	33.80	176.00	9141.83	22.40	-20.69	-22.01	0.34
75	9238	0.50	31.80	172.00	9236.82	21.74	-20.06	-21.61	0.11
76	9334	0.70	22.10	176.00	9332.82	20.81	-19.16	-21.17	0.23
77	9429	0.80	39.40	176.00	9427.81	19.71	-18.11	-20.53	0.26
78	9524	0.80	15.00	177.00	9522.80	18.51	-16.96	-19.93	0.36
79	9620	0.70	13.70	177.00	9618.79	17.27	-15.74	-19.62	0.11
80	9715	0.80	13.80	179.00	9713.79	16.04	-14.53	-19.33	0.11
81	9811	0.80	9.00	181.00	9809.78	14.71	-13.22	-19.06	0.07
82	9907	0.60	343.90	183.00	9905.77	13.58	-12.07	-19.10	0.38
83	10002	0.90	324.10	185.00	10000.76	12.54	-10.99	-19.67	0.41
84	10097	0.90	311.40	192.00	10095.75	11.53	-9.89	-20.67	0.21
85	10193	1.10	278.30	194.00	10191.74	11.02	-9.26	-22.15	0.63
86	10259	1.50	250.40	167.00	10257.72	11.33	-9.46	-23.59	1.12
87	10290	3.80	223.40	168.00	10288.68	12.30	-10.34	-24.67	8.24
88	10322	7.80	214.30	172.00	10320.52	15.02	-12.91	-26.63	12.79
89	10354	10.80	215.00	172.00	10352.09	19.49	-17.16	-29.57	9.38
90	10386	14.20	211.90	172.00	10383.33	25.57	-22.95	-33.37	10.83
91	10418	16.70	209.30	172.00	10414.17	33.24	-30.29	-37.69	8.10
92	10449	19.40	210.00	172.00	10443.64	41.94	-38.64	-42.45	8.74
93	10481	22.70	211.30	174.00	10473.50	52.26	-48.52	-48.31	10.41
94	10513	26.00	211.50	176.00	10502.65	64.04	-59.78	-55.19	10.32
95	10545	29.30	211.90	161.00	10530.99	77.26	-72.41	-62.99	10.33
96	10577	33.20	212.50	163.00	10558.34	91.97	-86.45	-71.84	12.23
97	10609	37.30	212.70	172.00	10584.47	108.27	-102.01	-81.79	12.82
98	10640	39.90	214.10	172.00	10608.70	125.22	-118.15	-92.44	8.85
99	10672	41.00	216.90	172.00	10633.05	143.03	-135.04	-104.50	6.64
100	10709	44.30	218.10	168.00	10660.26	164.08	-154.92	-119.76	9.19
101	10741	49.90	217.70	168.00	10682.04	183.67	-173.41	-134.15	17.52
102	10772	56.80	216.90	170.00	10700.53	204.60	-193.19	-149.21	22.35
103	10799	64.30	215.80	171.00	10713.79	224.59	-212.12	-163.13	28.00
104	10836	70.90	214.80	171.00	10727.89	254.00	-240.02	-182.88	18.01
105	10868	76.50	213.20	174.00	10736.86	280.76	-265.48	-200.04	18.15
106	10900	81.80	210.70	176.00	10742.89	308.67	-292.14	-216.66	18.25
107	10931	85.30	208.60	179.00	10746.37	336.58	-318.90	-231.89	13.14
108	10963	90.00	207.20	179.00	10747.68	365.94	-347.15	-246.85	15.32
109	10995	92.00	206.20	179.00	10747.12	395.59	-375.73	-261.22	6.99
110	11027	92.60	205.60	179.00	10745.84	425.39	-404.49	-275.19	2.65
111	11059	89.40	205.30	183.00	10745.28	455.29	-433.38	-288.94	10.04
112	11098	87.40	204.70	217.00	10746.37	491.83	-468.71	-305.41	5.35
113	11129	87.50	205.10	217.00	10747.75	520.89	-496.80	-318.45	1.33
114	11161	86.80	204.00	217.00	10749.34	550.93	-525.87	-331.73	4.07
115	11222	86.60	204.10	215.00	10752.85	608.37	-581.48	-356.55	0.37
116	11313	88.30	203.10	217.00	10756.90	694.34	-664.79	-392.94	2.17
117	11405	88.10	203.00	221.00	10759.79	781.58	-749.40	-428.95	0.24
118	11496	90.60	201.90	221.00	10760.82	868.20	-833.49	-463.69	3.00
119	11587	89.80	201.40	224.00	10760.50	955.21	-918.07	-497.26	1.04
120	11679	89.40	200.20	224.00	10761.15	1043.57	-1004.07	-529.93	1.37
121	11771	90.30	198.60	224.00	10761.39	1132.53	-1090.84	-560.49	2.00
122	11863	90.90	198.80	226.00	10760.42	1221.76	-1177.98	-589.98	0.69
123	11955	90.90	198.10	230.00	10758.98	1311.08	-1265.24	-619.10	0.76
124	12046	89.60	197.10	228.00	10758.58	1399.76	-1351.98	-646.61	1.80
125	12137	89.20	196.90	231.00	10759.53	1488.64	-1439.00	-673.22	0.49
126	12229	89.30	197.50	231.00	10760.74	1578.43	-1526.87	-700.42	0.66
127	12321	89.50	197.40	233.00	10761.70	1668.13	-1614.63	-728.00	0.24
128	12413	89.10	197.40	235.00	10762.83	1757.85	-1702.42	-755.51	0.43
129	12504	90.20	197.40	235.00	10763.38	1846.59	-1789.25	-782.73	1.21
130	12595	89.20	196.80	237.00	10763.86	1935.44	-1876.22	-809.48	1.28

Report #: **1**
Date: **2-Jan-12**



Ryan Job # **5279**
Kit # **44**

SURVEY REPORT

Customer: **Oasis Petroleum**
Well Name: **Jefferies 5301 43-12B**
Block or Section: **24 153N 101W**
Rig #: **Nabors 149**
Calculation Method: **Minimun Curvature Calculation**

MWD Operator: **Jace Pitre**
Directional Drillers: **Jason Standelin**
Survey Corrected To: **True North**
Vertical Section Direction: **184.63**
Survey Correction: **8.56**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
131	12686	90.00	195.30	235.00	10764.49	2024.64	-1963.67	-834.64	1.87
132	12778	90.20	195.00	239.00	10764.33	2115.09	-2052.47	-858.68	0.39
133	12869	91.90	195.40	240.00	10762.67	2204.53	-2140.27	-882.54	1.92
134	12961	90.20	195.10	235.00	10760.98	2294.93	-2229.02	-906.73	1.88
135	13053	89.40	194.30	237.00	10761.30	2385.51	-2318.00	-930.08	1.23
136	13144	88.30	193.50	240.00	10763.13	2475.31	-2406.32	-951.93	1.49
137	13235	88.40	193.40	242.00	10765.75	2565.19	-2494.78	-973.09	0.16
138	13327	88.10	192.60	239.00	10768.56	2656.17	-2584.38	-993.78	0.93
139	13423	86.80	195.90	239.00	10772.83	2750.71	-2677.32	-1017.38	3.69
140	13520	86.90	196.40	239.00	10778.16	2845.61	-2770.35	-1044.32	0.52
141	13615	90.10	190.30	237.00	10780.65	2939.43	-2862.70	-1066.23	7.25
142	13711	93.40	187.50	237.00	10777.72	3035.09	-2957.48	-1081.07	4.51
143	13807	91.60	186.40	239.00	10773.53	3130.92	-3052.68	-1092.68	2.20
144	13902	89.70	184.50	240.00	10772.45	3225.89	-3147.23	-1101.70	2.83
145	13998	89.90	181.60	240.00	10772.79	3321.84	-3243.09	-1106.81	3.03
146	14094	90.90	179.90	242.00	10772.12	3417.62	-3339.07	-1108.06	2.05
147	14190	90.40	179.10	246.00	10771.03	3513.23	-3435.06	-1107.23	0.98
148	14286	88.10	178.60	248.00	10772.28	3608.72	-3531.02	-1105.30	2.45
149	14381	89.80	178.30	244.00	10774.02	3703.15	-3625.97	-1102.73	1.82
150	14477	92.30	178.30	246.00	10772.27	3798.54	-3721.90	-1099.88	2.60
151	14572	92.00	179.10	248.00	10768.70	3892.97	-3816.81	-1097.73	0.90
152	14666	88.30	176.90	246.00	10768.46	3986.32	-3910.73	-1094.45	4.58
153	14760	90.50	177.10	246.00	10769.44	4079.47	-4004.59	-1089.53	2.35
154	14856	91.60	175.90	246.00	10767.68	4174.49	-4100.39	-1083.67	1.70
155	14952	91.20	175.80	249.00	10765.34	4269.34	-4196.11	-1076.73	0.43
156	15048	90.70	175.70	249.00	10763.74	4364.17	-4291.84	-1069.61	0.53
157	15144	90.60	174.90	251.00	10762.65	4458.89	-4387.51	-1061.75	0.84
158	15239	90.50	174.90	251.00	10761.74	4552.52	-4482.13	-1053.30	0.11
159	15334	90.40	174.90	253.00	10761.00	4646.15	-4576.75	-1044.86	0.11
160	15429	88.80	174.80	253.00	10761.66	4739.77	-4671.36	-1036.33	1.69
161	15524	90.60	174.40	251.00	10762.16	4833.31	-4765.93	-1027.39	1.94
162	15620	90.40	174.10	253.00	10761.32	4927.74	-4861.44	-1017.77	0.38
163	15714	89.80	173.90	255.00	10761.16	5020.12	-4954.93	-1007.95	0.67
164	15810	89.20	176.20	251.00	10761.99	5114.77	-5050.56	-999.66	2.48
165	15905	91.60	179.20	251.00	10761.33	5209.06	-5145.46	-995.85	4.04
166	16001	91.50	176.80	255.00	10758.73	5304.37	-5241.36	-992.50	2.50
167	16097	90.90	176.90	255.00	10756.72	5399.47	-5337.20	-987.23	0.63
168	16192	90.20	175.80	257.00	10755.81	5493.47	-5432.00	-981.18	1.37
169	16288	91.30	178.10	255.00	10754.55	5588.60	-5527.85	-976.07	2.66
170	16383	89.80	177.00	257.00	10753.64	5682.86	-5622.75	-972.01	1.96
171	16478	90.00	177.90	257.00	10753.81	5777.12	-5717.65	-967.79	0.97
172	16574	90.70	176.80	257.00	10753.22	5872.34	-5813.55	-963.35	1.36
173	16670	90.40	175.60	258.00	10752.30	5967.30	-5909.33	-956.99	1.29
174	16766	92.50	177.60	257.00	10749.87	6062.31	-6005.12	-951.30	3.02
175	16861	91.70	177.50	257.00	10746.39	6156.53	-6099.97	-947.24	0.85
176	16957	91.70	179.80	257.00	10743.54	6251.95	-6195.89	-944.98	2.39
177	17052	88.40	178.50	260.00	10743.46	6346.51	-6290.87	-943.57	3.73
178	17148	90.80	178.90	257.00	10744.13	6441.98	-6386.83	-941.39	2.53
179	17242	91.80	178.70	258.00	10742.00	6535.47	-6480.79	-939.42	1.08
180	17337	91.40	178.20	260.00	10739.34	6629.88	-6575.72	-936.85	0.67
181	17431	91.40	178.40	260.00	10737.05	6723.28	-6669.65	-934.07	0.21
182	17527	91.00	178.80	262.00	10735.04	6818.73	-6765.60	-931.72	0.59
183	17622	90.90	178.50	262.00	10733.46	6913.20	-6860.56	-929.48	0.33
184	17718	89.00	177.90	262.00	10733.54	7008.59	-6956.50	-926.47	2.08
185	17814	90.60	178.30	260.00	10733.88	7103.96	-7052.45	-923.28	1.72
186	17909	90.30	178.30	262.00	10733.13	7198.38	-7147.40	-920.47	0.32
187	18004	90.60	178.60	260.00	10732.39	7292.82	-7242.37	-917.90	0.45
188	18100	92.50	178.00	260.00	10729.79	7388.20	-7338.28	-915.05	2.08
189	18196	91.40	178.30	262.00	10726.52	7483.53	-7434.18	-911.95	1.19
190	18291	90.20	176.90	264.00	10725.20	7577.80	-7529.08	-907.97	1.94
191	18387	91.20	177.80	262.00	10724.02	7673.02	-7624.97	-903.54	1.40
192	18482	90.30	177.00	264.00	10722.78	7767.25	-7719.86	-899.23	1.27
193	18577	91.40	177.80	260.00	10721.37	7861.49	-7814.75	-894.92	1.43
194	18673	92.20	177.10	264.00	10718.36	7956.68	-7910.61	-890.65	1.11
195	18769	92.10	175.70	264.00	10714.75	8051.63	-8006.35	-884.63	1.46

Report #: **1**
 Date: **2-Jan-12**



Ryan Job # **5279**
 Kit # **44**

SURVEY REPORT

Customer: **Oasis Petroleum**
 Well Name: **Jefferies 5301 43-12B**
 Block or Section: **24 153N 101W**
 Rig #: **Nabors 149**
 Calculation Method: **Minimun Curvature Calculation**

MWD Operator: **Jace Pitre**
 Directional Drillers: **Jason Stadelin**
 Survey Corrected To: **True North**
 Vertical Section Direction: **184.63**
 Survey Correction: **8.56**
 Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
196	18864	92.30	179.10	260.00	10711.11	8145.79	-8101.16	-880.32	3.58
197	18960	92.20	177.40	262.00	10707.34	8241.12	-8197.04	-877.39	1.77
198	19056	92.30	179.90	262.00	10703.57	8336.51	-8292.93	-875.13	2.60
199	19152	89.80	179.90	262.00	10701.81	8432.16	-8388.91	-874.96	2.60
200	19247	89.40	179.20	264.00	10702.47	8526.79	-8483.90	-874.22	0.85
201	19342	91.70	180.30	264.00	10701.56	8621.43	-8578.89	-873.80	2.68
202	19437	91.30	178.40	266.00	10699.07	8715.99	-8673.85	-872.73	2.04
203	19532	90.50	177.80	266.00	10697.58	8810.36	-8768.78	-869.58	1.05
204	19625	91.90	178.80	264.00	10695.63	8902.77	-8861.72	-866.82	1.85
205	19721	91.70	177.60	266.00	10692.62	8998.12	-8957.62	-863.80	1.27
206	19817	90.90	177.50	267.00	10690.44	9093.36	-9053.51	-859.70	0.84
207	19914	91.60	178.40	266.00	10688.32	9189.68	-9150.42	-856.23	1.18
208	20010	90.90	177.70	267.00	10686.23	9285.02	-9246.34	-852.97	1.03
209	20110	92.50	178.10	253.00	10683.26	9384.28	-9346.23	-849.30	1.65
210	20206	91.90	178.10	257.00	10679.58	9479.59	-9442.10	-846.12	0.63
211	20301	90.80	178.10	258.00	10677.34	9573.95	-9537.02	-842.97	1.16
212	20397	92.00	177.80	257.00	10674.99	9669.26	-9632.93	-839.54	1.29
213	20492	90.80	178.10	258.00	10672.67	9763.59	-9727.84	-836.14	1.30
214	20587	91.30	179.10	257.00	10670.93	9858.05	-9822.79	-833.82	1.18
215	20683	92.00	179.30	258.00	10668.17	9953.58	-9918.74	-832.48	0.76
216	20779	90.00	179.20	258.00	10666.49	10049.13	-10014.72	-831.23	2.09
217	20875	88.80	179.40	258.00	10667.50	10144.71	-10110.70	-830.05	1.27
218	20970	90.50	179.50	258.00	10668.08	10239.32	-10205.69	-829.14	1.79
219	21066	90.90	180.10	258.00	10666.91	10334.97	-10301.68	-828.81	0.75
220	21162	90.70	180.40	258.00	10665.57	10430.68	-10397.67	-829.23	0.38
221	21206	89.90	179.80	258.00	10665.33	10474.54	-10441.67	-829.30	2.27
Projection	21250	89.90	179.80	258.00	10665.41	10518.38	-10485.67	-829.15	0.00

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
22220

[Handwritten signature]

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date April 3, 2012
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	
Approximate Start Date	

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Waiver from tubing/packer requirement

Well Name and Number Jefferies 5301 43-12B							
Footages 250 F S L		Qtr-Qtr 2510 F E L		Section SWSE	Township 12	Range 153 N	
Field Baker		Pool Bakken		County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a waiver from the tubing/pkr requirement included in NDAC 43-02-03-21: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.
The following assurances apply:

1. The well is equipped with new 29 & 32# casing at surface with an API burst rating of 11,220 psi
2. The frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure.
3. Damage to the casing during the frac would be detected immediately by monitoring equipment.
4. The casing is exposed to significantly lower rates and pressures during flow back than during the frac job.
5. The frac fluid and formation fluids have very low corrosion and erosion rates.
6. Production equipment will be installed as soon as possible after the well ceases flowing.
7. A 300# gauge will be installed on the surface casing during the flowback period.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>[Handwritten Signature]</i>	Printed Name Brandi Terry		
Title Regulatory Specialist	Date April 3, 2012		
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date <i>April 5, 2012</i>	
By <i>[Handwritten Signature]</i>	
Title PETROLEUM ENGINEER	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

22220
CTB# 222100-01

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date March 19, 2012	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other	Central production facility-commingle prod

Well Name and Number (see details)					
Footages		Qtr-Qtr	Section	Township	Range
F	L	F	L	12	153 N
Field Baker		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests approval to commingle oil and gas in a central production facility known as: Oasis 5301 13-24CTB with common ownership for the following wells:
Well File #22100 Achilles 5301 41-12B SW SW 12-153-101 API 33-053-03912
Well File #22220 Jefferies 5301 43-12B SW SE 12-153-101 API 33-053-03936
Well File #20864 Bray 5301 43-12H SW SE 12-153-101 API 33-053-03609
Please find the following attachments:
1. A schematic drawing of the facility which diagrams the testing, treating, routing, and transferring of production.
2. A plat showing the location of the central facility
3. Affidavit of title indicating common ownership

Oasis will allocate production measured at the central production facility to the various wells on the basis of isolated production tests utilizing oil, gas and water meters on a test separator at the central production facility. Oasis will measure the production from each well separately each month for a minimum of three days. Oasis believes that such allocation will result in an accurate determination of production from each well. Tank vapor gas is being recovered and burned by a 98% DRE enclosed combustor.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Brandi Terry</i>	Printed Name Brandi Terry		
Title Regulatory Specialist	Date March 19, 2012		
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 3-23-12	
By ORIGINAL SIGNED BY DARYL GRONFUR	
Title METER SPECIALIST	

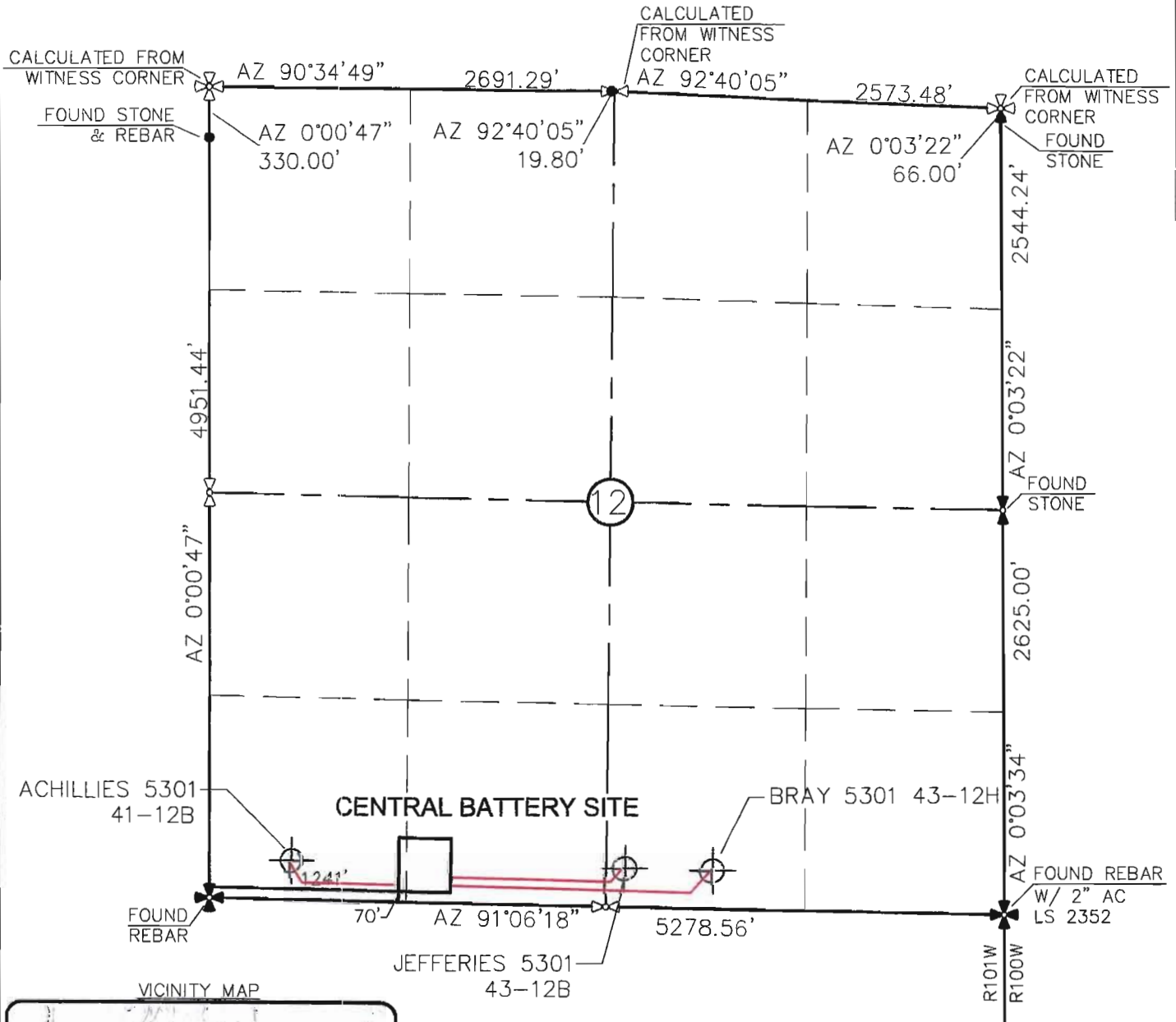
ADJACENT WELL LOCATIONS PLAT

OASIS PETROLEUM NORTH AMERICA, LLC

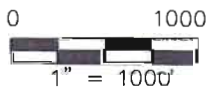
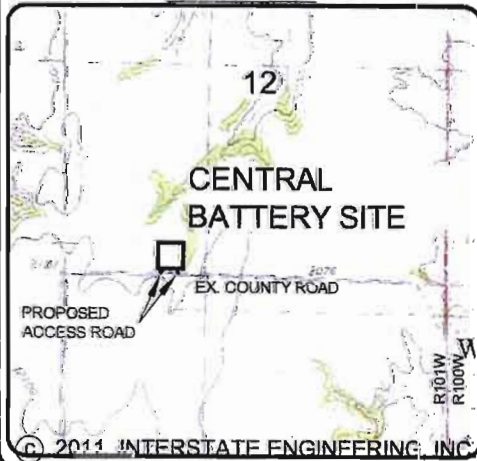
1001 FANNIN, SUITE 202 HOUSTON, TX 77002

CENTRAL BATTERY SITE

SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



VICINITY MAP



- ✖ - MONUMENT - RECOVERED
- ✖ - MONUMENT - NOT RECOVERED

1/1

SHEET NO.



Professional's you need, people you trust

Interstate Engineering, Inc.
P.O. Box 848
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.ieng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
ADJACENT WELLS PLAT
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: J.J.S. Project No.: S11-09-137
Checked By: D.K. Date: MARCH 2012

Revision No.	Date	By	Description

Oasis Petroleum Inc.

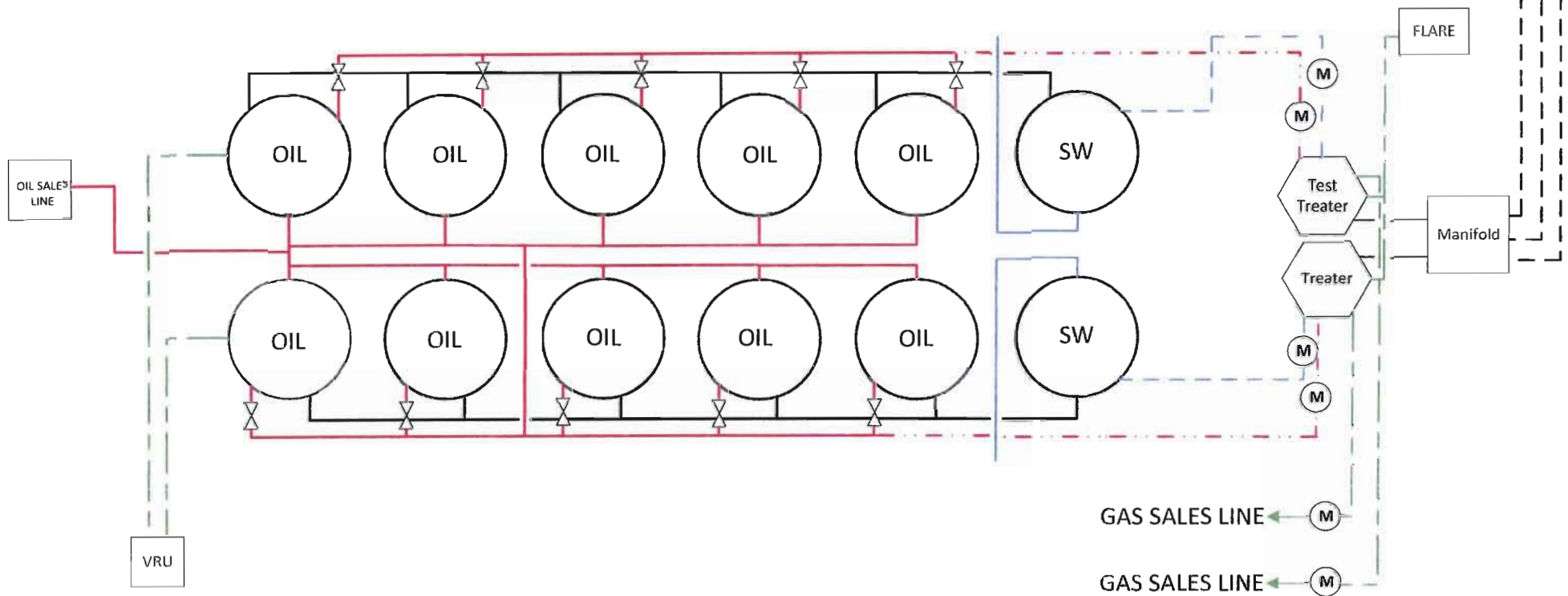
Brays 5301 43- 12H
SWSE 12-153-101
File #: 20864
API #: 3305303609

Oasis Petroleum Inc.

Jefferies 5301 43-12B
SWSE 12-153-101
File #: 22220
API #: 3305303936

Oasis Petroleum Inc.

Achilles 5301 41- 12B
SWSW 12-153-101
File #: 22100
API #: 3305303912



PRODUCTION COMMINGLING DIAGRAM - BRAY 5301 43-12H
JEFFERIES 5301 43-12B & ACHILLES 5301 41-12B

DATE	REV.	BY	APPR.	SCALE
March 14, 2012	0	LTZ		NA
LOCATION	FIELD			
NORTH DAKOTA	INDIAN HILLS - SOUTH			

STATE OF NORTH DAKOTA)
) ss.
COUNTY OF MCKENZIE)

Tom F. Hawkins, being duly sworn, states as follows:

1. I am the Vice President - Land and Contracts employed by Oasis Petroleum North America LLC with responsibilities in the State of North Dakota and I have personal knowledge of the matters set forth in this affidavit.
2. Sections 13 and 24, Township 153 North, Range 101 West, McKenzie County, North Dakota constitute a spacing unit in accordance with the applicable orders for the Bakken pool.
3. Three wells have been drilled in the spacing unit, which are the Bray 5301 43-12H, the Achilles 5301 41-12B and the Jefferies 5301 43-12B .
4. By Declaration of Pooled Unit dated August 26, 2011, filed in McKenzie County, North Dakota, document number 422312, all oil and gas interests within the aforementioned spacing unit were pooled.
5. All Working Interests, Royalty Interests and Overriding Royalty Interests in the Bray 5301 43-12H, the Achilles 5301 41-12B and the Jefferies 5301 43-12B wells are common.

Dated this 14th day of March, 2012

Tom F. Hawkins

STATE OF TEXAS)
) ss.
COUNTY OF HARRIS)

Subscribed to and sworn before me this 14th day of March, 2012



Lauren Shields
Notary Public
State of Texas
My Commission Expires: 10-17-2012

**SUNDRY NOTICE AND REPORTS ON WELLS - FORM**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.
22220

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.



MA

☒ Notice of Intent	Approximate Start Date March 6, 2012	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☒ Reclamation
		☐ Other	Reserve pit reclamation

Well Name and Number Jefferies 5301 43-12B					
Footages	Qtr-Qtr	Section	Township	Range	
250 F S L 2510 F E L SWSE 12 153 N 101 W					
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Excel Industries, Inc			
Address P.O. Box 159	City Miles City	State MT	Zip Code 59301

DETAILS OF WORK

Oasis Petroleum North America LLC plans to reclaim the reserve pit for the above referenced wells as follows:
The NDIC field inspector, Mark Binnes (NDIC) was notified on 2/29/2012. The surface owner was notified on 3/1/2012
Surface owners: Larry Heen, 14033 45th Street NW, Williston, ND 58801
Drain all fluid from pit, mix cuttings with flyash and lime to solidify. Cap with clay and slope and contour location to ensure proper drainage.

Company Oasis Petroleum North America LLC	Telephone Number 281-404-9491
Address 1001 Fannin, Suite 1500	
City Houston	State TX
Zip Code 77002	
Signature <i>Brandi Terry</i>	Printed Name Brandi Terry
Title Regulatory Specialist	Date March 1, 2012
Email Address bterry@oasispetroleum.com	

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 3-1-12	
By <i>Carl V...</i>	
Title <i>...</i>	

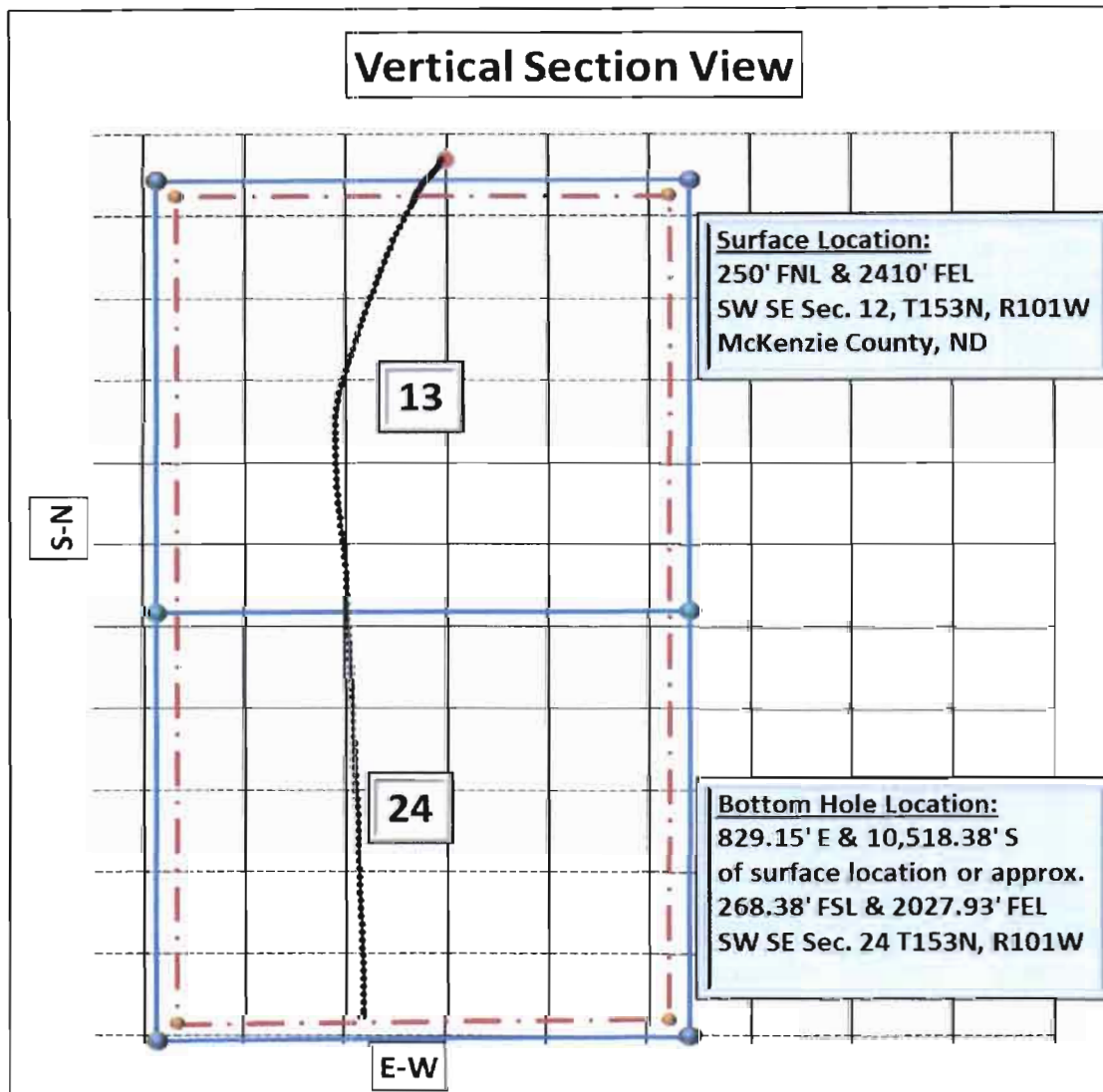
22220

44

MAR 2012
RECEIVED
ND OIL & GAS
DIVISION

Oasis Petroleum North America, LLC

Jefferies 5301 43-12B



Services Performed For:

Mike Box

Oasis Petroleum North America, LLC

1001 Fannin, Suite 202

Houston, TX 77002

Onsite Geology Performed by:

K. Pearson, T. Bushedorf

RPM Geologic, Inc

geology@rpmconsultinginc.com

(303) 595-7625

Well Information

Operator:	Oasis Petroleum North America, LLC	API #:	33-053-03936-00-00
Address:	1001 Fannin Suite 202 Houston, TX 77002	ND Well File #:	22220
Well Name:	Jefferies 5301 43-12B	Surface Location:	SW SE Sec. 12 T153N R101W
Field/ Prospect:	Camp Field	Footages:	250' FSL & 2,410' FEL
Elevation:	GL: 2,094' KB: 2,118'	County, State:	McKenzie County, North Dakota
Spud Date:	January 31, 2012	Basin:	Williston
		Well Type:	Horizontal Middle Bakken

Contractor:	Nabors #149	Chemical Company	Fluid Control
Toolpushers:	Larry Erie, Dwight Knutson	Mud Engineer	Ryan Buckley, Judd Burman
Field Supervisors:	Eli Puckett, Mark Lawler	H₂S MONITORING:	NA
Directional Drilling	RPM Consulting, Inc. Mark Lawler, Eli Puckett, Jason Straindlin	MWD	Ryan Energy Jase Pitre, Daniel Ogden

Wellsite Geologist	Krista E. Pearson Travis Bushendorf	Rock Sampling:	30' from 8,360' to 10,800' 10' from 10,800' to 11,140' 30' from 11,140' to 21,250' (TD)
Prospect Geologist	Mike Box	Gas Detector	Bloodhound Gas Detection
Sample Examination:	Binocular microscope & fluoroscope	Sample Cuts:	EnTron Solvent
Horizontal Target	Middle Bakken Porosity		

Key Offset Wells:

St Mary Land & Exploration	Lindvig 1-11-3C	SESE Sec. 11 T153N R101W	McKenzie County, ND
Oasis Petroleum North America	Achillies 5301 41-12B	SWSW Sec. 12 T153N R101W	McKenzie County, ND
Oasis Petroleum North America	Bray 5301 41-12H	SWSE Sec. 12 T153N R101W	McKenzie County, ND

Pumps:	#1 & #2: National 10P-100 - 5.5" liners Output: 0.0837 bbl/stroke (95% efficiency)		
Mud Type:	Fresh water surface to 2,190' Diesel invert mud 2,190' to 11,110' (curve landing) Salt water from 11,110' to 22,250' (TD)		
Casing:	Surface: 9 5/8" 36# J-55 set to 2,070' Intermediate: 7" casing set to 11,093'		
Hole Size:	13 1/2" from conductor pipe at 105' to 2,090'	8 3/4" to 11,110'	6" to 21,250' (TD)
Total Drilling Days:	22 days		

<u>Horizontal Target:</u>	<i>Middle Bakken "B"</i>	<u>BOTTOM HOLE LOCATION:</u>
<u>Shut-In Point / Date:</u>	<i>10,270' / 7 February 2012</i>	<i>829.15' E & 10,518.38' S</i>
<u>Total Depth/ Date:</u>	<i>21,250' / 21 February 2012</i>	<i>of surface location or approximately</i>
		<i>268.38' FSL & 2,027.93' FEL</i>
		<i>SWSE Section 24, T153N, R101W</i>
<u>Ending Vertical Section</u>	<i>10,518.38'</i>	
<u>Ending Azimuth</u>	<i>179.80°</i>	
<u>Status of Well:</u>	<i>Awaiting Completion</i>	
<u>Exposure to Formation:</u>	<i>100%</i>	

Summary

- The *Jefferies 5301 43-12B* was spud 31 January 2012 in the Baker Field in McKenzie County, North Dakota.
- Bit #1 was used to drill the vertical section from the conductor pipe at 105' to 2,190'. Surface casing was set at 2,170' and isolated with 9 5/8" 36# J-55 casing pipe. Bit #2 was used to drill the remainder of the vertical hole to a total depth of 10,270'.
- Bit #3 was used to drill the first 449' of the curve and was replaced with a tricone bit due to inadequate build rates. Bit #4 drilled the final 391' of the curve which allowed for the build rates needed to reach the landing target.
- Intermediate casing was set to 11,093' MD / ~ 10,746' TVD and isolated with 7" casing pipe. Bit #5 was used to drill out of intermediate casing and drilled the entire 10,140' lateral.
- The target zone consisted of a 9' thick vertical section of the Middle Member of the Bakken Formation situated 10' to 19' feet below the Upper Shale of the Bakken Formation. The target zone was identified at the top and bottom by low gamma readings.
- Background gas levels ranged from ~2,700 to ~3,400 units while drilling the lateral. Gas was diverted through the gas buster producing an average flare of 2' to 6' throughout the lateral.
- The Middle Member of the Bakken Formation observed in cutting samples consisted of two primary facies observed while drilling in the target zone: 1) light gray silty sandstone with visible oil staining and 2) light gray silty sandstone with trace amounts of patchy oil stain.
- Drilling operations were ceased on 21 February 2012 at 11:33 hours at an approximate bottom hole location of 829.15' E & 10,518.38' S of surface location or approximately 268.38' FSL & 2,027.93' FEL in SWSE Section 24 T153N R101W.

Respectfully submitted,

Krista E. Pearson

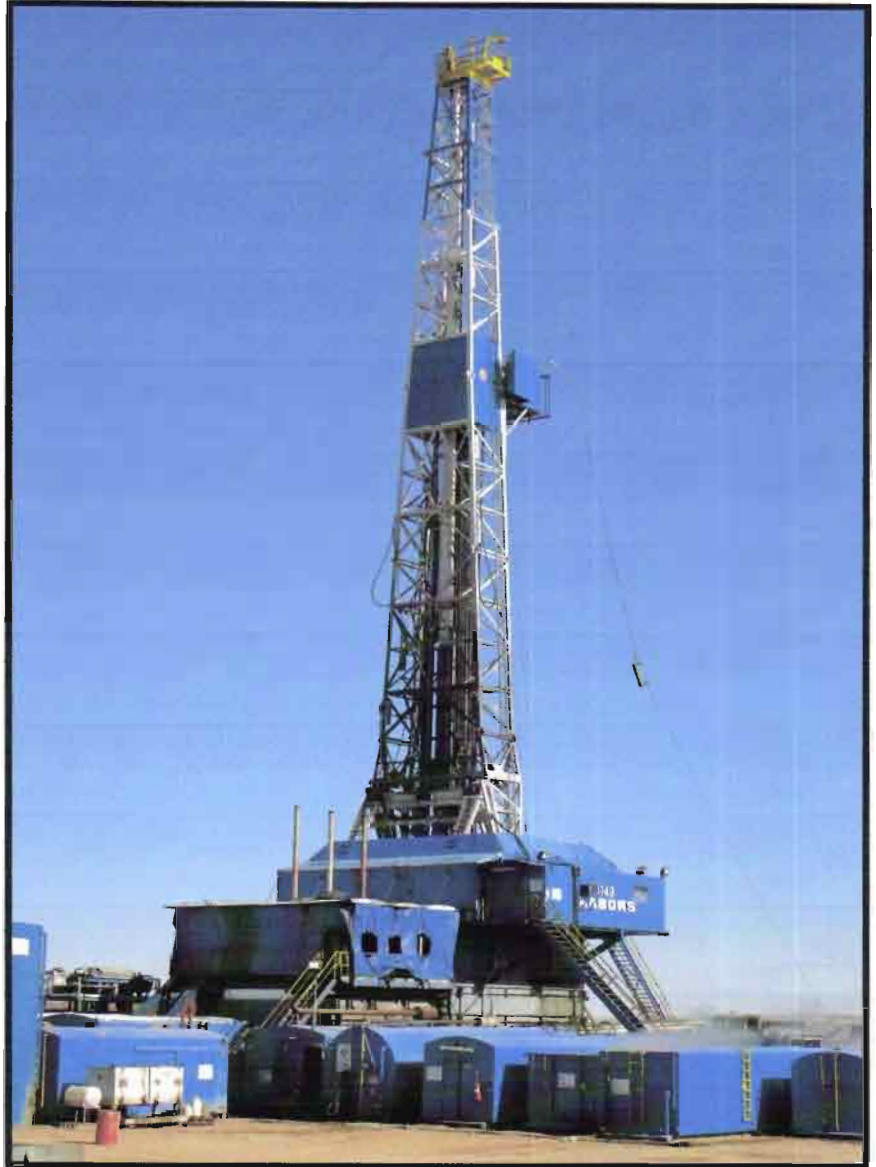
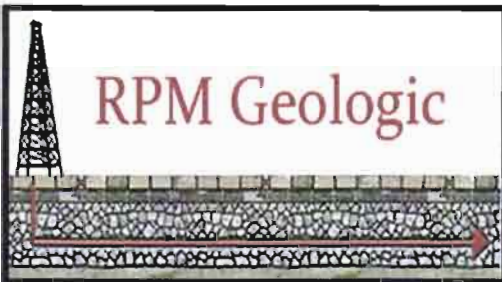
RPM Geologic, Inc.

08/2012

OASIS
PETROLEUM

24 Hr. Emergency Contact # 1-866-584-8016

JEFFERIES 5301-43-12B
SWSE 12-153N-101W
McKENZIE CNTY, ND
P# 22220 API# 33-053-03936



Well Evaluation

Synopsis

The *Jefferies 5301 43-12B* [SWSE Section 12, T153N, R101W] is a horizontal well located in the Baker Field in McKenzie County, North Dakota approximately 6 miles south of Williston. The prognosis proposed a single lateral leg trending due south to be drilled through sections 13 and 24 from the SWSE corner of Section 12 to the SE SW corner of Section 24. The lateral portion of the *Jefferies 5301 43-12B* targeted the porous silty sandstone of the Middle Member of the Bakken Formation, approximately 10' to 19' below the Upper Shale of the Bakken Formation.

Geologic Assessment

Methods

Geologic support of the *Jefferies 5301 43-12B* was provided by experienced RPM Well Site Geologists. Gas and chromatograph levels were measured using iBall Instruments Bloodhound (Bloodhound) real time gas detector and chromatograph system. The Bloodhound gas detection system uses non-dispersive infrared and chemical sensor gas detection to evaluate gases liberated from the formation by the drill bit and carried to the surface by the drilling fluid.

The Bloodhound was interfaced with a RigWatch Drilling Recorder system. RigWatch provided rate of penetration (ROP), on-off bottom and pump strokes to the Bloodhound and received total gas information from the Bloodhound for viewing on location and remotely.

Under the direction of RPM well site geologists, rig crews caught lagged drill cutting samples at 30' intervals from 8,300' through 10,780'. Samples were logged at 10' intervals during the curve from 10,780' to 11,050'. Throughout the lateral section, samples were continuously logged at 30' intervals until total depth 21,250'.

Sampled drill cuttings were examined wet and dry under a binocular microscope using plain (broad spectrum) and transmitted light. Cuttings were evaluated for hydrocarbon "cut" by immersion in *Entron* critical cleaning solvent and inspected under a UV fluoroscope. Alizarin red stain and ten percent hydrochloric acid were used to determine the calcareous and dolomitic content of rocks and cementing.

Control Wells

Two completed wells were used as control wells for the *Jefferies 5301 43-12B*. During the curve, the gamma-ray data and e-log data from the *Lindvig 11-1HR* and the *Achilles 5301 41-12B* were used to determine the curve landing depth for the *Jefferies 5301 43-12B*.

The *Lindvig 11-1HR* [SESE Section 11, T153N, R101W] is located ~ 0.5 miles east of *Jefferies 5301 43-12B*. The *Lindvig 11-1HR* was spud on 11 December 1982 by Gulf Oil Corporation under the original well name *N. Alexander #1-11-3C*. This well is currently operated by SM Energy Company. The *Lindvig 11-1HR* was drilled to the Red River Formation with a total depth 14,461'. Production was established in both the Madison and Red River intervals. Cumulative production from the Madison Group has been recorded as 203,232 bbls of oil, 242,165 bbls of water and 242,165 MCF gas. The production status for the Red River Formation has been listed as dry.

The *Achilles 5301 41-12B* [SWSW Section 12, T153N, R101W] is located within the same section approximately one half mile west of the *Jefferies 5301 43-12B*. The *Achilles 5301 41-12B* was spud on 3 January 2012 by Oasis Petroleum North America, LLC. The *Achilles 5301 41-12B* is a horizontal well drilled through the Middle Member of the Bakken Formation. Initial production data for the *Achilles 5301 41-12B* had not been disclosed at the time of this report.

To define the initial landing target, Table 1 was constructed to calculate the distance of formation tops and Lodgepole markers to the target depth determined from the both offset wells. The comparison showed distance to the landing target from the Lodgepole Formation averaged 738' between the *Lindvig 11-1HR* and the *Achilles 5301 41-12B*.

The initial kick off point (KOP) was projected at 10,300' (-8,182'). The bottom hole assembly for the curve build was picked up at 10,270' (-8,152') on 7 February 2012. The Lodgepole Formation top was picked from MWD gamma data at 10,010' (-7,892). Utilizing the Upper Bakken Shale isopach map and the distance to target from both control wells, the landing target for the *Jefferies 5301 43-12B* was set at 10,750' (-8,632') total vertical depth (TVD).

During the curve progression, Lodgepole markers A through F were used to help determine the landing target and aid directional drillers to calculate the curve build. The close proximity of the control wells in addition to the marker picks determined the landing target would remain at 10,750' TVD.

Control Wells						
Operator: Well Name: Location: Elevation:	SM Energy Company Lindvig 11-1HR SESE Sec. 11 T153N R101W KB: 2.108'			Oasis Petroleum North America, LLC Achilles 5301 41-12B SWSW Sec. 12 T153N R101W KB: 2.119'		
Formation/Marker	E-Log/GR	MSL Datum	Distance to target	E-Log/GR	MSL Datum	Distance to target
Charles Salt	8.509'	-6.401'	2.226'	8.552'	-6.433'	2.182'
Base of Charles Salt	9.207'	-7.099'	1.528'	9.219'	-7.100'	1.515'
Lodgepole	9.991'	-7.883'	744'	10.002'	-7.883'	732'
LPA	10.411'	-8.303'	324'	10.411'	-8.292'	323'
LPB	10.479'	-8.371'	256'	10.460'	-8.341'	274'
LPC	10.503'	-8.395'	232'	10.497'	-8.378'	237'
LPD	10.561'	-8.453'	174'	10.558'	-8.439'	176'
LPE	10.607'	-8.499'	128'	10.610'	-8.491'	124'
LPF	10.660'	-8.552'	75'	10.657'	-8.538'	77'
False Bakken	10.700'	-8.592'	35'	10.701'	-8.582'	33'
Scallion	10.703'	-8.595'	32'	10.702'	-8.583'	32'
Upper Bakken Shale	10.709'	-8.601'	26'	10.708'	-8.589'	26'
Middle Bakken	10.726'	-8.618'	9'	10.727'	-8.608'	7'
Target	10.735'	-8.627'	-	10.734'	-8.615'	-

Table 1. Comparison of distances from formation tops and Lodgepole Formation markers to landing target from two control wells utilized to determine kick off point and target TVD at curve landing.

Vertical Operations

Overview

RPM well site geologists arrived at site on 5 February 2012 and began logging the vertical section at 8,300' shortly before picking the top of the limestone marker of the Kibbey Formation at 8,390' (-6,272').

The *Jefferies 5301 43-12B* was spud on 1 February 2012 using Nabors #149 drilling rig. Nabors #149 was used to drill the surface from the conductor at 105' to 2,190' using fresh water. Surface casing was set to 2,170' and isolated with 9 5/8" 36# J-55 casing pipe.

Bit #1 (13.5" JZ HC605) was used to drill from 105'to 2,190' in 13 hours with an average ROP of 160.4 feet per hour. Bit #2 was used to drill the 8,080' vertical section from surface casing to KOP at 10,270' (-8,152') in 79.5 hours with an average ROP of 101.6 feet per hour.

Diesel invert drilling fluid with a mud weight ranging from 9.5 to 10.3 ppg was used during the vertical hole from surface casing through the curve build and landing at 11,110' MD / ~10,748' TVD.

Lithology

The top of the limestone marker from the **Kibbey Formation** [Mississippian Big Snowy Group] was logged at 8,390' (-6,272'), 17' low to the *Lindvig 11-1HR*. Samples from this interval were described as (Figure 1):

LIMESTONE: wackestone, medium gray, tan, finely crystalline, hard, blocky, earthy to micro crystalline

SANDSTONE: white to clear, fine grained, friable, subrounded to rounded, moderately to well sorted, earthy texture, calcareous cement



Figure 1. Fine grained sandstone (left) and limestone (right) as seen in sample while drilling in the Kibbey Formation.

The top of the **Charles Salt** [Mississippian Madison Group] was logged at 8,523' (-6,405'), 4' low to the *Lindvig 11-1HR*. Samples from this interval (Figure 2) were described as:

SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture

SANDSTONE: white to clear, trace gray, fine to medium grained, friable, subrounded to rounded, moderately to well sorted, earthy texture, calcareous cement

ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture

LIMESTONE: mudstone, light gray, microcrystalline, subblocky, firm, crystalline texture, argillaceous in part, trace intercrystalline porosity

DOLOMITE: mudstone, light blue, microcrystalline, subblocky, firm, microsucrosic texture

DOLOMITIC LIMESTONE: mudstone, light gray to blue, light brown to gray, medium gray, microcrystalline, subblocky, firm, microsucrosic texture



Figure 2. Subhedral grains of salt (left), anhydrite stained with invert drilling mud (center), and light blue dolomite (right) observed in the Charles Formation.

The top of the **Base Last Salt** was drilled at 9,224' (-7,106'), 5' low to the *Lindvig 11-1HR*. Samples from this interval (Figure 3) were described as:

LIMESTONE: mudstone, light gray to brown, trace light yellow, microcrystalline, subblocky, firm to hard, earthy texture, rare calcite, slightly argillaceous

DOLOMITIC LIMESTONE: mudstone to wackestone, light brown to gray, blue, microcrystalline to very fine crystalline, subblocky, firm, earthy to sucrosic texture, trace fossils fragments, trace intergranular porosity

ANHYDRITE: very light gray, gray to blue, microcrystalline, firm, earthy texture



Figure 3. Light to medium gray anhydrite observed below the Base Last Salt.

The top of the **Mission Canyon Formation** of the Madison Group [Mississippian] was penetrated at 9,451' (-7,333'), 10' low to the *Lindvig 11-1HR*. Samples from this interval (Figure 4) were described as:

LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm, earthy texture, trace algal matts, trace calcite

LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite



Figure 4. Limestone observed while drilling in the Mission Canyon Formation.

The top of the **Lodgepole Formation** of the Madison Group (Mississippian) was logged at 10,010' (-7,892), 3' high to the *Lindvig 11-1HR*. Samples collected from the **Lodgepole Formation** (Figure 5) were described as:

LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

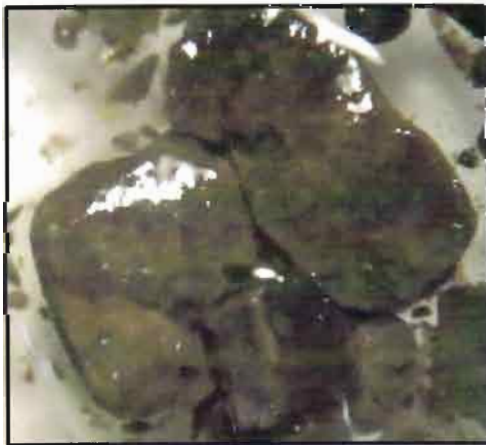


Figure 5. Microcrystalline limestone (left) and limestone with calcite (right) observed while drilling in the Lodgepole Formation.

Directional Operations

RPM geologists worked closely with onsite MWD and Directional Drilling technicians to formulate steering decisions to maximize exposure of borehole in the pay zone. MWD gamma-ray information and penetration rates were closely observed to aid in steering decisions and dip degree estimations. Ryan Energy provided equipment and personnel for MWD and Directional Drilling services.

Curve Build Section

Overview

During the curve progression, isopach data and gamma signatures from the *Lindvig 11-1HR* and the *Achilles 5301 41-12B* were compared to gamma readings from the MWD tool below the Lodgepole Formation. Six possible Lodgepole markers were identified to determine the target landing point.

Operator:	Oasis Petroleum, Inc.				
Well Name:	Jefferies 5301 43-12B				
Location:	250' FSL & 2.410' FWL				
	Section 12. T153N. R101W				
Elevation:	GL: 2.093'	Sub: 25'		KB: 2.118'	
Formation / Marker	MWD Gamma Pick	MSL Datum	Distance to Target	Distance to Target Lindvig 11-1HR	Distance to Target Achilles 5301 41-12B
Kibbey Limestone	8.390'	-6.272'	2.360'	-	-
Charles Salt	8.523'	-6.405'	2.227'	2.226'	2.182'
Base Last Salt	9.224'	-7.106'	1.526'	1.528'	1.515'
Mission Canyon	9.451'	-7.333'	1.299'	-	-
Lodgepole	10.010'	-7.892'	740'	744'	732'
LPA	10.421'	-8.303'	329'	324'	323'
LPB	10.491'	-8.373'	259'	256'	274'
LPC	10.513'	-8.395'	237'	232'	237'
LPD	10.575'	-8.457'	175'	174'	176'
LPE	10.619'	-8.501'	131'	128'	124'
LPF	10.672'	-8.554'	78'	75'	77'
False Bakken	10.718'	-8.600'	32'	35'	33'
Upper Bakken Shale	10.725'	-8.607'	25'	26'	26'
Middle Bakken	10.740'	-8.622'	10'	9'	7'
Target	10.750'	-8.632'	-	-	-

Table 2. Table utilized by RPM Geologists to calculate, track and determine the final landing target for the Lindvig 11-1HR as formation tops and marker picks were made available from gamma ray data provided by Ryan Energy.

The curve was started at the kick off point of 10,300' using a PDC bit. Bit #3 (Halliburton FXD55M) was used from 10,270' to 10,560' for 9 hours. Bit #3 was rerun after a trip out of the hole to adjust the mud motor from 2.3° to 2.45° due to inadequate build rates. At 10,719' the bottom hole assembly was replaced a 2.35° motor and bit #4 which was a more aggressive tricone bit. This bottom hole assembly provided the build rates needed to land on target. The curve was landed at 90°. While drilling 65' of rathole, the inclination increased to 92.00°, followed by 92.60° during the next two surveys. Directional slid down to correct with the next consecutive surveys recorded as 89.40° and 87.40°.

The curve was landed in the Middle Member of the *Bakken Formation* ~ 6' below the base of the Upper Bakken Shale. The curve was completed on 10 November 2012 at 11,110' MD / ~10,748' TVD. Intermediate casing (7") was set to 11,093' MD.

Lithology

The top of the **False Bakken** marker was logged at 10,810' MD / ~ 10,718' TVD. A weak gas show of 332 units was recorded while drilling through the False Bakken marker.

The Upper Shale of the Bakken Formation [Mississippian – Devonian] was drilled at 10,823' MD / ~ 10,725' TVD. The top of the Upper Shale is 5' low compared to offset well, *Lindvig 11-1HR*. The *Upper Shale* was described as black, carbonaceous and petroliferous shale (Figure 6) characterized by gamma-ray values in excess of 250 API counts. Due to a previously unrecognized gas trap glitch, gas levels were recorded with a peak of 481' units while drilling through the Upper Bakken Shale. Samples were described as:

SHALE: black, firm, subblocky, waxy texture, very carbonaceous, very slightly calcareous, petroliferous, visible oil stain, no visible porosity



Figure 6. Upper Bakken Shale.

The Middle Member of the Bakken Formation was picked at 11,888' MD / ~10,740' TVD, 5' low to *Lindvig 11-1HR*. Samples from Middle Member of the Bakken Formation observed during the curve build section were described as:

SILTY SANDSTONE: light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately sorted, occasional calcite fracture fill, trace nodules pyrite, dolomitic cement, visible intergranular porosity

SILTY SANDSTONE: light brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain

Lateral Section

Overview

Drilling fluid consisting of a salt water program (9.7 to 9.9 ppg) while drilling the lateral section. The drilling mud was successful in preserving stable hole conditions, minimizing washout through the salt intervals and maintaining hydrostatic balance.

One 6" Security DBS PDC bit was used to drill the lateral section. Bit #5 was used to drill out of intermediate casing to 13,534' MD in 35.5 hours averaging 68.3 feet per hour. Bit #5 was run a second time after a trip out of the hole to replace the mud motor. The new motor was adjusted to 1.83°. Bit #5RR1 drilled to 20,085' MD in 88 hours with an average drill rate of 74.4 feet per hour. Bit #5 was run a third time after a trip out of the hole to replace the motor. Bit #5RR2 drill to total depth of 21,250'.

Target Zone

The target zone was a nine foot vertical section ranging from 10' to 19' below the Upper Bakken Shale. A type log (Figure 7) was adapted from a nearby offset well demonstrating the general gamma character. The target zone was bound by a low gamma marker (~ 50 to ~ 55 API) near the top of the zone. The bottom of the zone was also bound by a low gamma signature (~ 50 to ~ 58 API). The middle of the target zone was characterized by "ratty" gamma which ranged from ~ 50 to ~80 API counts.

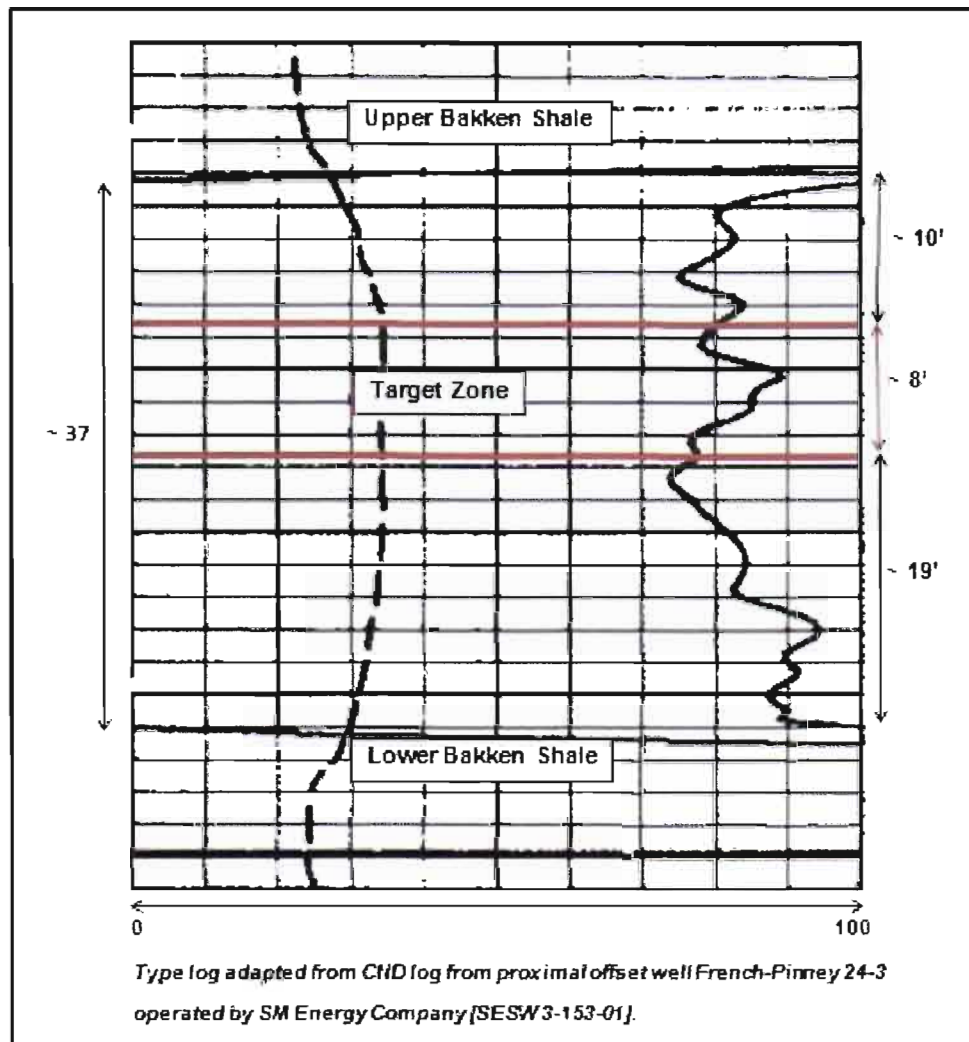


Figure 7. Type log adapted from compensated neutron density log from proximal well exhibiting gamma ray signature for Middle Member of the Bakken Formation. The eight foot target zone is delineated between 10' and 18' below the Upper Bakken Shale.

Lithology

The Middle Member of the Bakken Formation observed in cutting samples consisted of two primary facies observed while drilling in the target zone: 1) light gray silty sandstone with visible oil staining and 2) light gray silty sandstone with trace amounts of patchy oil stain. Figures 8 illustrates the primary facies observed while drilling through the Middle Bakken Formation.

SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain

SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain

SLTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity

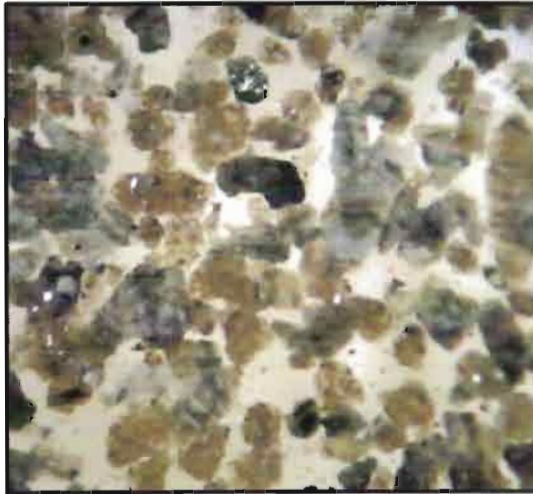


Figure 8: Photograph exhibiting common oil staining observed sample of the Middle Bakken near the top of the target zone (left). The lower part of the target zone demonstrates patchy oil staining (right).

Geosteering

Geosteering decisions were made by carefully observing MWD gamma ray data and aligning the well bore with calculated TVD projections based on calculated dip of formation. Table 3 demonstrates gamma points picked from repeat gamma signatures used to calculate the dip of formation.

Jefferies 5301 43-12B				CALCULATED DIP OF FORMATION				
Marker I Gamma API	Marker II Gamma API	Measured Depth I	Measured Depth II	TVD I	TVD II	Δ MD	Δ TVD	Calculated dip of formation
52	52	11.918	11.311	10,759.56	10,756.81	607	2.75	-0.29°
71	71	13.328	12.784	10,768.56	10,764.23	544	4.33	-0.46°
48	46	13.760	13.456	10,774.64	10,775.58	304	-0.94	+0.17°
53	51	14.111	13.843	10,771.96	10,773.12	268	-1.16	+0.26°
51	51	14.721	13.843	10,769.03	10,773.12	878	-4.09	+0.45°
52	50	16.827	15.740	10,747.64	10,761.39	1,087	-13.75	+0.72°
51	58	17.770	16.812	10,733.73	10,748.18	958	-14.45	+0.86°
82	79	19.862	18.471	10,689.46	10,722.92	1,391	-33.46	+1.37°

Table 3: Calculations used to estimated dip of formation using gamma ray data from repeat gamma ray signatures.

Gas and Hydrocarbon Shows

Gas shows were weak through the Charles Formation. Total gas levels elevated significantly near the Base Last Salt with an average background gas of ~270 units with recurring peaks of ~330 to ~450 units. A peak of 528 units was recorded in the lower Mission Canyon Formation.

During the curve, the average background gas was ~200 units. A weak gas show of 481 units was recorded while drilling through the Upper Bakken Shale. Repairs were made to gas trap equipment before drilling resumed in the lateral. The average background gas recorded during the lateral ranged from ~2,800 to ~3,400 units. Gas was diverted and a steady 2' to 6' flare was produced. Trip gas produced significant flares of up to approximately 12'. The Bloodhound gas chromatograph showed components of C1 through C4 throughout the entire lateral (Figure 9).

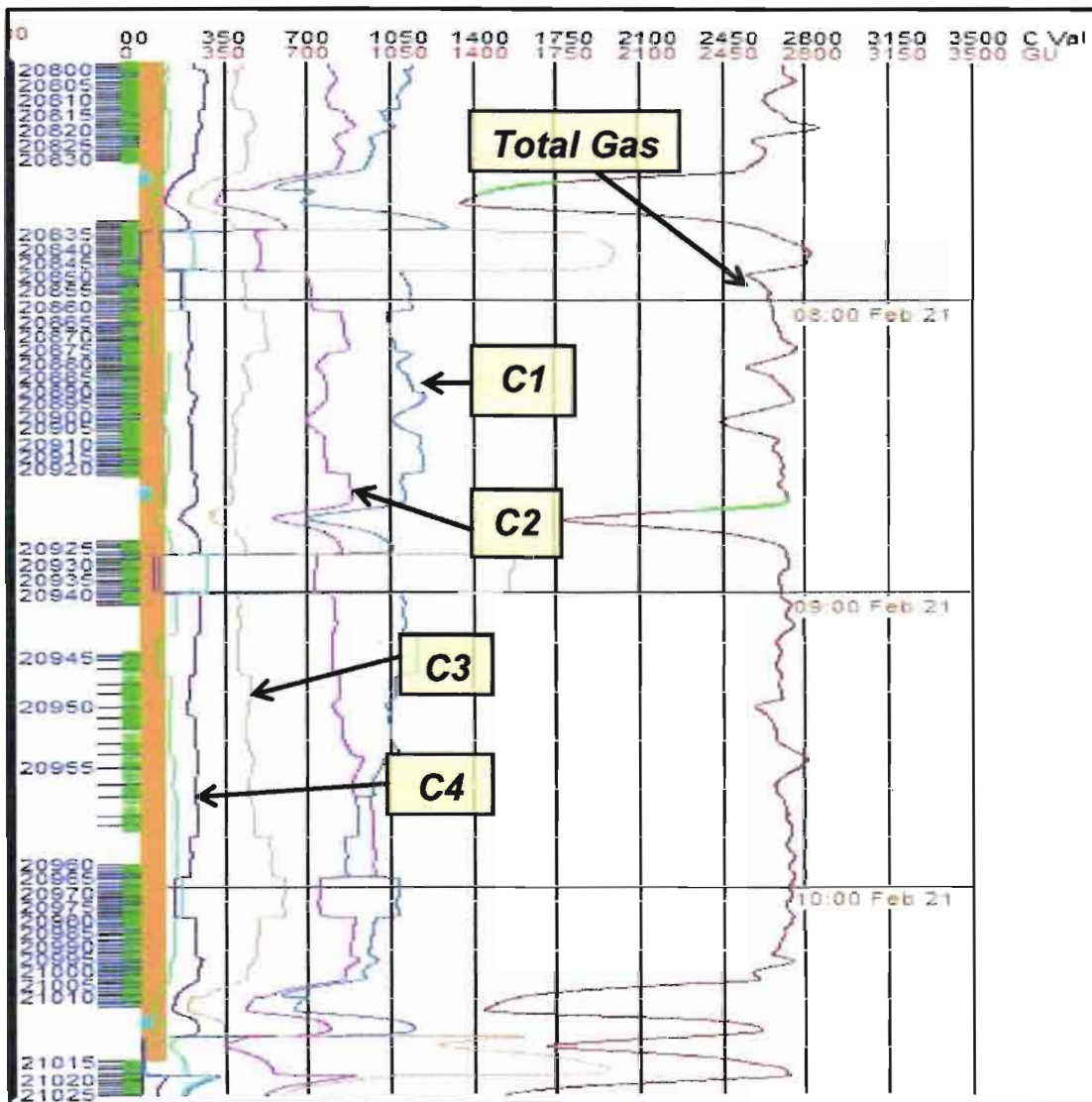


Figure 9. Screen shot of Bloodhound Chromatograph showing formation gas of up to approximately 2,700 units. Total gas units and percentages of gas components are illustrated.

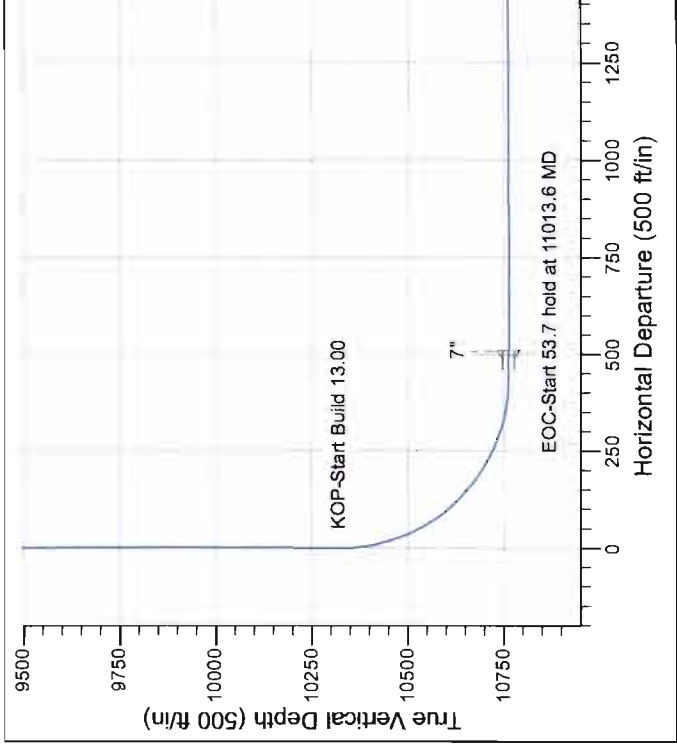
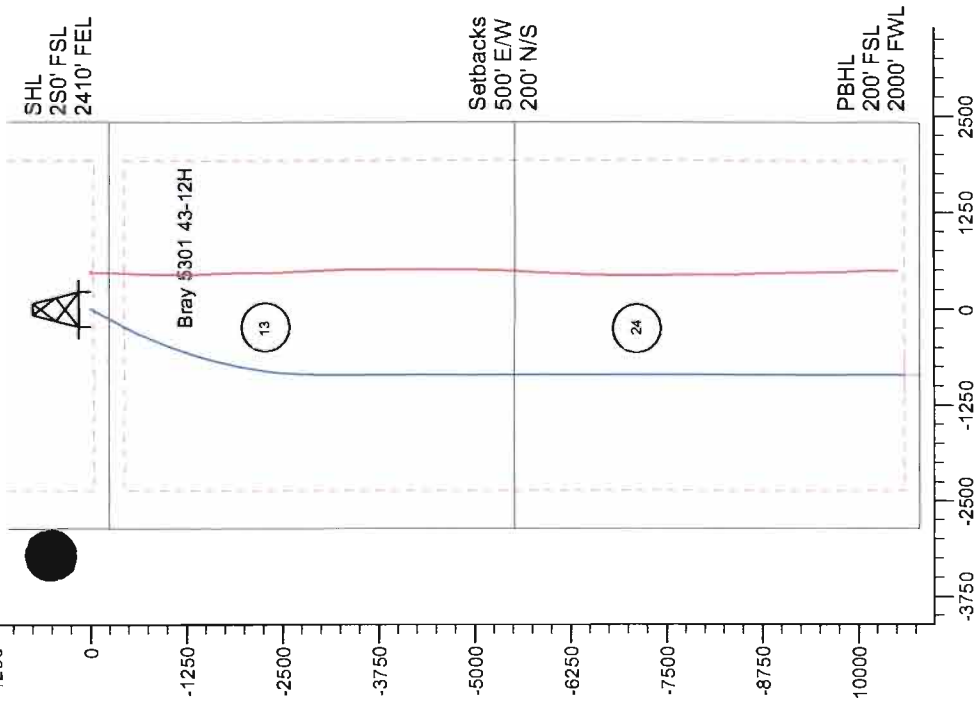


SITE DETAILS: 153N-101W-13/24
Site Centre Latitude: 48° 4' 51.360 N
Longitude: 103° 36' 43.250 W
Positional Uncertainty: 0.0
Convergence: -2.32
Local North: True

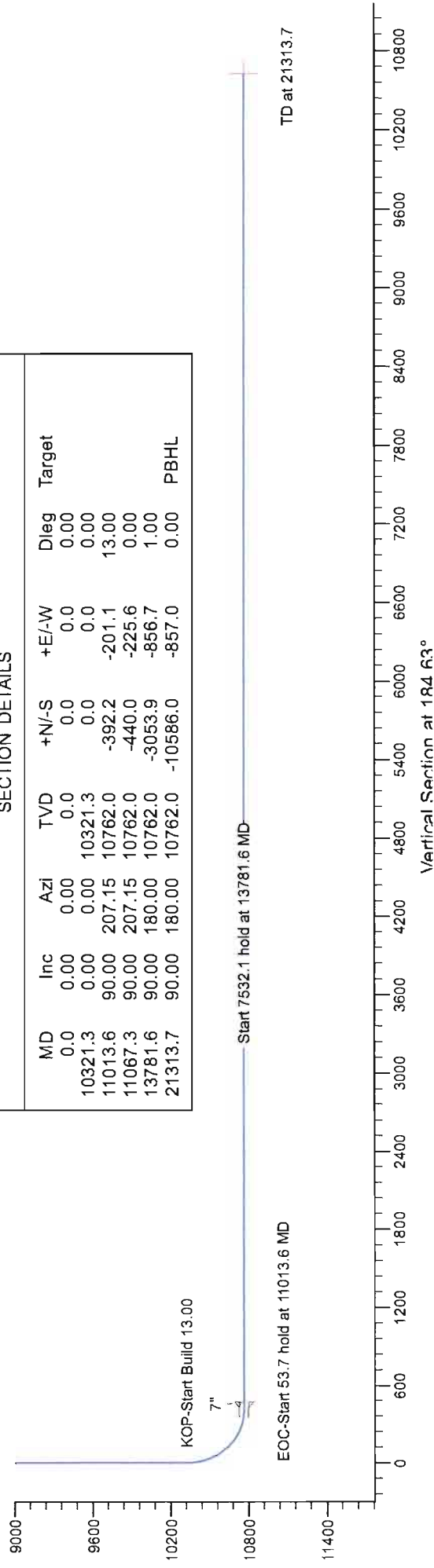
Project: Indian Hills
Site: 153N-101W-13/24
Well: Jefferies 5301 43-12B
Wellbore: OH
Design: Plan #1

Azimuths to True North
Magnetic North: 8.56°
Magnetic Field
Strength: 56724.3snT
Dip Angle: 73.09°
Date: 12/9/2011
Model: IGRF200510

CASING DETAILS			
TVD	MD	Name	Size
2170.0	2170.0	9 5/8"	9.625
10762.0	11085.0	7"	7.000



SECTION DETAILS						
MD	Inc	Azi	TVD	+N/-S	+E/-W	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00
10321.3	0.00	0.00	10321.3	0.0	0.0	0.00
11013.6	90.00	207.15	10762.0	-392.2	-201.1	13.00
11067.3	90.00	207.15	10762.0	-440.0	-225.6	0.00
13781.6	90.00	180.00	10762.0	-3053.9	-856.7	1.00
21313.7	90.00	180.00	10762.0	-10586.0	-857.0	0.00



TD at 21313.7

Daily Activity

Day	Date 2012	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
1	31-Jan	105'	613'	1	-	-	-	-	-	-	-	Rig up to spud, pre-spud safety meeting, drill from 105' to 250'; repair rig, work on blower for top drive, drill from 250' to 485'; wireline survey at 0 degrees, drill from 485' to 718'.	Surface
2	1-Feb	718'	1,472'	1	-	-	-	-	-	-	-	Repair shaker motors, drill from 718' to 935', repair shaker motors, rig service, drill from 935' to 2,190'. Circulate bottoms up, wiper trip, level derrick, wiper trip, circulate bottoms up, wireline survey at 2,190' at 1.24 degrees, TOH to run casing, lay down 8" BHA, safety meeting, rig up casing crew equipment.	Pierre Shale
3	2-Feb	2,190'	0'	-	-	-	-	-	-	-	-	Run surface casing, tag up with casing head 20' off the conductor, land casing at 2,170', cement surface casing, nipple up and pressure test BOP's.	Pierre Shale
4	3-Feb	2,190'	3,533'	-	-	-	-	-	-	-	-	Pressure test BOP's, pick up directional tools, TIH to float equipment; displace fresh water out of the hole with invert, drill out cement and float equipment, drill from 2,190' to 5,723'.	Dakota
5	4-Feb	5,723'	2,035'	-	-	-	-	-	-	-	-	Drill from 5,723' to 6,709', rig service, grease crown and blocks, drill from 6,709' to 7,758'.	Minnelusa Group

Daily Activity

Day	Date 2012	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
6	5-Feb	7,758'	1,430'	2	30	-	40	2400	78	78	475	Drill from 7,758' to 8,425', rig service, drill from 8,425' to 9,188'.	Charles
7	6-Feb	9,188'	1,082'	2	35	-	40	2500	82	82	481	Drill from 9,188' to 9,665', service rig, drill from 9,665' to 10,270', spot pill, pumped dry job, TOOH.	Lodgepole
8	7-Feb	10,270'	210'	3	30	40	40	2175	79	79	463	TOOH, lay down BHA, bit and motor, MWD tool, pick up BHA, bit and motor/pick up and scribe MWD tool, TIH and test MWD tool, drill from 10,270' to 10,480'.	Lodgepole
9	8-Feb	10,480'	239'	3/3RR1	25	35	40	2650	79	79	463	Drill from 10,480' to 10,560', circulate and cond, TOOH, lay down BHA, pull MWD tool, TIH, drill from 10,592' to 10,719', spot bill/pumped dry job, TOOH.	Lodgepole
10	9-Feb	10,719'	95'	4	42	36	40	2800	77	77	446	TOOH, lay down BHA, MWD tool, mud motor and bit. Cut drilling line, service rig, pick up BHA, mud motor and bit. Scribe MWD tool, TIH, drill from 10719' to 10,814'.	Lodgepole
11	10-Feb	10,814'	296'	4	88	54	20	2760	78	78	457	Drill from 10814' to 11,110', circulate bottoms up, TOOH 10 stands, TIH 10 stands, circulate bottoms up, rig up/down to run casing, lay down drill pipe.	Middle Bakken

Daily Activity

Day	Date 2012	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
12	11-Feb	11,110'	0'	-	-	-	-	-	-	-	-	Drill from 10,814' to 11,110', circulate bottoms up. TOOH 10 stands, TIH 10 stands, circulate bottoms up, rig up/down to run casing, cement casing, change out pipe rams.	Middle Bakken
13	12-Feb	11,110'	0'	-	-	-	-	-	-	-	-	Changed out pipe rams and flushed out lines with salt water, test BOP's, disassemble/reassemble rams, test BOP's, pick up BHA, TIH 10 stands, install/remove wear bushing, pre job safety on rigging up laydown truck and picking up drill pipe, pick up drill pipe.	Middle Bakken
14	13-Feb	11,110'	1,475'	5	20	34	40	2250	78	0	229	Change rotating head/rubber, tag cement @ 10,964' / float @ 11,000' / shoe @ 11,093', drill from 11,110' to 11,456', service rig, drill from 11,456' to 12,585'.	Middle Bakken
15	14-Feb	12,585'	949'	5	15	42	40	2500	0	82	240	Drill from 12,585' to 13,474', service rig, drill from 13,474' to 13,534', TOOH.	Middle Bakken
16	15-Feb	13,534'	284'	5RR1	35	40	40	2000	0	76	231	Change rotating head, install trip nipple, TOH, lay down BHA, MWD, stabilizers and mud motor, pick up BHA, MWD, mud motor, adjust mud motor, surface test, TIH, circulate kill mud, drill from 13,534' to 13,818'.	Middle Bakken

Daily Activity

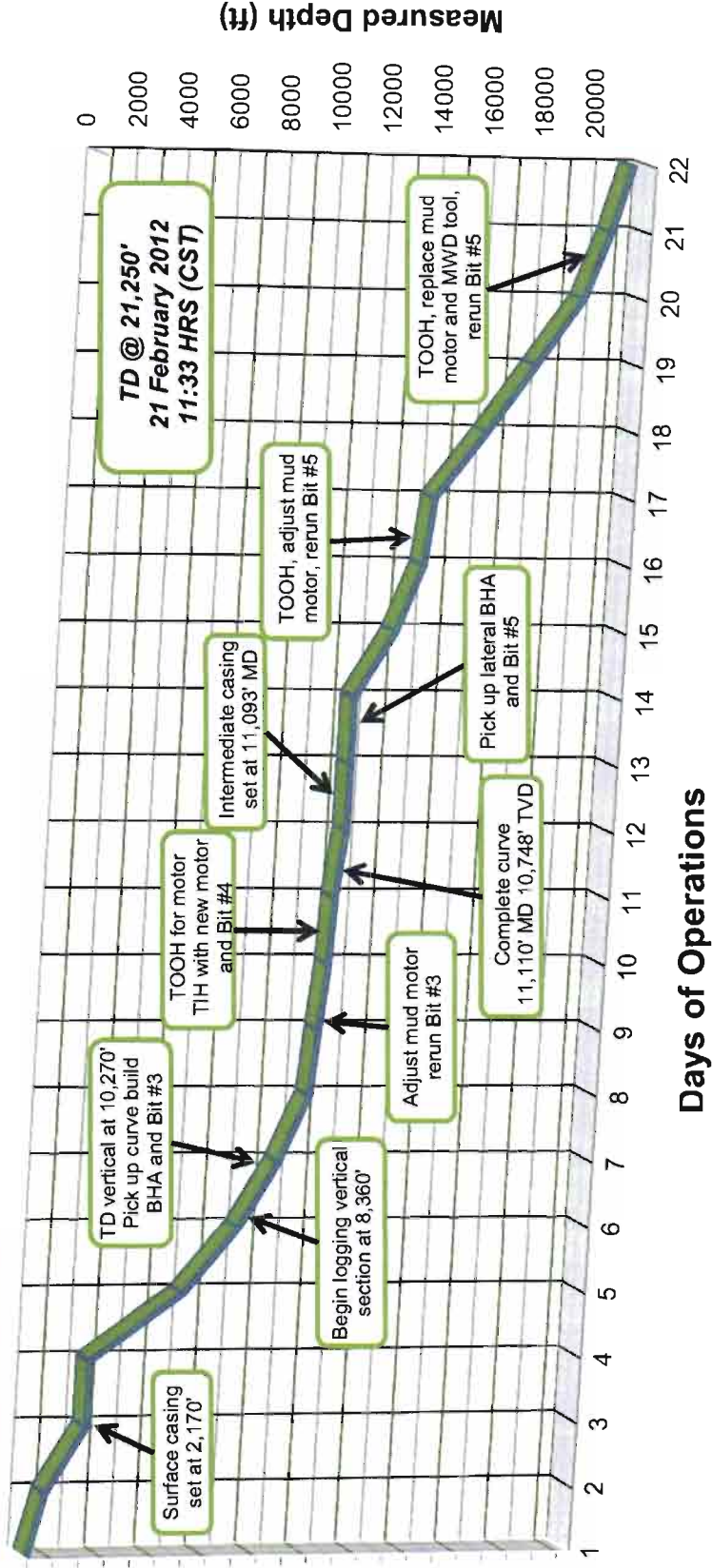
Day	Date 2012	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
17	16-Feb	13,818'	1,891'	5RR1	40	35	40	2000	0	79	231	Drilling from 13,818' to 14,520', service rig, drill from 14,520' to 15,709'.	Middle Bakken
18	17-Feb	15,709'	1,765'	5RR1	30	40	35	2700	0	79	231	Drill from 15,709' to 16,553', rig service, drill from 16,553' to 17,474'.	Middle Bakken
19	18-Feb	17,474'	1,721'	5RR1	30	40	35	2700	0	79	231	Drill from 17,474' to 18,334', rig service, fuction test floor valves, drill from 18,334' to 19,195'.	Middle Bakken
20	19-Feb	19,195'	917'	5RR1	10	60	33	3000	0	78	229	Drill from 19,195' to 20,112', spot pill, circulate and pump pill, TOH, change rotating head rubber and install flow nipple, conduct flow check, TOH, remove dart valve from string, lay down BHA.	Middle Bakken
21	20-Feb	20,112'	711'	5RR2	10	60	33	3000	0	78	229	Pick up BHA, scribe MWD, TIH, change rotating head, install new rubber, TIH, circulate and condition bottoms up, drill from 20,112' to 20,823'.	Middle Bakken
22	21-Feb	20,823'	427'	5RR2	15	35	40	2500	0	82	240	Drill from 20,823' to 21,250', circulate, TOH for reamer run.	Middle Bakken



Daily Progress

Oasis Petroleum North America, LLC
Jefferies 5301 43-12B

Spud: 31 January 2012



Bit Record

Bit #	Size	Make	Model	Serial #	Jets	Depth In	Depth Out	Footage	Hours	Mean ROP (ft/hr)	Accum. Hours
1	13 1/2	JZ	HC605	H25022	4 x 20	105'	2,190'	2,085'	13	160.4	13.0
2	8 3/4	Smith	MDSi616	JF5128	6 x 14	2,190'	10,270'	8,080'	79.5	101.6	92.5
3	8 3/4	Halliburton	FXD55M	11848919	5 x 18	10,270'	10,560'	290'	9	32.2	101.5
3RR1	8 3/4	Halliburton	FXD55M	11848919	5 x 18	10,560'	10,719'	159'	5.5	28.9	107.0
4	8 3/4	Smith	F30T	PT8568	3 x 18	10,719'	11,110'	391'	19	20.6	126.00
5	6	Security	DBS	1875684	6 x 18	11,110'	13,534'	2,424'	35.5	68.3	161.50
5RR1	6	Security	DBS	1875684	6 x 18	13,534'	20,085'	6,551'	88	74.4	249.50
5RR2	6	Security	DBS	1875684	6 x 18	20,085'	21,250'	1,165'	17	68.5	266.50

Daily Mud Data

Day	Date 2012	Depth (sample)	Mud WT (ppg)	VIS (sec)	PV (cP)	YP (lbs/ 100 ft ²)	Gels (lbs/ 100 ft ²)	600/300	HTHP (cc/30min)	NAP/ H ₂ O (ratio)	Cake (APT/ HT)	Solids (%)	pH	Alk	Cl- (mg/l)	ES (v)	Daily Loss (bbls)
1	31-Jan	300'	8.3	28	2	1	1/0/0	5/3	-	0/100	-	<1	8	-	4K	-	-
2	1-Feb	1,625'	8.8	32	2	1	1/0/0	5/3	-	0/98	-	2	8	-	6K	-	-
3	2-Feb	Displace freshwater drilling mud with diesel invert at surface casing (9 5/8") set to 2,170'.															
4	3-Feb																
5	4-Feb																
6	5-Feb	8,254'	9.5	53	20	7	8/13	47/27	4	79.2/20.7	4	13	-	2.8	45K	602	76
7	6-Feb	9,398'	10.0	58	24	8	9/17	56/32	3.2	78.8/21.2	4	15	-	2.5	45K	723	97
8	7-Feb	10,270'	10.0	59	24	7	8/15	55/31	3.2	81.2/18.8	4	15	-	2.7	51K	784	11
9	8-Feb	10,560'	10.1	63	26	8	10/0	60/34	4	81.2/18.8	4	15	-	2.4	50K	925	64
10	9-Feb	10,719'	10.3	95	29	7	11/20	65/36	4	81.2/18.8	4.00	15	-	3.1	49K	0	0
11	10-Feb	10,966'	10.3	71	28	9	11/21	65/37	3.2	82.4/17.6	4.00	13	-	2.7	49K	1163	4
12	11-Feb	Displace diesel invert with salt water at intermediate casing (7") set to 11,093'.															
13	12-Feb																
14	13-Feb	11,330'	9.8	29	1	1	0/0/1	3/2	-	-	-	10	8	-	172K	-	-
15	14-Feb	12,977'	9.8	28	2	1	0/0/1	5/3	-	-	-	5	8	-	184K	-	-
16	15-Feb	13,543'	9.8	28	1	1	0/0/1	3/2	-	-	-	10	7.5	-	192K	-	-
17	16-Feb	14,175'	9.7	27	1	1	0/0/1	3/2	-	-	-	9	8.5	-	170K	-	-
18	17-Feb	15,948'	9.8	27	2	1	0/0/1	5/3	-	-	-	8	8	-	165K	-	-
19	18-Feb	18,022'	9.8	29	2	1	0/0/1	5/3	-	-	-	8	8	-	180K	-	-
20	19-Feb	19,687'	9.8	26	2	1	0/0/1	5/3	-	-	-	8	8	-	177K	-	-
21	20-Feb	20,111'	9.9	30	2	1	0/0/1	5/3	-	-	-	8	8	-	182K	-	-

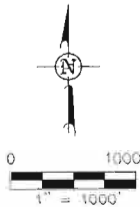
WELL LOCATION PLAT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
 "JEFFERIES 5301 43-12B"
 250 FEET FROM SOUTH LINE AND 2410 FEET FROM EAST LINE
 SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

STAKED ON 12/6/11
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 13 WITH AN ELEVATION OF 2090.8'

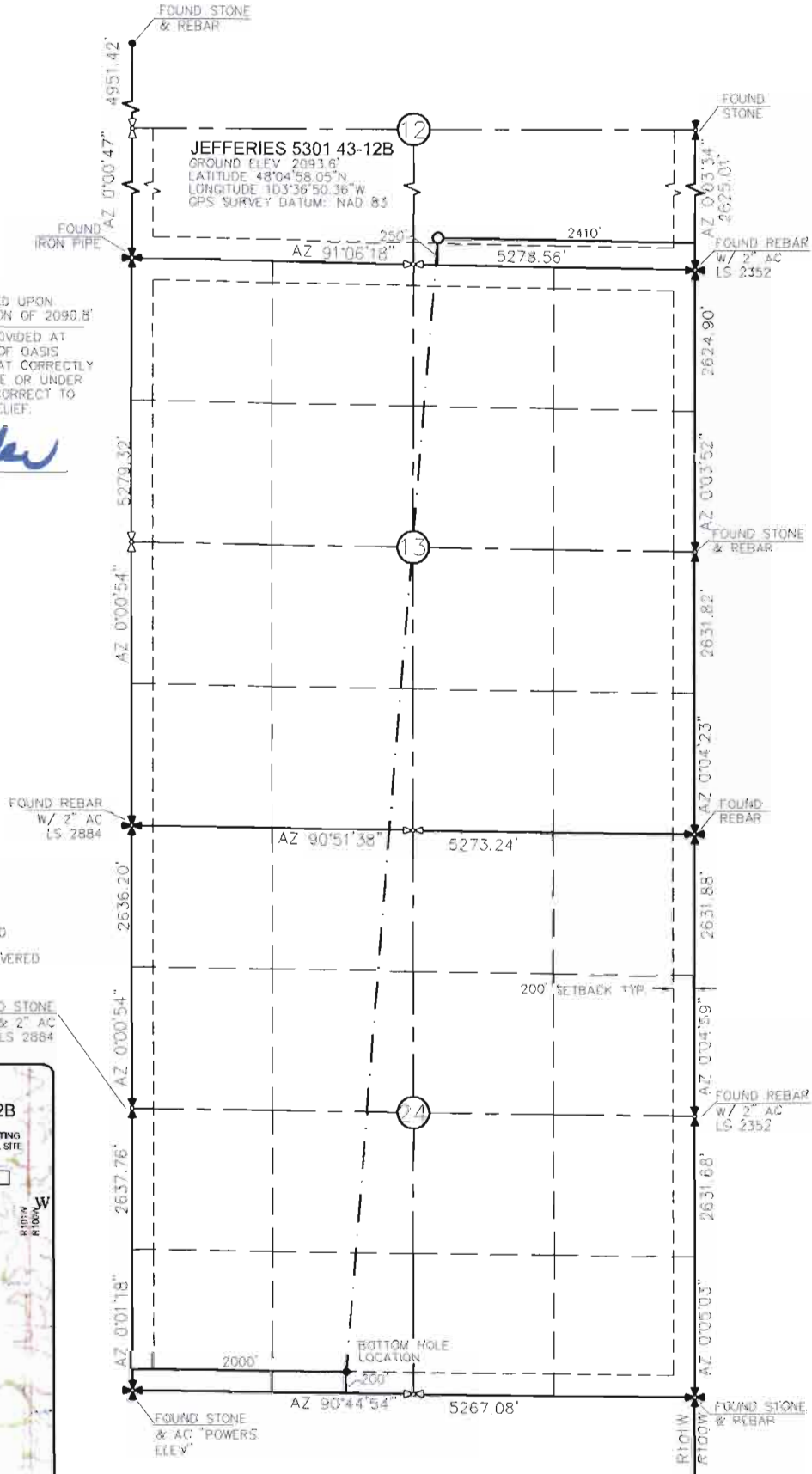
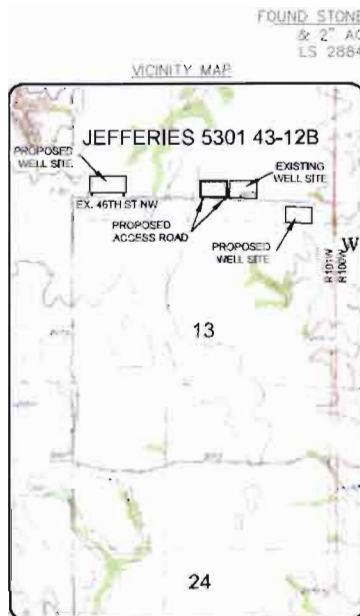
THIS SURVEY AND PLAT IS BEING PROVIDED AT
 THE REQUEST OF FABIAN KJØRSTAD OF OASIS
 PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY
 REPRESENTS WORK PERFORMED BY ME OR UNDER
 MY SUPERVISION AND IS TRUE AND CORRECT TO
 THE BEST OF MY KNOWLEDGE AND BELIEF.

Carl S. Vender

CARL S. VENDER LS 1222



- ✕ - MONUMENT - RECOVERED
- ✕ - MONUMENT - NOT RECOVERED



© 2011, INTERSTATE ENGINEERING, INC.

1/8



INTERSTATE
ENGINEERING

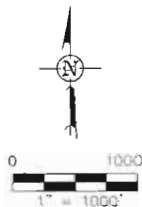
Interstate Engineering, Inc.
 P.O. Box 646
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617

OASIS PETROLEUM NORTH AMERICA, LLC
 WELL LOCATION PLAT
 SECTION 12, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA

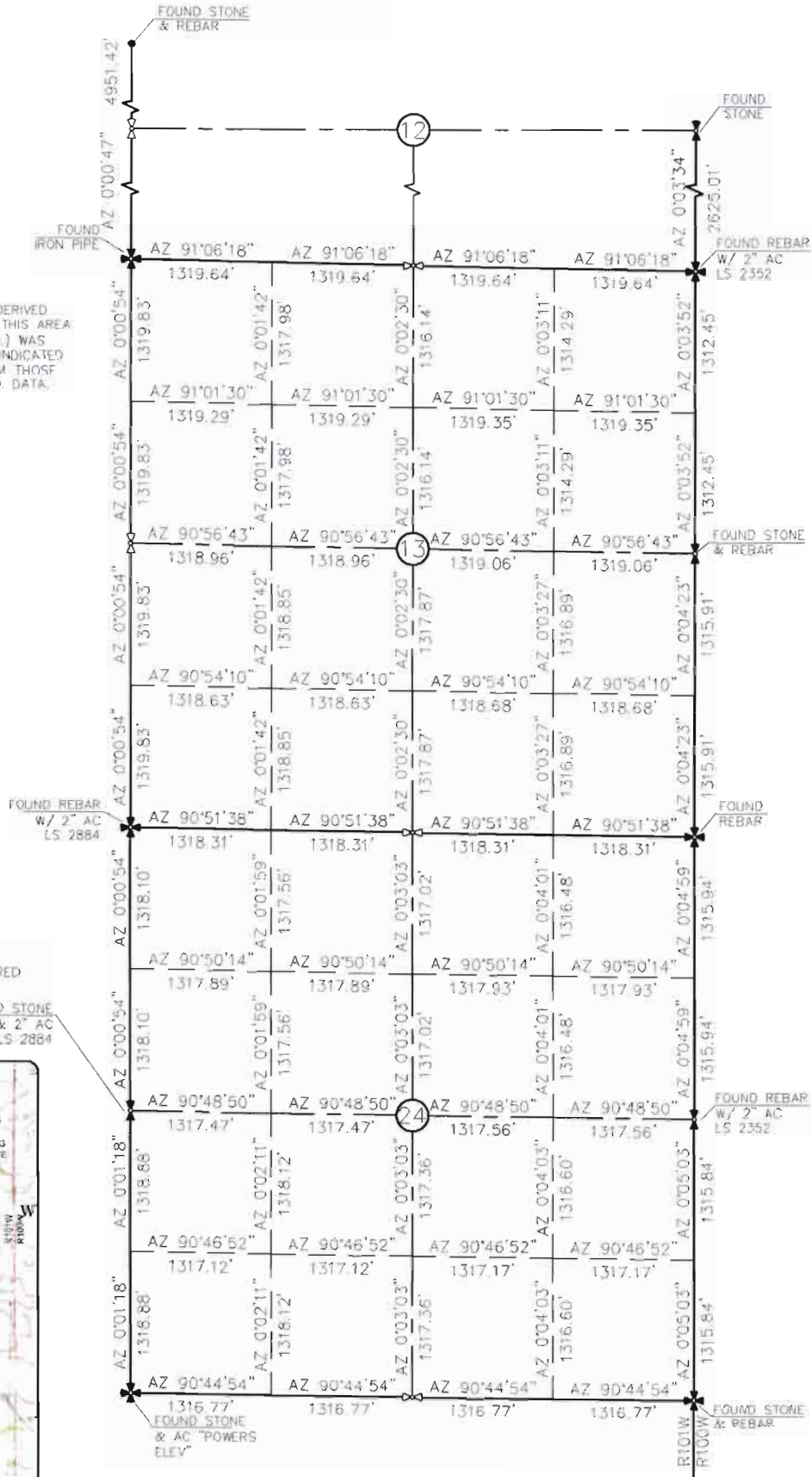
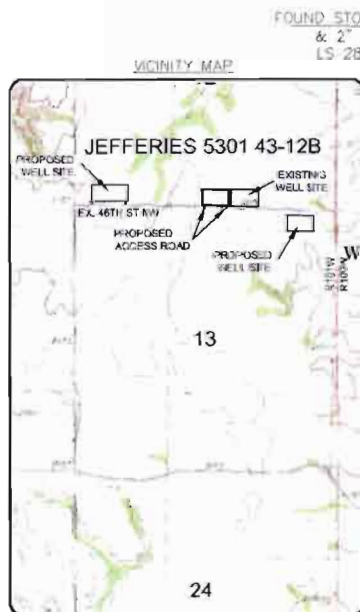
Revision No.	Date	By	Checked/Reviewed

SECTION BREAKDOWN
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
 "JEFFERIES 5301 43-12B"
 250 FEET FROM SOUTH LINE AND 2410 FEET FROM EAST LINE
 SECTIONS 12, 13, & 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

ALL BEARINGS ARE BASED ON G.P.S. DERIVED BEARINGS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1900. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA.



- MONUMENT - RECOVERED
- MONUMENT - NOT RECOVERED



© 2011, INTERSTATE ENGINEERING, INC.

2/8

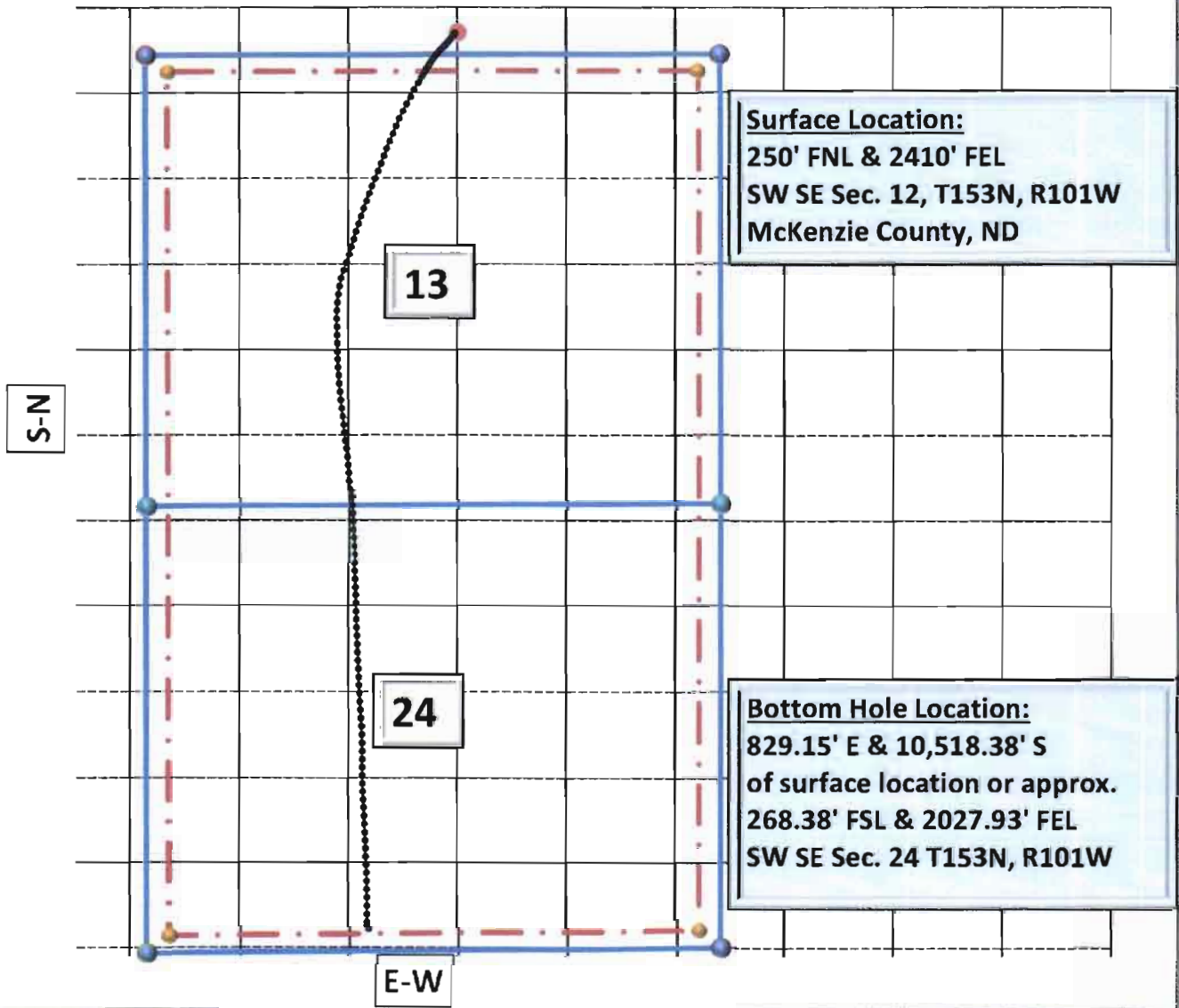


Interstate Engineering, Inc.
 P.O. Box 648
 425 East Adams Street
 Sidney, Montana 59226
 Ph: (406) 439-5617

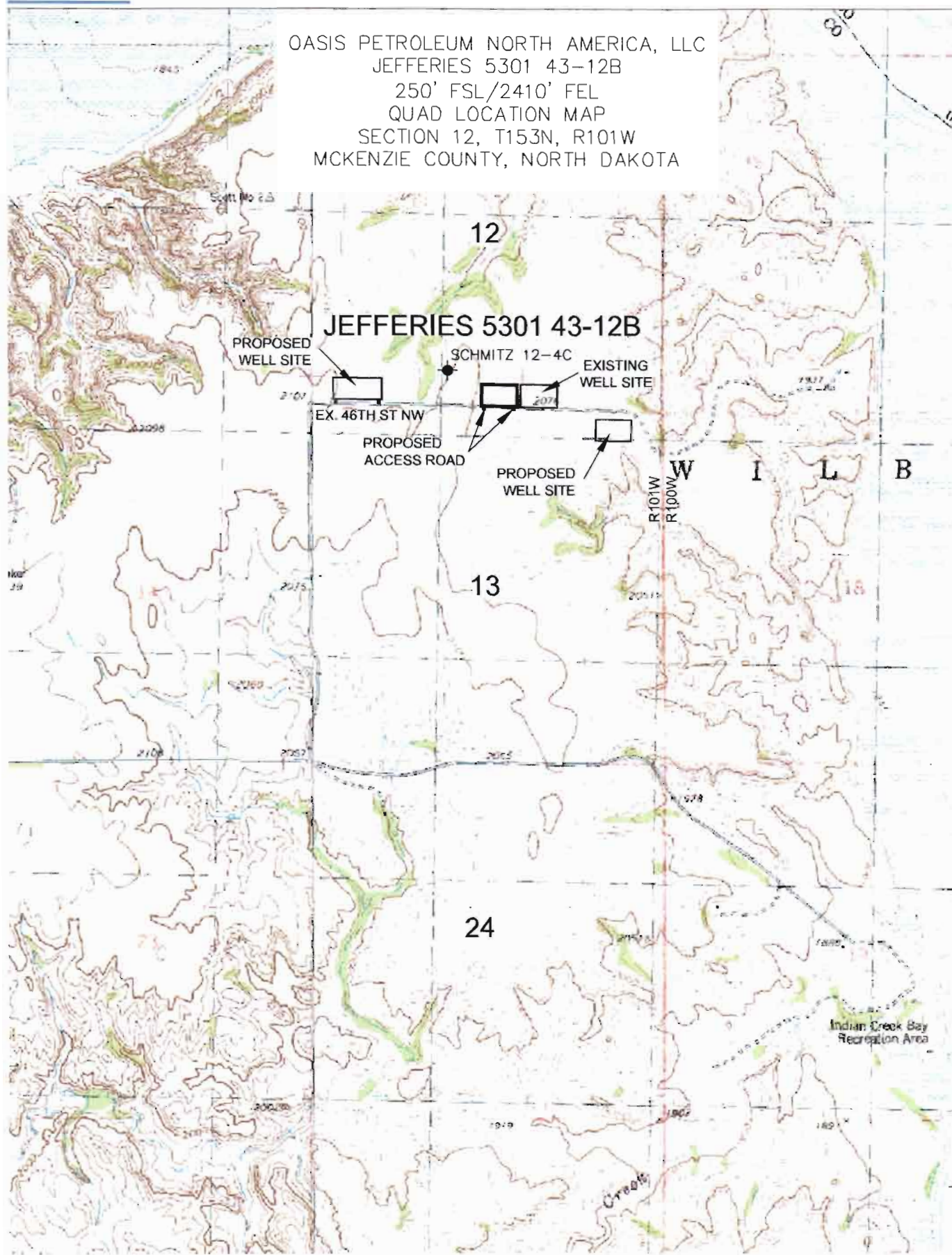
OASIS PETROLEUM NORTH AMERICA, LLC
 SECTION BREAKDOWN
 SECTIONS 12, 13, & 24, T153N, R101W
 MCKENZIE COUNTY, NORTH DAKOTA

Author	Drawn	By	Check/Rev

Vertical Section View



OASIS PETROLEUM NORTH AMERICA, LLC
JEFFERIES 5301 43-12B
250' FSL/2410' FEL
QUAD LOCATION MAP
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA



© 2011, INTERSTATE ENGINEERING, INC.

5/8



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617

OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

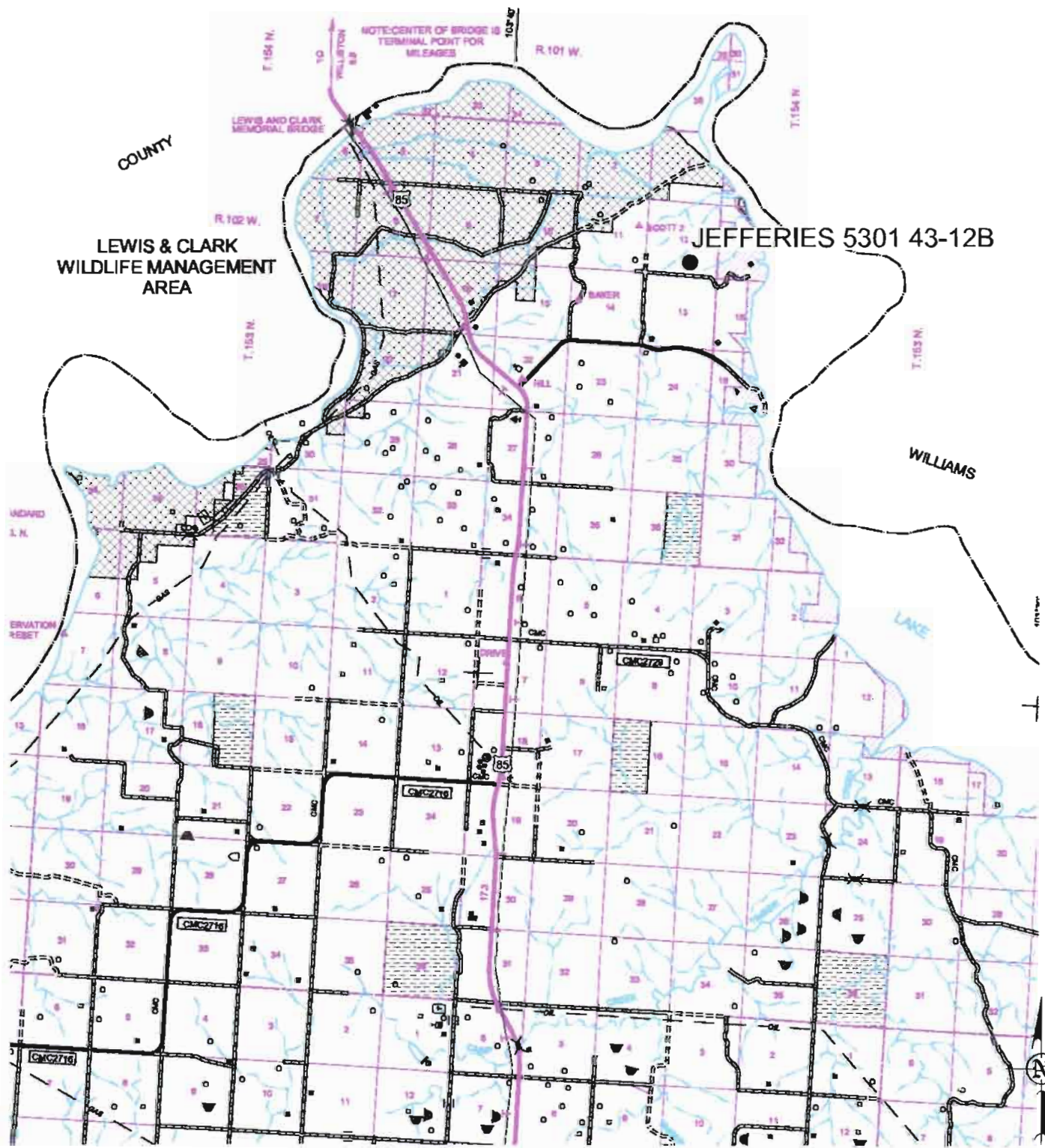
Revision No.	Date	By	Description

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002

"JEFFERIES 5301 43-12B"

250 FEET FROM SOUTH LINE AND 2410 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2011, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

6/8

INTERSTATE
ENGINEERING

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617

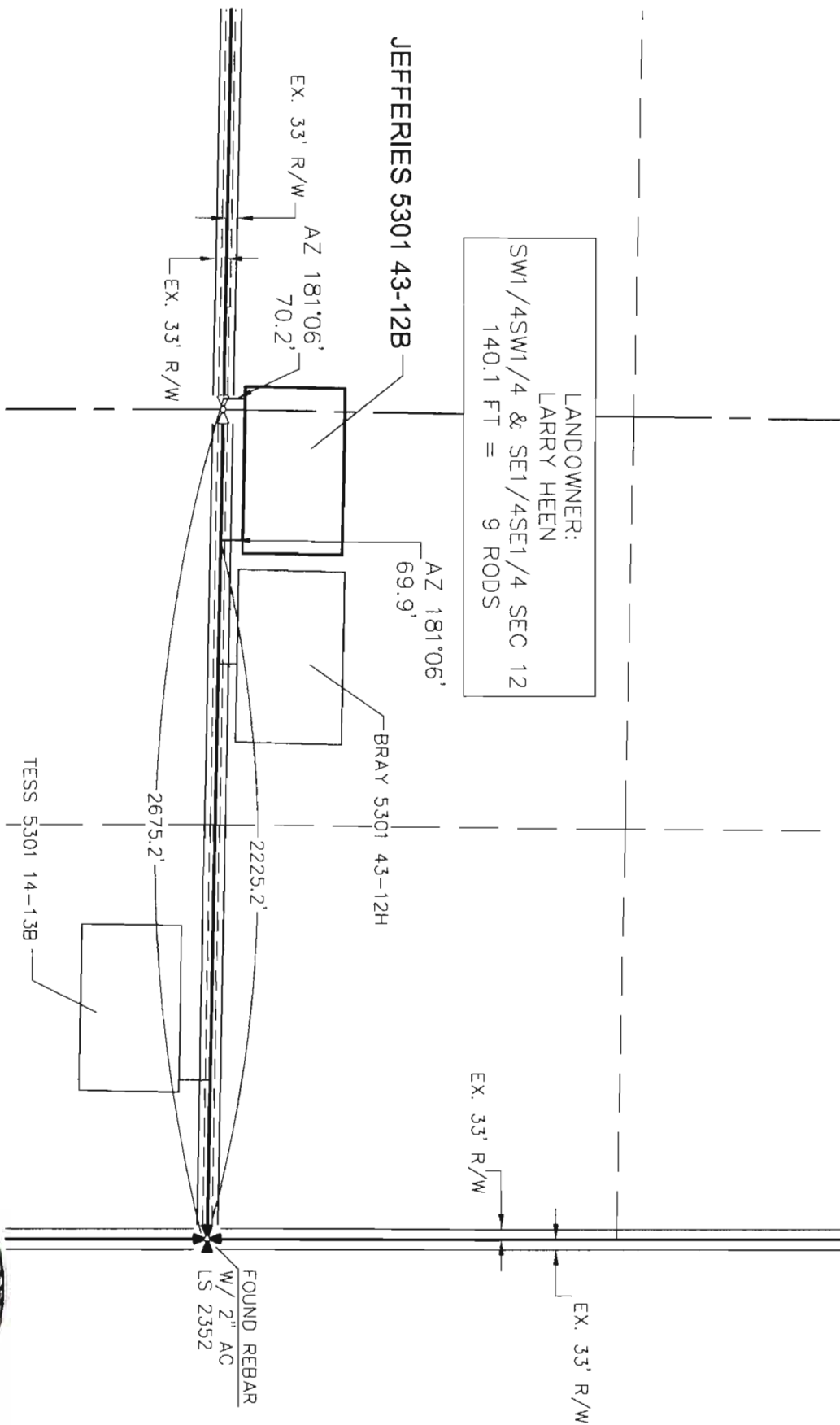
OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Revision No.	Date	By	Description

ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"
250 FEET FROM SOUTH LINE AND 2410 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

LANDOWNER:
LARRY HEEN
SW1/4SW1/4 & SE1/4SE1/4 SEC 12
140.1 FT = 9 RODS



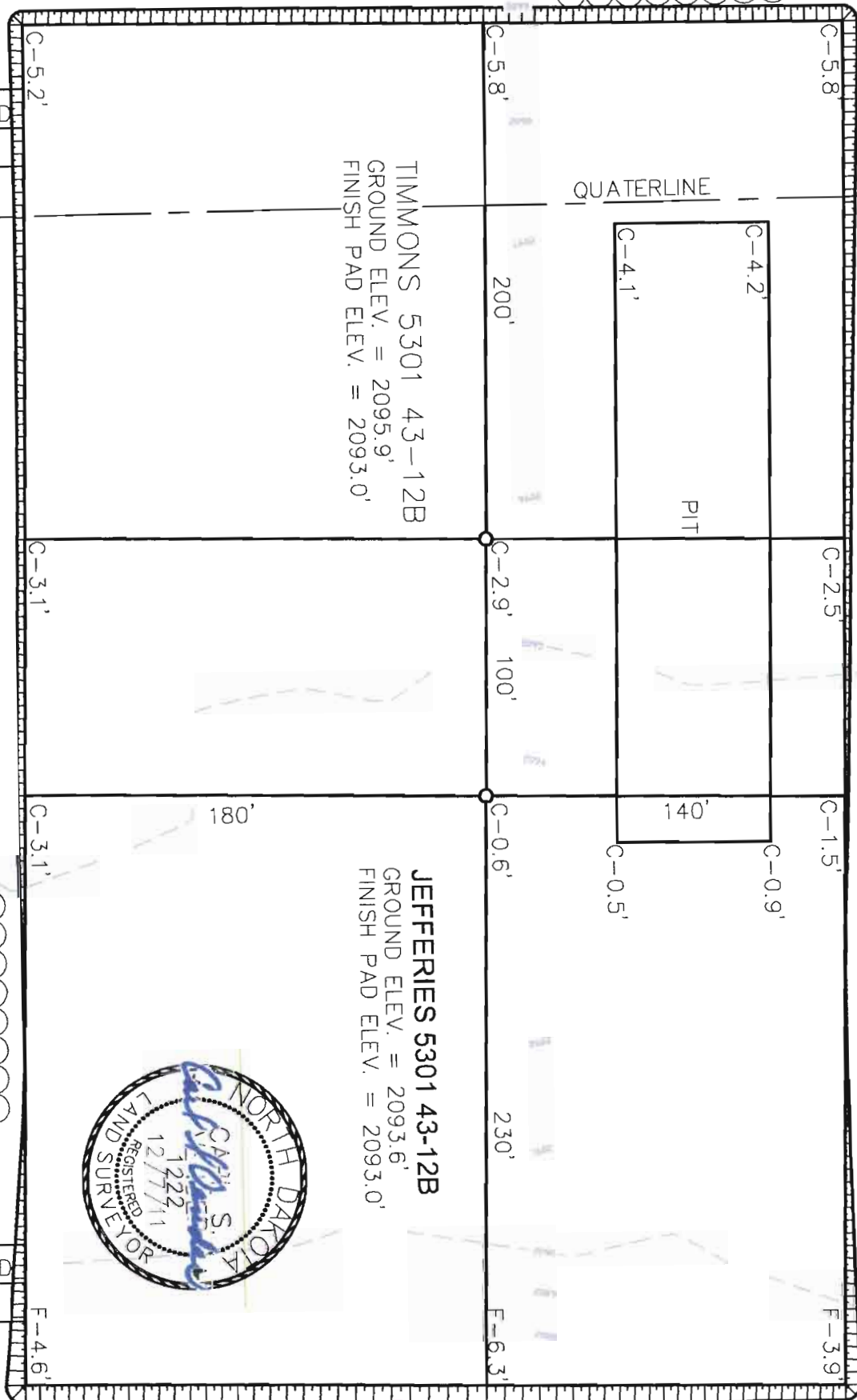
NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.



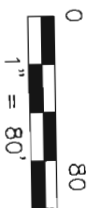
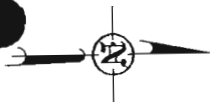
PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"

250 FEET FROM SOUTH LINE AND 2410 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.



DRILLING PLAN									
PROSPECT/FIELD	Indian Hills Horizontal Middle Bakken			COUNTY/STATE	McKenzie Co. ND				
OPERATOR	Quest Operating			RIG	Natura 149				
WELL NO.	5301-43-125			LEASE	Jeffries				
LOCATION	SWSE 12-153N-101W			Surface Location (survey plat):	2007 N		2410 1st		
EST. T.D.	21,313'			GROUND ELEV:	2093' Finished Pad Elev.		Sub Hieght: 26'		
TOTAL LATERAL:				10,228' (est)		KB ELEV:		2112'	
PROGNOSIS:				Based on 2 118' 50m/s)					
LOGS:				Type Interval					
OH Logs:				File to omit					
CBL/GR:				Above top of cement/GR to base of casing					
MWD GR:				KOP to lateral TD					
DEVIATION:									
Surf:				3 deg. max., 1 deg / 100'; svy every 500'					
Prod:				5 deg. max., 1 deg / 100'; svy every 100'					
DSTS:				None planned					
CORES:				None planned					
MUDLOGGING:									
Two-Man:				8,327'					
				~200' above the Charles (Kibbey) to Casing point; Casing point to TD					
				30' samples at direction of wellsite geologist; 10' through target @ curve land					
BOP:				11" 5000 psi blind, pipe & annular					
Dip Rate: -0.25" or .02" / 100' DOWN first 4000' then +0.55" or .93" / 100' UP									
Max. Anticipated BHP: 4670									
Surface Formation: Glacial till									
MUD:									
Interval									
Type									
WT									
Vis									
WL									
Remarks									
Surface									
Intermediate									
Liner									
Casing:									
Size									
Wt ppf									
Hole									
Depth									
Cement									
WOC									
Remarks									
Surface									
Intermediate									
Production									
Production Liner:									
PROBABLE PLUGS, IF REQ'D:									
OTHER:									
MD									
TVD									
FNL/FSL									
FEL/FWL									
S-T-R									
AZ									
Survey Company:									
Build Rate:									
13 deg / 100'									
Comments:									
DRILL TO KOP.									
DRILL CURVE TO 90 DEG AND 7" CASING POINT									
SET 7" CASING. DRILL MIDDLE BAKKEN LATERAL.									
MWD Surveys will be taken every 100' in vertical hole, and a minimum of every 30' while building curve and every 90' while drilling lateral.									
MWD GR to be run from KOP to Lateral TD.									
GR must be run to ground surface.									
Geology: MRB 12-9-2011									
Prepared by:									
Engineering: L. Strong 12/9/2011									

Formation / Marker Depths

Operator:	Oasis Petroleum, Inc.				
Well Name:	Jefferies 5301 43-12B				
Location:	250' FSL & 2,410' FWL Section 12, T153N, R101W				
Elevation:	GL: 2,093'	Sub: 25'		KB: 2,118'	
Formation / Marker	MWD Gamma Pick	MSL Datum	Distance to Target	Dip to Lindvig 11-	Dip to Achilles 5301
Kibbey Limestone	8,390'	-6,272'	2,360'	17' low	10' low
Charles Salt	8,523'	-6,405'	2,227'	4' low	25' low
Base Last Salt	9,224'	-7,106'	1,526'	5' low	8' low
Mission Canyon	9,451'	-7,333'	1,299'	10' low	29' low
Lodgepole	10,010'	-7,892'	740'	9' low	26' low
LPA	10,421'	-8,303'	329'	PARALLEL	11' low
LPB	10,491'	-8,373'	259'	2' low	32' low
LPC	10,513'	-8,395'	237'	PARALLEL	17' low
LPD	10,575'	-8,457'	175'	4' low	18' low
LPE	10,619'	-8,501'	131'	3' low	10' low
LPF	10,672'	-8,554'	78'	2' low	16' low
False Bakken	10,718'	-8,600'	32'	7' low	21' low
Upper Bakken Shale	10,725'	-8,607'	25'	5' low	21' low
Middle Bakken	10,740'	-8,622'	10'	5' low	18' low
Target	10,750'	-8,632'	-	8' low	15' low

Control Wells

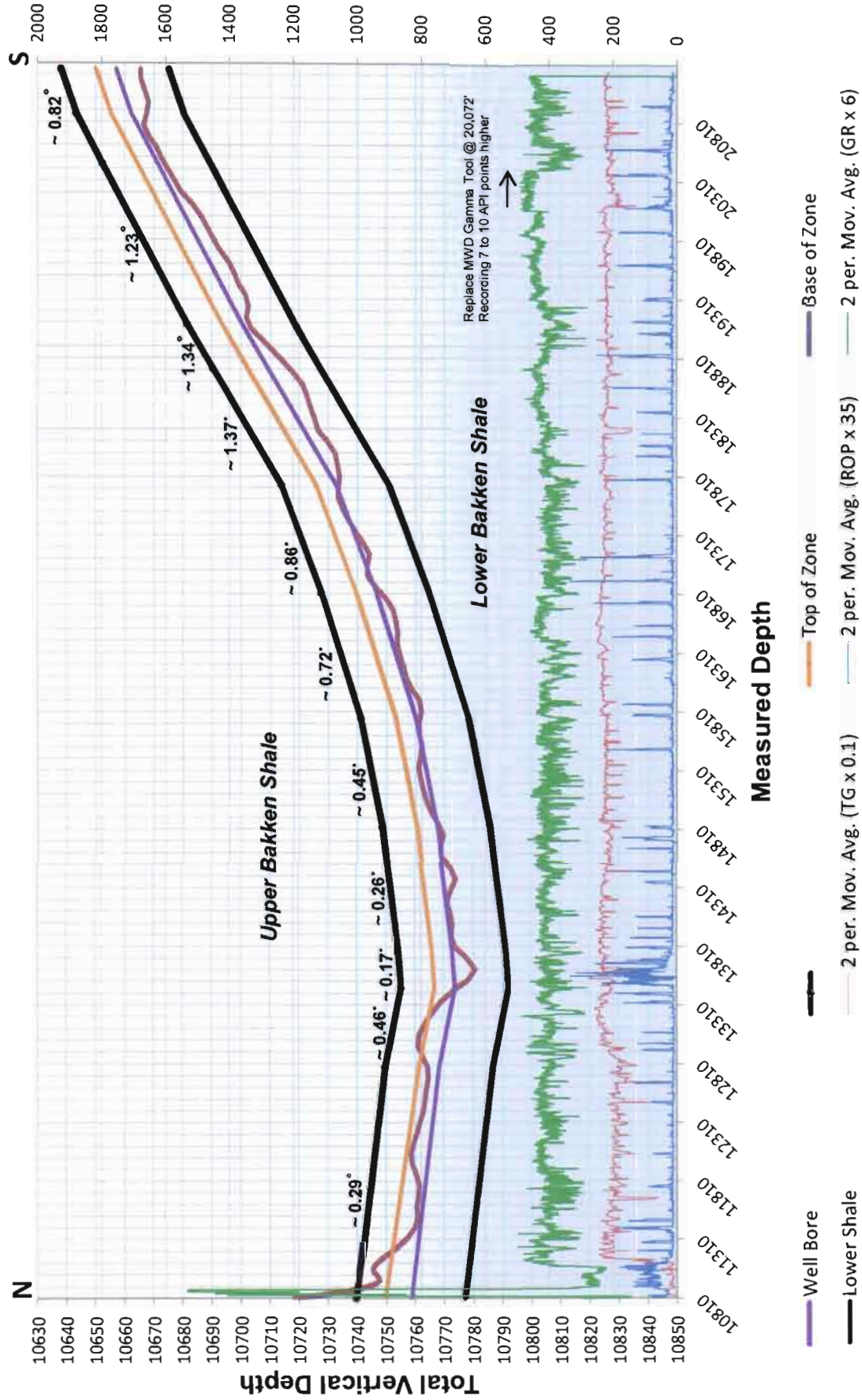
Operator:	SM Energy Company			Oasis Petroleum North America, LLC		
Well Name:	Lindvig 11-1HR			Achilles 5301 41-12B		
Location:	SESE Sec. 11 T153N R101W			SWSW Sec. 12 T153N R101W		
Elevation:	KB: 2,108'			KB: 2,119'		
Formation/Marker	E-Log/GR	MSL Datum	Distance to target	E-Log/GR	MSL Datum	Distance to target
Charles Salt	8,509'	-6,401'	2,226'	8,552'	-6,433'	2,182'
Base of Charles Salt	9,207'	-7,099'	1,528'	9,219'	-7,100'	1,515'
Lodgepole	9,991'	-7,883'	744'	10,002'	-7,883'	732'
LPA	10,411'	-8,303'	324'	10,411'	-8,292'	323'
LPB	10,479'	-8,371'	256'	10,460'	-8,341'	274'
LPC	10,503'	-8,395'	232'	10,497'	-8,378'	237'
LPD	10,561'	-8,453'	174'	10,558'	-8,439'	176'
LPE	10,607'	-8,499'	128'	10,610'	-8,491'	124'
LPF	10,660'	-8,552'	75'	10,657'	-8,538'	77'
False Bakken	10,700'	-8,592'	35'	10,701'	-8,582'	33'
Scallion	10,703'	-8,595'	32'	10,702'	-8,583'	32'
Upper Bakken Shale	10,709'	-8,601'	26'	10,708'	-8,589'	26'
Middle Bakken	10,726'	-8,618'	9'	10,727'	-8,608'	7'
Target	10,735'	-8,627'	-	10,734'	-8,615'	-



Surface: 250' FSL & 2,410' FEL
NW NW Sec. 12, T153N, R101W
McKenzie County, ND

Jefferies 5301 43-12B Horizontal Cross Section

Bottom Hole Location
829.15' E & 10,518.38' S
of surface location or approx.
268.38' FSL & 2,027.93' FEL
SWSE Section 24 T153N R101W





Operator: Oasis Petroleum	
Well : Rowley 6093 43-23H	
MWD Providers	Weatherford
Directional Supervision:	RPM

Section:	23	QQ:	NE NW	County:	Burke	State:	ND
Township:	160	N/S:	N	Footages:	180	FN/SL:	S
Range:	93	E/W:	W		1350	FE/WL:	E

Vertical Section Plane:	0.00
-------------------------	------

#	MD	Inc.	Az.	T.V.D.	Ver. Sect.	Coordinates		DLS
						+N / -S	+E / -W	
Tie	2042.00	0.00	0.00	2042.00	0.00	0.00	0.00	0.00
1	2190.00	1.67	257.30	2189.98	-0.47	-0.47	-2.10	1.13
2	2285.00	1.49	264.99	2284.94	-0.89	-0.89	-4.68	0.29
3	2381.00	1.38	269.32	2380.91	-1.01	-1.01	-7.08	0.16
4	2476.00	1.43	271.99	2475.88	-0.98	-0.98	-9.41	0.09
5	2571.00	1.30	271.16	2570.86	-0.92	-0.92	-11.67	0.14
6	2666.00	1.38	275.80	2665.83	-0.78	-0.78	-13.89	0.14
7	2756.00	1.22	284.85	2755.81	-0.43	-0.43	-15.89	0.29
8	2850.00	1.31	300.16	2849.79	0.37	0.37	-17.79	0.37
9	2945.00	1.38	285.47	2944.76	1.22	1.22	-19.83	0.37
10	3040.00	1.36	280.24	3039.73	1.73	1.73	-22.04	0.13
11	3134.00	1.03	274.99	3133.71	2.00	2.00	-23.98	0.37
12	3228.00	1.02	273.77	3227.70	2.13	2.13	-25.66	0.03
13	3323.00	0.97	277.39	3322.68	2.29	2.29	-27.30	0.08
14	3417.00	1.10	289.29	3416.67	2.69	2.69	-28.94	0.27
15	3512.00	1.46	288.22	3511.64	3.37	3.37	-30.95	0.38
16	3607.00	1.53	295.71	3606.61	4.29	4.29	-33.24	0.22
17	3701.00	1.29	288.83	3700.58	5.18	5.18	-35.38	0.31
18	3795.00	0.95	290.89	3794.56	5.80	5.80	-37.11	0.36
19	3890.00	1.17	291.10	3889.55	6.43	6.43	-38.75	0.23
20	3984.00	1.14	291.19	3983.53	7.11	7.11	-40.51	0.03
21	4079.00	0.90	292.55	4078.51	7.74	7.74	-42.08	0.25
22	4173.00	1.05	297.17	4172.50	8.42	8.42	-43.53	0.18
23	4268.00	0.86	300.26	4267.49	9.17	9.17	-44.92	0.21
24	4362.00	0.23	322.39	4361.48	9.68	9.68	-45.65	0.69
25	4457.00	0.29	31.14	4456.48	10.04	10.04	-45.64	0.31
26	4551.00	0.49	74.84	4550.48	10.34	10.34	-45.13	0.37
27	4646.00	0.88	143.65	4645.47	9.86	9.86	-44.30	0.88
28	4740.00	1.01	131.31	4739.46	8.74	8.74	-43.25	0.26
29	4835.00	0.94	111.38	4834.45	7.90	7.90	-41.90	0.36
30	4929.00	1.34	121.32	4928.43	7.05	7.05	-40.24	0.47
31	5023.00	1.17	131.18	5022.41	5.84	5.84	-38.58	0.29

32	5118.00	1.39	153.12	5117.38	4.18	4.18	-37.33	0.56
33	5212.00	0.97	124.39	5211.36	2.71	2.71	-36.16	0.76
34	5307.00	0.88	135.73	5306.35	1.73	1.73	-34.98	0.21
35	5401.00	0.69	127.07	5400.34	0.88	0.88	-34.03	0.24
36	5496.00	0.68	133.08	5495.34	0.15	0.15	-33.16	0.08
37	5590.00	0.16	128.95	5589.33	-0.32	-0.32	-32.65	0.55
38	5685.00	0.13	184.12	5684.33	-0.51	-0.51	-32.56	0.14
39	5779.00	0.25	170.07	5778.33	-0.82	-0.82	-32.53	0.14
40	5874.00	0.31	154.33	5873.33	-1.25	-1.25	-32.38	0.10
41	5968.00	0.36	142.81	5967.33	-1.72	-1.72	-32.09	0.09
42	6063.00	0.24	169.43	6062.33	-2.15	-2.15	-31.88	0.19
43	6157.00	0.13	175.54	6156.33	-2.45	-2.45	-31.83	0.12
44	6252.00	0.44	115.54	6251.33	-2.72	-2.72	-31.49	0.41
45	6346.00	0.69	153.33	6345.32	-3.38	-3.38	-30.91	0.46
46	6440.00	1.05	123.56	6439.31	-4.36	-4.36	-29.94	0.60
47	6535.00	1.39	94.26	6534.29	-4.93	-4.93	-28.07	0.74
48	6629.00	1.98	73.82	6628.25	-4.56	-4.56	-25.37	0.89
49	6724.00	0.33	48.48	6723.23	-3.92	-3.92	-23.59	1.78
50	6818.00	0.08	14.83	6817.23	-3.68	-3.68	-23.37	0.28
51	6912.00	0.47	345.77	6911.23	-3.24	-3.24	-23.45	0.43
52	7007.00	0.86	21.12	7006.22	-2.20	-2.20	-23.29	0.58
53	7101.00	0.98	102.53	7100.21	-1.71	-1.71	-22.25	1.28
54	7196.00	1.26	131.97	7195.19	-2.59	-2.59	-20.68	0.66
55	7290.00	1.16	125.38	7289.17	-3.83	-3.83	-19.14	0.18
56	7385.00	1.22	140.83	7384.15	-5.17	-5.17	-17.71	0.34
57	7479.00	1.13	132.23	7478.13	-6.57	-6.57	-16.39	0.21
58	7573.00	1.14	147.23	7572.12	-7.98	-7.98	-15.20	0.32
59	7668.00	1.04	132.10	7667.10	-9.35	-9.35	-14.05	0.32
60	7762.00	1.02	136.58	7761.08	-10.53	-10.53	-12.84	0.09
61	7856.00	0.91	150.48	7855.07	-11.79	-11.79	-11.90	0.27
62	7951.00	0.66	155.99	7950.06	-12.94	-12.94	-11.30	0.27
63	8045.00	1.10	162.88	8044.05	-14.30	-14.30	-10.82	0.48
64	8139.00	1.02	166.54	8138.03	-15.98	-15.98	-10.36	0.11
65	8234.00	0.87	172.82	8233.02	-17.51	-17.51	-10.07	0.19
66	8329.00	0.64	156.14	8328.01	-18.72	-18.72	-9.77	0.33
67	8386.00	0.76	147.42	8385.01	-19.33	-19.33	-9.43	0.28
68	8412.00	0.72	150.21	8411.01	-19.61	-19.61	-9.26	0.21
69	8443.00	0.84	10.68	8442.01	-19.56	-19.56	-9.12	4.72
70	8475.00	3.86	357.89	8473.97	-18.25	-18.25	-9.12	9.52
71	8506.00	7.34	358.31	8504.82	-15.23	-15.23	-9.21	11.23
72	8537.00	10.85	0.48	8535.43	-10.33	-10.33	-9.25	11.37
73	8569.00	14.35	1.17	8566.65	-3.35	-3.35	-9.14	10.95
74	8600.00	17.99	0.00	8596.42	5.28	5.28	-9.06	11.79
75	8632.00	21.38	359.88	8626.55	16.06	16.06	-9.08	10.59

76	8663.00	24.71	359.63	8655.07	28.19	28.19	-9.13	10.75
77	8695.00	28.29	359.03	8683.70	42.47	42.47	-9.30	11.22
78	8726.00	32.14	358.89	8710.48	58.06	58.06	-9.59	12.42
79	8757.00	34.94	357.03	8736.32	75.17	75.17	-10.21	9.62
80	8789.00	38.09	356.89	8762.04	94.19	94.19	-11.22	9.85
81	8820.00	42.05	357.29	8785.76	114.11	114.11	-12.23	12.80
82	8852.00	46.81	357.59	8808.60	136.48	136.48	-13.22	14.89
83	8883.00	51.71	356.68	8828.83	159.94	159.94	-14.40	15.96
84	8915.00	56.87	356.56	8847.50	185.86	185.86	-15.94	16.13
85	8946.00	61.76	356.40	8863.31	212.47	212.47	-17.57	15.78
86	8978.00	65.57	356.38	8877.51	241.08	241.08	-19.38	11.91
87	9009.00	69.43	357.08	8889.37	269.67	269.67	-21.01	12.63
88	9041.00	74.72	357.02	8899.21	300.07	300.07	-22.58	16.53
89	9072.00	80.03	356.76	8905.98	330.26	330.26	-24.22	17.15
90	9131.00	88.67	359.15	8911.79	388.88	388.88	-26.30	15.19
91	9226.00	90.07	357.88	8912.83	483.84	483.84	-28.76	1.99
92	9322.00	91.05	357.81	8911.90	579.76	579.76	-32.37	1.02
93	9417.00	91.96	358.26	8909.40	674.67	674.67	-35.63	1.07
94	9513.00	92.09	358.85	8906.01	770.58	770.58	-38.05	0.63
95	9608.00	90.35	359.48	8903.99	865.55	865.55	-39.43	1.95
96	9704.00	90.91	359.67	8902.93	961.54	961.54	-40.15	0.62
97	9800.00	89.37	359.99	8902.70	1057.53	1057.53	-40.43	1.64
98	9895.00	89.58	359.59	8903.57	1152.53	1152.53	-40.78	0.48
99	9991.00	89.51	358.52	8904.33	1248.51	1248.51	-42.36	1.12
100	10086.00	89.93	358.46	8904.79	1343.48	1343.48	-44.86	0.45
101	10182.00	89.72	357.76	8905.09	1439.42	1439.42	-48.03	0.76
102	10277.00	90.28	358.66	8905.09	1534.37	1534.37	-51.00	1.12
103	10372.00	90.77	358.69	8904.21	1629.35	1629.35	-53.20	0.52
104	10465.00	91.33	358.04	8902.51	1722.29	1722.29	-55.85	0.92
105	10559.00	90.42	359.24	8901.08	1816.25	1816.25	-58.08	1.60
106	10652.00	90.35	359.89	8900.45	1909.25	1909.25	-58.78	0.70
107	10745.00	90.28	359.07	8899.94	2002.24	2002.24	-59.63	0.88
108	10838.00	89.79	357.99	8899.88	2095.21	2095.21	-62.01	1.28
109	10931.00	88.60	356.94	8901.19	2188.10	2188.10	-66.13	1.71
110	11024.00	89.79	358.73	8902.50	2281.02	2281.02	-69.64	2.31
111	11117.00	89.86	358.12	8902.78	2373.99	2373.99	-72.20	0.66
112	11210.00	90.14	358.26	8902.78	2466.94	2466.94	-75.13	0.34
113	11303.00	89.79	358.11	8902.84	2559.89	2559.89	-78.08	0.41
114	11396.00	89.65	357.95	8903.29	2652.84	2652.84	-81.28	0.23
115	11490.00	90.63	358.20	8903.06	2746.78	2746.78	-84.43	1.08
116	11585.00	90.56	358.52	8902.07	2841.74	2841.74	-87.15	0.34
117	11681.00	90.63	357.88	8901.08	2937.69	2937.69	-90.17	0.67
118	11776.00	91.53	359.44	8899.29	3032.64	3032.64	-92.39	1.90
119	11872.00	91.61	359.18	8896.66	3128.60	3128.60	-93.54	0.28

120	11968.00	90.63	358.78	8894.78	3224.56	3224.56	-95.25	1.10
121	12063.00	91.19	359.88	8893.27	3319.54	3319.54	-96.36	1.30
122	12159.00	90.42	358.97	8891.92	3415.52	3415.52	-97.33	1.24
123	12255.00	90.98	357.95	8890.75	3511.48	3511.48	-99.91	1.21
124	12350.00	91.26	358.00	8888.89	3606.40	3606.40	-103.26	0.30
125	12446.00	90.63	359.19	8887.31	3702.36	3702.36	-105.62	1.40
126	12541.00	90.28	359.11	8886.56	3797.35	3797.35	-107.03	0.38
127	12637.00	91.47	0.11	8885.09	3893.33	3893.33	-107.68	1.62
128	12733.00	90.00	359.36	8883.86	3989.32	3989.32	-108.12	1.72
129	12828.00	90.35	359.92	8883.57	4084.31	4084.31	-108.72	0.70
130	12924.00	90.56	0.13	8882.81	4180.31	4180.31	-108.68	0.31
131	13019.00	90.56	0.57	8881.88	4275.30	4275.30	-108.10	0.46
132	13115.00	90.42	0.40	8881.06	4371.30	4371.30	-107.29	0.23
133	13211.00	90.21	0.67	8880.53	4467.29	4467.29	-106.39	0.36
134	13306.00	90.56	0.62	8879.89	4562.28	4562.28	-105.32	0.37
135	13402.00	90.91	0.13	8878.66	4658.27	4658.27	-104.69	0.63
136	13497.00	90.98	1.08	8877.09	4753.25	4753.25	-103.69	1.00
137	13593.00	91.33	1.04	8875.16	4849.22	4849.22	-101.91	0.37
138	13688.00	90.07	2.27	8874.00	4944.17	4944.17	-99.17	1.85
139	13784.00	89.79	2.01	8874.11	5040.10	5040.10	-95.58	0.40
140	13879.00	90.07	1.71	8874.23	5135.05	5135.05	-92.50	0.43
141	13975.00	89.65	0.86	8874.46	5231.02	5231.02	-90.35	0.99
142	14071.00	90.07	1.29	8874.70	5327.01	5327.01	-88.55	0.63
143	14166.00	91.05	1.59	8873.77	5421.97	5421.97	-86.16	1.08
144	14262.00	90.07	0.44	8872.83	5517.95	5517.95	-84.46	1.57
145	14357.00	90.35	1.18	8872.48	5612.94	5612.94	-83.12	0.83
146	14453.00	89.30	0.75	8872.78	5708.92	5708.92	-81.50	1.18
147	14548.00	89.58	359.69	8873.70	5803.92	5803.92	-81.13	1.15
148	14644.00	91.19	0.05	8873.06	5899.91	5899.91	-81.35	1.72
149	14739.00	92.30	359.59	8870.17	5994.86	5994.86	-81.65	1.26
150	14835.00	91.75	359.75	8866.77	6090.80	6090.80	-82.20	0.60
151	14931.00	91.74	359.90	8863.85	6186.76	6186.76	-82.50	0.16
152	15026.00	91.05	0.85	8861.54	6281.72	6281.72	-81.87	1.24
153	15122.00	90.07	0.83	8860.60	6377.71	6377.71	-80.47	1.02
154	15217.00	89.79	0.06	8860.72	6472.70	6472.70	-79.73	0.86
155	15313.00	89.93	1.07	8860.95	6568.70	6568.70	-78.78	1.06
156	15408.00	90.35	0.21	8860.72	6663.69	6663.69	-77.72	1.01
157	15504.00	91.33	359.27	8859.31	6759.68	6759.68	-78.16	1.41
158	15599.00	92.23	359.04	8856.36	6854.62	6854.62	-79.56	0.98
159	15695.00	91.19	0.02	8853.50	6950.57	6950.57	-80.34	1.49
160	15791.00	91.33	359.90	8851.38	7046.55	7046.55	-80.41	0.19
161	15886.00	91.61	359.75	8848.95	7141.52	7141.52	-80.70	0.33
162	15981.00	91.75	359.90	8846.16	7236.48	7236.48	-80.99	0.22
163	16077.00	92.02	359.03	8843.00	7332.42	7332.42	-81.89	0.95

164	16172.00	90.70	0.07	8840.75	7427.39	7427.39	-82.63	1.77
165	16268.00	91.12	359.55	8839.23	7523.37	7523.37	-82.95	0.70
166	16364.00	90.49	0.53	8837.88	7619.36	7619.36	-82.88	1.21
167	16459.00	90.35	0.23	8837.18	7714.36	7714.36	-82.25	0.35
168	16555.00	90.91	359.78	8836.12	7810.35	7810.35	-82.25	0.75
169	16650.00	89.86	359.89	8835.49	7905.35	7905.35	-82.52	1.11
170	16746.00	89.93	0.03	8835.66	8001.35	8001.35	-82.59	0.16
171	16842.00	91.33	359.80	8834.61	8097.34	8097.34	-82.73	1.48
172	16937.00	91.89	0.12	8831.94	8192.30	8192.30	-82.80	0.68
173	17033.00	89.37	357.90	8830.88	8288.27	8288.27	-84.45	3.50
174	17128.00	90.14	358.19	8831.29	8383.21	8383.21	-87.69	0.87
175	17224.00	89.93	357.86	8831.23	8479.15	8479.15	-91.00	0.41
176	17320.00	90.63	357.99	8830.76	8575.09	8575.09	-94.48	0.74
177	17415.00	91.33	358.25	8829.14	8670.02	8670.02	-97.60	0.79
178	17511.00	90.14	357.29	8827.90	8765.94	8765.94	-101.33	1.59
179	17606.00	90.49	357.31	8827.38	8860.83	8860.83	-105.81	0.37
180	17702.00	91.26	357.62	8825.92	8956.72	8956.72	-110.05	0.86
181	17797.00	89.51	357.03	8825.28	9051.61	9051.61	-114.48	1.94
182	17893.00	89.86	356.61	8825.81	9147.46	9147.46	-119.81	0.57
183	17988.00	90.28	356.89	8825.69	9242.31	9242.31	-125.20	0.53
184	18084.00	90.28	356.24	8825.22	9338.14	9338.14	-130.95	0.68
185	18178.00	90.84	355.72	8824.30	9431.90	9431.90	-137.54	0.81
186	18275.00	90.77	355.61	8822.94	9528.61	9528.61	-144.87	0.13
187	18370.00	91.12	355.45	8821.37	9623.31	9623.31	-152.27	0.41
188	18466.00	90.56	355.31	8819.96	9718.99	9718.99	-160.00	0.60
189	18561.00	90.28	355.14	8819.27	9813.66	9813.66	-167.91	0.34
190	18657.00	90.56	355.06	8818.56	9909.30	9909.30	-176.11	0.30
191	18753.00	90.84	355.24	8817.39	10004.95	10004.95	-184.23	0.35
192	18794.00	91.12	355.43	8816.69	10045.81	10045.81	-187.56	0.83
Proj.	18850.00	91.11	355.43	8815.60	10101.62	10101.62	-192.02	0.02

Jefferies 5301 43-12B

LITHOLOGY

Rig crews caught lagged drill cutting samples at 30' intervals from 8,360' through 10,780'. Samples were logged at 10' intervals during the curve from 10,780' to 11,050'. Samples were continuously logged at 30' intervals throughout the lateral section.

Electric geophysical log, sample and/or MWD gamma ray markers and tops are included in the sample descriptions below for reference. Samples were examined wet and dry under a binocular microscope and checked for hydrocarbon cut fluorescence with *Entron* critical cleaning solvent in approximately 30' intervals. Sample descriptions begin just above the limestone marker in the Kibbey Formation. Drilling fluids consisted diesel invert during vertical and curve build sections and salt water during lateral to TD.

Drilling in Kibbey Formation [Mississippian Big Snowy Group]

8360-8390 SANDSTONE: white to clear, fine grained, friable, subrounded to rounded, moderately to well sorted, earthy texture, calcareous cement; SHALE: rounded to orange, firm, subblocky, earthy to waxy texture, calcareous cement

Kibbey Limestone Marker

8,390' TVD (-6,272')

8390-8426 LIMESTONE: wackestone, medium gray, tan, finely crystalline, hard, blocky, earthy to micro crystalline

8426-8450 SANDSTONE: white to clear, fine grained, friable, subrounded to rounded, moderately to well sorted, earthy texture, calcareous cement

8450-8480 SANDSTONE: white to clear, fine grained, friable, subrounded to rounded, moderately to well sorted, earthy texture, calcareous cement, trace salt

8480-8510 SANDSTONE: white to clear, trace gray, fine to medium grained, friable, subrounded to rounded, moderately to well sorted, earthy texture, calcareous cement

8510-8523 SANDSTONE: white to clear, light to medium gray, fine to medium grained, friable, subrounded to rounded, moderately to well sorted, earthy texture, calcareous cement

Charles Salt Formation

8,523' TVD (-6,405')

8523-8546 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture

8546-8568 SANDSTONE: white to clear, trace gray, fine to medium grained, friable, subrounded to rounded, moderately to well sorted, earthy texture, calcareous cement; ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture

8568-8600 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture

8600-8630 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture

8630-8660 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture; trace
ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture

8660-8690 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture; trace
ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture

8690-8720 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture; trace
ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture

8720-8750 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture; trace
ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture

8750-8780 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture; trace
ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture

8780-8810 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture; trace
ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture, trace LIMESTONE

8810-8828 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture; trace
ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture, trace LIMESTONE

8828-8866 LIMESTONE: mudstone to wackestone, medium gray, medium brown, microcrystalline, blocky, firm to hard, earthy texture, trace to moderately anhydrite

8866-8890 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture; trace
ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture, trace LIMESTONE

8890-8920 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture; trace
ANHYDRITE: white to off white, microcrystalline, soft, amorphous texture, trace LIMESTONE

8920-8950 LIMESTONE: mudstone, light gray, microcrystalline, subblocky, firm, crystalline texture, argillaceous in part, trace intercrystalline porosity

8950-8980 LIMESTONE: mudstone, light gray, microcrystalline, subblocky, firm, crystalline texture, argillaceous in part, trace intercrystalline porosity with interbedded DOLOMITE: mudstone, light blue, microcrystalline, subblocky, firm, microsucrosic texture

8980-9016 LIMESTONE: mudstone, light gray, microcrystalline, subblocky, firm, crystalline texture, argillaceous in part

9016-9042 SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture

9042-9080 LIMESTONE: mudstone, light gray, microcrystalline, subblocky, firm, crystalline texture, argillaceous in part, trace intergranular porosity with interbedded DOLOMITE: mudstone, light blue, microcrystalline, subblocky, firm, microsucrosic texture, trace intergranular porosity

9080-9110 DOLOMITIC LIMESTONE: mudstone, light gray to blue, light brown to gray, medium gray, microcrystalline, subblocky, firm, microsucrosic texture; LIMESTONE: mudstone, light gray, medium gray, microcrystalline, subblocky, firm, crystalline texture, argillaceous in part, trace intergranular porosity

9110-9148 LIMESTONE: as above; ANHYDRITE: medium to light gray, microcrystalline, soft, slightly firm, amorphous texture; DOLOMITE: mudstone, light blue, microcrystalline, subblocky, firm, microsucrosic texture

9148-9180 LIMESTONE: mudstone, light gray, medium gray, microcrystalline, subblocky, firm, crystalline texture, argillaceous in part, trace intergranular porosity; ANHYDRITE: medium to light gray, microcrystalline, soft, slightly firm, amorphous texture

9180-9224 ANHYDRITE: medium to light gray, microcrystalline, subblocky, firm, amorphous texture; SALT: translucent, milky white, cryptocrystalline, hard, subhedral, crystalline texture

Base Charles Salt

9,224' TVD (-7,106')

9224-9260 LIMESTONE: mudstone, light gray to brown, microcrystalline, subblocky, firm to hard, earthy texture, rare calcite, slightly argillaceous, trace intergranular porosity

9260-9290 LIMESTONE: mudstone, light gray to brown, trace light yellow, microcrystalline, subblocky, firm to hard, earthy texture, rare calcite, slightly argillaceous, with interbedded DOLOMITIC LIMESTONE: mudstone to wackestone, light brown to gray, blue, microcrystalline to very fine crystalline, subblocky, firm, earthy to sucrosic texture, trace fossils fragments, trace intergranular porosity

9290-9320 LIMESTONE: mudstone, light gray to light brown, microcrystalline to cryptocrystalline, subblocky to subplatey, firm to hard, sucrosic to earthy texture, slightly dolomitic; trace ANHYDRITE: white to off white, microcrystalline, subblocky, firm, amorphous texture

9320-9350 LIMESTONE: mudstone, light gray to light brown, microcrystalline to cryptocrystalline, subblocky to subplatey, firm to hard, sucrosic to earthy texture, slightly dolomitic

9350-9380 LIMESTONE: mudstone, light to medium brown, microcrystalline, subplatey, hard, crystalline texture, possible intergranular porosity; ANHYDRITE: white to off white, microcrystalline, subblocky, firm, amorphous texture subplatey, hard, crystalline texture

9378-9410 ANHYDRITE: very light gray, gray to blue, microcrystalline, firm, earthy texture with interbedded LIMESTONE: mudstone, light to medium brown, microcrystalline, subplatey, hard, crystalline texture, slightly dolomitic

9410-9451 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm, crystalline texture, trace calcite, slightly dolomitic

Mission Canyon Formation

9,451' TVD (-7,333')

9451-9480 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm, crystalline texture, trace calcite, slightly dolomitic

9480-9510 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm, earthy texture, trace algal matts, trace calcite

9510-9540 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm, earthy texture, trace algal matts, trace calcite

9540-9570 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm, earthy texture, trace algal matts, trace calcite

9570-9600 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm, earthy texture, trace calcite

9600-9630 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm, earthy texture, trace calcite, argillaceous in part

9660-9690 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm to hard, earthy texture, trace calcite, argillaceous in part

9690-9720 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm to hard, earthy texture, trace calcite, argillaceous in part

9720-9750 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm to hard, earthy texture, trace calcite, argillaceous in part

9750-9780 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplaty, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

9780-9810 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

9810-9840 LIMESTONE: mudstone to wackestone, brown, medium to dark gray, white in part, microcrystalline to very fine crystalline, subblocky to subplaty, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite



9870-9900 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplaty, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

9900-9930 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplaty, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

9930-9960 Limestone: mudstone to wackestone, dark gray, medium to dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplaty, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

9960-9990 LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplaty, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

9990-10020 LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplaty, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

9990-10050 LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplaty, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

9990-10080 LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplaty, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite



Lodgepole Formation

10,010' TVD (-7,892')

10110-10140 LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10140-10170 LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10170-10200 LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, firm to hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10200-10240 LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10240-10270 LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to subplatey, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10270-10300 LIMESTONE: mudstone to wackestone, dark gray, dark brown, white in part, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10300-10350 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10300-10350 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10400-10450 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10450-10500 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10500-10550 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10550-10600 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10600-10650 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10650-10700 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10700-10750 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

10750-10810 LIMESTONE: mudstone to wackestone, dark gray, medium to dark brown, microcrystalline to very fine crystalline, subblocky to blocky, hard, earthy texture, trace calcite, argillaceous in part, trace pyrite

Lodgepole False Bakken Marker**10,810' MD / 10,718' TVD**

10810-10812 SHALE: very dark brown to brown, blocky, firm, waxy texture, very slightly calcareous

10812-10823 LIMESTONE: crystalline carbonaceous, light to medium gray, off white, microcrystalline, soft to friable, earthy to microsucrosic texture, common disseminated pyrite, occasional calcite, trace fossil fragments, no visible oil stain, no visible porosity

Bakken Formation, Upper Shale**10,823' MD / 10,725' TVD**

10823-10888 SHALE: black, firm, subblocky, waxy texture, very carbonaceous, very slightly calcareous, petroliferous, visible oil stain, no visible porosity

Bakken Formation, Middle Member**10,888' MD / 10,740' TVD**

10888-10950 SILTY SANDSTONE: light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately sorted, occasional calcite fracture fill, trace nodules pyrite, dolomitic cement, visible intergranular porosity

10950-11000 SILTY SANDSTONE: light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately sorted, trace calcite fracture fill, trace nodules pyrite, dolomitic cement, visible intergranular porosity

11000-11050 SILTY SANDSTONE: light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately sorted, trace calcite fracture fill, trace nodules pyrite, dolomitic cement, visible intergranular porosity

11050-11100 SILTY SANDSTONE: light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately sorted, trace calcite fracture fill, trace nodules pyrite, dolomitic cement, visible intergranular porosity

11100-11150 SILTY SANDSTONE: light brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain

11150-11200 SILTY SANDSTONE: light brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain

11200-11250 SILTY SANDSTONE: light brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain

11250-11300 SILTY SANDSTONE: light brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain

11300-11350 SILTY SANDSTONE: light brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain

11350-11400 SILTY SANDSTONE: light brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain

11950-12000 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light to medium brown oil stain, instant diffuse cut, moderately streaming cut



12050-12100 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light to medium brown oil stain, instant diffuse cut, moderately streaming cut

12100-12150 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light to medium brown oil stain, instant diffuse cut, moderately streaming cut

12150-12200 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light to medium brown oil stain, instant diffuse cut, moderately streaming cut

12200-12250 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light to medium brown oil stain, instant diffuse cut, moderately streaming cut



12300-12350 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light to medium brown oil stain, instant diffuse cut, moderately streaming cut

12350-12400 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light to medium brown oil stain, instant diffuse cut, moderately streaming cut

12400-12450 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light to medium brown oil stain, instant diffuse cut, moderately streaming cut

12450-12500 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light to medium brown oil stain, instant diffuse cut, moderately streaming cut

12550-12600 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, very sandy, dolomitic cement, visible intergranular porosity, visible light to medium brown oil stain, instant diffuse cut, moderately streaming cut

12600-12650 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, very fast white streaming to blooming cut

12650-12700 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, very fast white streaming to blooming cut

12700-12750 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, very fast white streaming to blooming cut

12750-12800 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, very fast white streaming to blooming cut

12800-12850 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

12850-12900 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common nodules pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

12900-12950 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common nodules pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

12950-13000 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common nodules pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

13000-13050 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common nodules pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

13050-13100 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common nodules pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

13100-13150 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common nodules pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

13150-13200 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common nodules pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

13250-13300 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

13300-13350 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

13350-13400 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, sandy, dolomitic cement, visible intergranular porosity, visible light brown oil stain, instant white streaming cut

13400-13450 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, rare light brown oil stain, very slightly white to yellow diffuse cut

13450-13500 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, rare light brown oil stain, very slightly white to yellow diffuse cut

13550-13600 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, rare light brown oil stain, very slightly white to yellow diffuse cut

13600-13650 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, rare light brown oil stain, very slightly white to yellow diffuse cut

13650-13700 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, rare light brown oil stain, very slightly white to yellow diffuse cut

13700-13750 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, rare light brown oil stain, very slightly white to yellow diffuse cut

13750-13800 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, rare light brown oil stain, very slightly white to yellow diffuse cut

13800-13850 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, common disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, rare light brown oil stain, very slightly white to yellow diffuse cut

1

1

1

15250-15300 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

15

15

15

15

15

15

15

15

15
SU

15

15

15

15

16050-16100 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16100-16150 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16150-16200 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16200-16250 SILTY SANDSTONE: light to medium brown, light gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16250-16300 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut



16350-16400 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16400-16450 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16450-16500 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16500-16550 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16550-16600 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16600-16650 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16750-16800 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16800-16850 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16850-16900 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16900-16950 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

16950-17000 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

17050-17100 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

17100-17150 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

17150-17200 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

17200-17250 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

17250-17300 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

17300-17350 SILTY SANDSTONE: light to medium brown, light to medium gray, white in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace light brown oil stain, slightly white to yellow diffuse cut

17450-17500 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

17500-17550 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

17550-17600 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

17600-17650 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

17650-17700 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace nodules pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

17750-17800 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

17800-17850 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut.

17850-17900 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

17900-17950 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

17950-18000 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut.

18000-18050 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut.

18150-18200 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

18200-18250 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

18250-18300 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

18300-18350 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

18350-18400 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

18450-18500 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

18500-18550 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

18550-18600 SILTY SANDSTONE: light to medium gray, white in part, rare light brown, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, trace patchy oil stain, fast white streaming to blooming cut

18600-18650 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut

18650-18700 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut

18700-18750 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut



18

18

18

19

19

19

19

19

19

19

19

19



19550-19600 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty dolomitic cement.

19600-19650 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded firm moderately to poorly sorted trace disseminated pyrite silty dolomitic cement

19650-19700 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded firm moderately to poorly sorted trace disseminated pyrite silty dolomitic cement

19700-19750 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded firm moderately to poorly sorted trace disseminated pyrite silty dolomitic cement

19750-19800 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded firm moderately to poorly sorted trace disseminated pyrite silty dolomitic cement

19800-19850 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded firm moderately to poorly sorted trace disseminated pyrite silty dolomitic cement

19850-19900 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded firm moderately to poorly sorted trace disseminated pyrite silty dolomitic cement

19900-19950 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded firm moderately to poorly sorted trace disseminated pyrite silty dolomitic cement

19950-20000 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded firm moderately to poorly sorted trace disseminated pyrite silty dolomitic cement

20000-20050 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty dolomitic cement

20050-20100 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded firm moderately to poorly sorted trace disseminated pyrite silty dolomitic cement

20100-20112 SILTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement.

20100-20150 SLTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded firm moderately to poorly sorted trace disseminated pyrite silty dolomitic cement

●

●

20850-20900 SLTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut, very poor samples

20900-20950 SLTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut, very poor samples

20950-21000 SLTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut, very poor samples

21000-21050 SLTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut, very poor samples

21050-21100 SLTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut, very poor samples

21100-21150 SLTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut, very poor samples

21150-21200 SLTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut, very poor samples

21200-21250 SLTY SANDSTONE: light to medium gray, white in part, light brown in part, very fine grained, subrounded to rounded, firm, moderately to poorly sorted, trace disseminated pyrite, silty, dolomitic cement, visible intergranular porosity, patchy oil stain, fast white streaming to blooming cut, very poor samples

TD at 21,250' reached on 21 February 2012 at 11:33 HRS (CST)



SUNDRY NOTICE AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.
22220



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date February 17, 2011
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Program	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Change well status to CONFIDENTIAL

Well Name and Number Jefferies 5301 43-12B					
Footages 250 F S L 2510 F E L		Qtr-Qtr SWSE	Section 12	Township 153 N	Range 101 W
Field Baker	Pool Bakken		County McKenzie		

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective immediately, we request CONFIDENTIAL STATUS for the above referenced well.

Ends 8-21-2012

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>Brittany Kunnemann</i>		Printed Name Brittany Kunnemann	
Title Operations Assistant		Date February 17, 2012	
Email Address bkunnemann@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 2-22-2012	
By <i>David Tabor</i>	
Title Engineering Technician	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 1**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
22220

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date February 1, 2012
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Waiver to rule Rule 43-02-03-31

Well Name and Number Jefferies 5301 43-12B					
Footages	Qtr-Qtr	Section	Township	Range	
250 F S L 2410 F E L	SWSE	12	153 N	101 W	
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

The SM Energy/Lindvig 1-11HR (NDIC 9309) located less than a mile from the subject well

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9461	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>Kaitlin Bass</i>		Printed Name Kaitlin Bass	
Title Operations Assistant		Date February 1, 2012	
Email Address kbass@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 2-2-2012	
By <i>RC</i>	
Title Richard A. Suggs Geologist	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
22220

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date 1/1258/2012
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☒ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other Change surface location	

Well Name and Number Jefferies 5301 43-12B					
Footages 250 F S L 2410 F E L		Qtr-Qtr SWSE	Section 12	Township 153 N	Range 101 W
Field Baker		Pool Bakken		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

We respectfully request permission to revise the APD issued for this well to include the following change:
The surface location will change to 250' FSL & 2510' FEL, Sec. 12-153N-101W. Correlating to this surface location the new 7" casing point will be 207' FNL & 2500' FWL, Sec. 13-153N-101W at an azimuth of 209.65°. The new bottom hole location will be 200' FSL & 2000' FWL Sec. 24 T153N R101W at an azimuth of 180°.

The following statements remain true:
Notice has been provided to the owner of any permanently occupied dwelling within 1320 feet.
This well is not located within 500 feet of an occupied dwelling.

Attached is the new well plat, drill plan, directional plan, and directional plot.

Company Oasis Petroleum Inc.		Telephone Number (281) 404-9461	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>Kaitlin Bass</i>		Printed Name Kaitlin Bass	
Title Operations Assistant		Date January 23, 2012	
Email Address kbass@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 1-25-11	
By <i>Anthony L. Lube</i>	
Title Petroleum Resource Specialist	

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"
250 FEET FROM SOUTH LINE AND 2510 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

Revision No.	Date	By	Description
REV 1	1/19/12	JJS	SWAPPED WELL LOCATIONS

DRILLING PLAN									
PROSPECT/FIELD	Madison Hills Horizontal Middle Bakken			COUNTY/STATE	McKenzie Co., N.D.				
OPERATOR	Oasis Operating			RIG	Nacore 139				
WELL NO.	0401 43-123			LEASE	Julesburg				
LOCATION	GRVSE 12-153N-101W			Surface Location (survey plat): 25:3 5.1		25:3 5.1			
EST. T.D.	21,310			GROUND ELEV:		2095		Finished Pad Elev	
TOTAL LATERAL:		10,210' (est)		KB ELEV:		2112		Sub Hieght: 75	
PROGNOSIS: Based on 2,110' KB (est)				LOGS: Type Interval					
MARKER DEPTH (Surf Loc) DATUM (Surf Loc)				OH Logs: File to omit					
Pierre NDIC MAP 1,968 150' Greenhorn 4,634 -2,516' Mowry 5,030 -2,912' Dakota 5,481 -3,343' Rierdon 6,377 -4,259' Dunham Salt 8,896 -4,778' Dunham Salt Base 6,963 -4,845' Spearfish 6,908 -4,890' Pine Salt 7,212 -5,094' Pine Salt Base 7,337 -5,210' Opeche Salt 7,385 -5,247' Opeche Salt Base 7,444 -5,306' Broom Creek (Top of Minnelusa Gp) 7,625 -5,507' Amsden 7,668 -5,550' Tyler 7,844 -5,726' Otter (Base of Minnelusa Gp.) 8,031 -5,913' Kibbey 8,380 -6,262' Charles Salt 8,527 -6,409' UB 9,150 -7,032' Base Last Salt 9,227 -7,109' Ratcliffe 9,275 -7,157' Mission Canyon 9,451 -7,333' Lodgepole 10,025 -7,907' False Bakken 10,740 -8,613' Upper Bakken 10,755 -8,622' Middle Bakken 10,762 -8,631' Middle Bakken Sand Target 10,762 -8,631' Base Middle Bakken Sand Target 10,771 -8,639' Lower Bakken 10,791 -8,659' Three Forks 10,818 -8,700'				CBL/GR: Above top of cement/GR to base of casing MWD GR: KOP to lateral TD DEVIATION: Surf: 3 deg. max., 1 deg / 100', svry every 500' Prod: 5 deg. max., 1 deg / 100', svry every 100' DST'S: None planned CORES: None planned MUDLOGGING: Two-Man: 8,327' ~200' above the Charles (Kibbey) to Casing point; Casing point to TD 30' samples at direction of wellsite geologist; 10' through target @ curve land BOP: 11" 5000 psi blind, pipe & annular					
Dip Rate: -0.25" or .51' / 100' DOWN first 4000' then +0.55" or .53' / 100' LP				Surface Formation: Glacial till					
Max. Anticipated BHP: 4975									
MUD:		Interval		Type		WT		Vis	
Surface		0' - 2,170'		FW/Gel - Lime Sweeps		8.6 - 8.9		28-34	
Intermediate		2,170' - 11,100'		Invert		9.6-10.4		40-60	
Liner		11,100' - 21,310'		Salt Water		9.3-10.4		28-34	
CASING:		Size		Wt ppf		Hole		Depth	
Surface:		9-5/8"		36#		13-1/2"		2,170'	
Intermediate:		7"		29/32#		8-3/4"		11,100'	
Production:		4.5"		11.6#		6"		21,310'	
						TOL @ 10,275'		4,961'	
PROBABLE PLUGS, IF REQ'D:									
OTHER:		MD		TVD		FNL/FSL		FEL/FWL	
Surface:		2,170'		2,170'		250' FSL		2510' FEL	
KOP:		10,321'		10,321'		250' FSL		2510' FEL	
EOC:		11,813'		10,762'		30' FNL		2843' FWL	
Casing Point:		11,100'		10,762'		207' FNL		2500' FWL	
Middle Bakken Lateral TD:		21,310'		10,762'		200' FSL		2000' FWL	
								S-T-R	
								12-T153N-R101W	
								12-T153N-R101W	
								13-T153N-R101W	
								13-T153N-R101W	
								24-T153N-R101W	
								AZI	
								Survey Company:	
								Build Rate: 10 deg / 100'	
								209.6	
								209.6	
								180.0	
Comments:									
DRILL TO KOP DRILL CURVE TO 90 DEG AND 7" CASING POINT SET 7" CASING DRILL MIDDLE BAKKEN LATERAL MWD Surveys will be taken every 100' in vertical hole, and a minimum of every 30' while building curve and every 90' while drilling lateral MWD GR to be run from KOP to Lateral TD. GR must be run to ground surface.									
Geology: MARB 12/9/2011 Prepared by: Engineering: L. Strong 1/18/2012									



Project: Indian Hills
 Site: 153N-101W-13/24
 Well: Jefferies 5301 43-12B
 Wellbore: Jefferies 5301 43-12B
 Design: Plan #1



Azimuths to True North
 Magnetic North: 8.56°

Magnetic Field
 Strength: 56724.3snT
 Dip Angle: 73.09°
 Date: 12/9/2011
 Model: IGRF200510

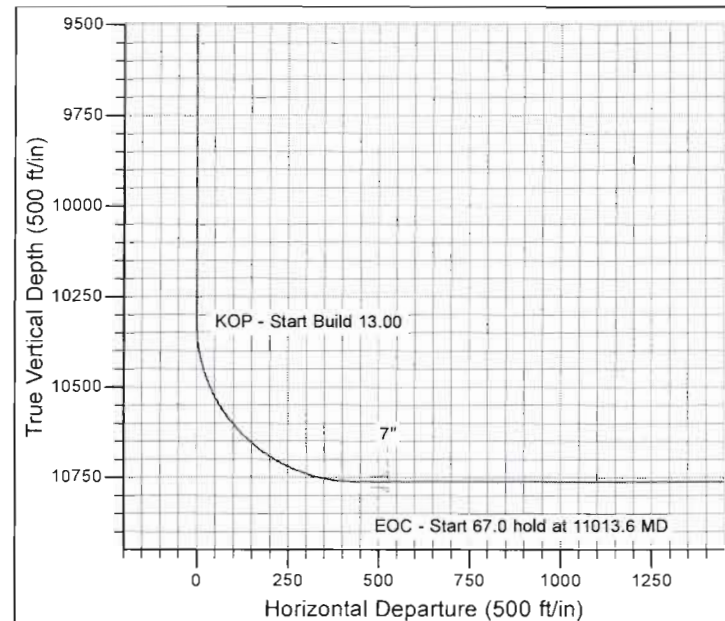
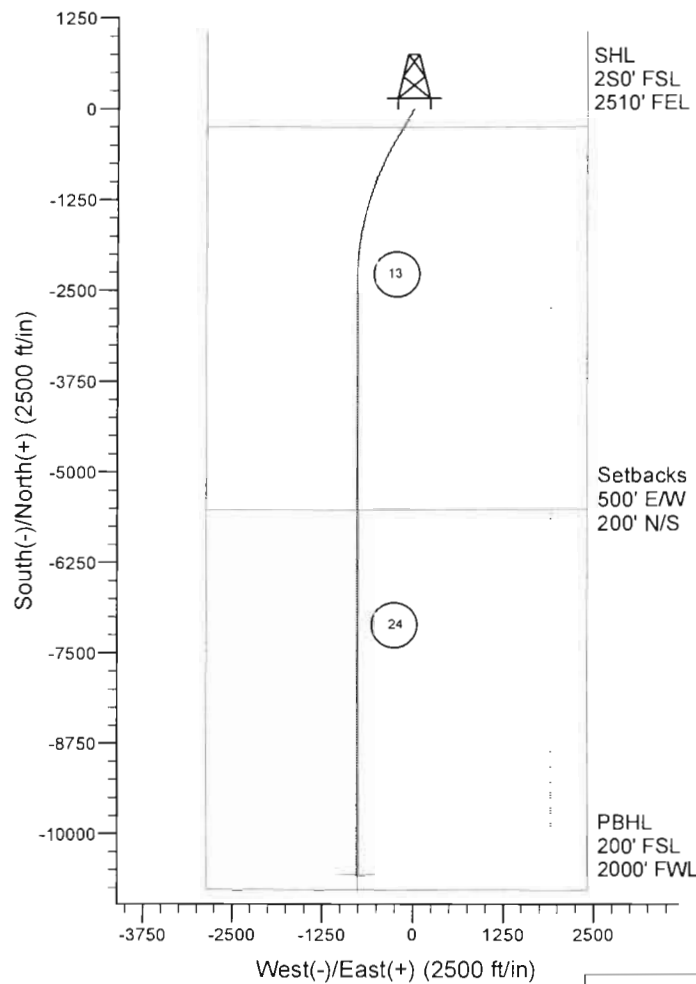
CASING DETAILS

TVD	MD	Name	Size
2170.0	2170.0	9 5/8"	9.625
10762.0	11100.0	7"	7.000

SITE DETAILS: 153N-101W-13/24

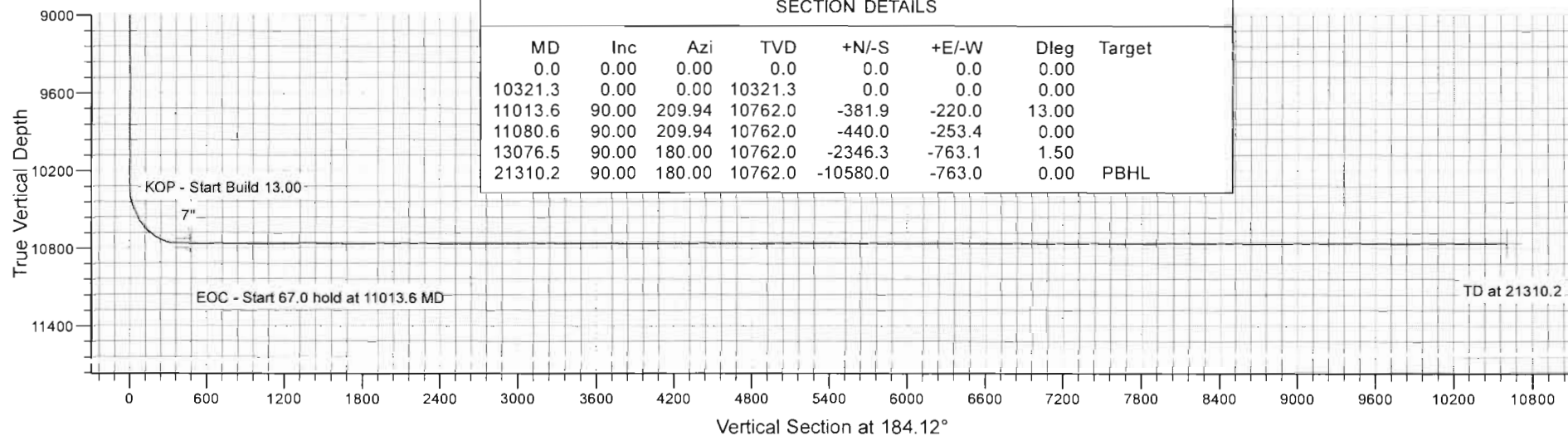
Site Centre Latitude: 48° 4' 57.960 N
 Longitude: 103° 36' 43.250 W

Positional Uncertainty: 0.0
 Convergence: -2.32
 Local North: True



SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
10321.3	0.00	0.00	10321.3	0.0	0.0	0.00	
11013.6	90.00	209.94	10762.0	-381.9	-220.0	13.00	
11080.6	90.00	209.94	10762.0	-440.0	-253.4	0.00	
13076.5	90.00	180.00	10762.0	-2346.3	-763.1	1.50	
21310.2	90.00	180.00	10762.0	-10580.0	-763.0	0.00	PBHL



Oasis

Indian Hills

153N-101W-13/24

Jefferies 5301 43-12B

Jefferies 5301 43-12B

Jefferies 5301 43-12B

Plan: Plan #1

Standard Planning Report

24 January, 2012

Microsoft
Planning Report

Database: EDM 5000.1 9.0 Production DB
Company: Oasis
Project: Indian Hills
Site: 153N-101W-13/24
Well: Jefferies 5301 43-12B
Wellbore: Jefferies 5301 43-12B
Design: Plan #1

Local Co-ordinate Reference: Well Jefferies 5301 43-12B
TVD Reference: WELL @ 2118.0ft (Original Well Elev)
MD Reference: WELL @ 2118.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-101W-13/24		
Site Position:		Northing:	125,067.66 m
From:	Lat/Long	Easting:	368,214.56 m
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in
		Latitude:	48° 4' 57.96" N
		Longitude:	103° 36' 51.830 W
		Grid Convergence:	-2.32°

Well	Jefferies 5301 43-12B		
Well Position	+N/-S	11.2 ft	Northing: 125,078.23 m
	+E/-W	-582.5 ft	Easting: 368,037.28 m
Position Uncertainty	0.0 ft	Wellhead Elevation:	Latitude: 48° 4' 58.07" N
			Longitude: 103° 36' 51.830 W
			Ground Level: 2,093.0 ft

Wellbore	Jefferies 5301 43-12B		
Magnetics	Model Name	Sample Date	Declination (°)
	IGRF200510	12/9/2011	8.56
			Dip Angle (°) 73.09
			Field Strength (nT) 56,724

Design	Plan #1		
Audit Notes:			
Version:	Phase:	PLAN	Tie On Depth: 0.0
Vertical Section:	Depth From (TVD)	+N/-S (ft)	+E/-W (ft)
	0.0	0.0	0.0
			Direction (°) 184.12

Plan Sections										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
10,321.3	0.00	0.00	10,321.3	0.0	0.0	0.00	0.00	0.00	0.00	
11,013.6	90.00	209.94	10,762.0	-381.9	-220.0	13.00	13.00	0.00	209.94	
11,080.6	90.00	209.94	10,762.0	-440.0	-253.4	0.00	0.00	0.00	0.00	
13,076.5	90.00	180.00	10,762.0	-2,346.3	-763.1	1.50	0.00	-1.50	-90.00	
21,310.2	90.00	180.00	10,762.0	-10,580.0	-763.0	0.00	0.00	0.00	0.00	Jefferies 5301 43-12B

Microsoft
Planning Report

Database: EDM 5000.1 9.0 Production DB
Company: Oasis
Project: Indian Hills
Site: 153N-101W-13/24
Well: Jefferies 5301 43-12B
Wellbore: Jefferies 5301 43-12B
Design: Plan #1

Local Co-ordinate Reference: Well Jefferies 5301 43-12B
TVD Reference: WELL @ 2118.0ft (Original Well Elev)
MD Reference: WELL @ 2118.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	0.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.00	0.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.00	0.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.00	0.00	400.0	0.0	0.0	0.0	0.00	0.00	0.00
500.0	0.00	0.00	500.0	0.0	0.0	0.0	0.00	0.00	0.00
600.0	0.00	0.00	600.0	0.0	0.0	0.0	0.00	0.00	0.00
700.0	0.00	0.00	700.0	0.0	0.0	0.0	0.00	0.00	0.00
800.0	0.00	0.00	800.0	0.0	0.0	0.0	0.00	0.00	0.00
900.0	0.00	0.00	900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,000.0	0.00	0.00	1,000.0	0.0	0.0	0.0	0.00	0.00	0.00
1,100.0	0.00	0.00	1,100.0	0.0	0.0	0.0	0.00	0.00	0.00
1,200.0	0.00	0.00	1,200.0	0.0	0.0	0.0	0.00	0.00	0.00
1,300.0	0.00	0.00	1,300.0	0.0	0.0	0.0	0.00	0.00	0.00
1,400.0	0.00	0.00	1,400.0	0.0	0.0	0.0	0.00	0.00	0.00
1,500.0	0.00	0.00	1,500.0	0.0	0.0	0.0	0.00	0.00	0.00
1,600.0	0.00	0.00	1,600.0	0.0	0.0	0.0	0.00	0.00	0.00
1,700.0	0.00	0.00	1,700.0	0.0	0.0	0.0	0.00	0.00	0.00
1,800.0	0.00	0.00	1,800.0	0.0	0.0	0.0	0.00	0.00	0.00
1,900.0	0.00	0.00	1,900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,968.0	0.00	0.00	1,968.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,000.0	0.00	0.00	2,000.0	0.0	0.0	0.0	0.00	0.00	0.00
2,100.0	0.00	0.00	2,100.0	0.0	0.0	0.0	0.00	0.00	0.00
2,170.0	0.00	0.00	2,170.0	0.0	0.0	0.0	0.00	0.00	0.00
9 5/8"									
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.0	0.00	0.00	0.00
2,300.0	0.00	0.00	2,300.0	0.0	0.0	0.0	0.00	0.00	0.00
2,400.0	0.00	0.00	2,400.0	0.0	0.0	0.0	0.00	0.00	0.00
2,500.0	0.00	0.00	2,500.0	0.0	0.0	0.0	0.00	0.00	0.00
2,600.0	0.00	0.00	2,600.0	0.0	0.0	0.0	0.00	0.00	0.00
2,700.0	0.00	0.00	2,700.0	0.0	0.0	0.0	0.00	0.00	0.00
2,800.0	0.00	0.00	2,800.0	0.0	0.0	0.0	0.00	0.00	0.00
2,900.0	0.00	0.00	2,900.0	0.0	0.0	0.0	0.00	0.00	0.00
3,000.0	0.00	0.00	3,000.0	0.0	0.0	0.0	0.00	0.00	0.00
3,100.0	0.00	0.00	3,100.0	0.0	0.0	0.0	0.00	0.00	0.00
3,200.0	0.00	0.00	3,200.0	0.0	0.0	0.0	0.00	0.00	0.00
3,300.0	0.00	0.00	3,300.0	0.0	0.0	0.0	0.00	0.00	0.00
3,400.0	0.00	0.00	3,400.0	0.0	0.0	0.0	0.00	0.00	0.00
3,500.0	0.00	0.00	3,500.0	0.0	0.0	0.0	0.00	0.00	0.00
3,600.0	0.00	0.00	3,600.0	0.0	0.0	0.0	0.00	0.00	0.00
3,700.0	0.00	0.00	3,700.0	0.0	0.0	0.0	0.00	0.00	0.00
3,800.0	0.00	0.00	3,800.0	0.0	0.0	0.0	0.00	0.00	0.00
3,900.0	0.00	0.00	3,900.0	0.0	0.0	0.0	0.00	0.00	0.00
4,000.0	0.00	0.00	4,000.0	0.0	0.0	0.0	0.00	0.00	0.00
4,100.0	0.00	0.00	4,100.0	0.0	0.0	0.0	0.00	0.00	0.00
4,200.0	0.00	0.00	4,200.0	0.0	0.0	0.0	0.00	0.00	0.00
4,300.0	0.00	0.00	4,300.0	0.0	0.0	0.0	0.00	0.00	0.00
4,400.0	0.00	0.00	4,400.0	0.0	0.0	0.0	0.00	0.00	0.00
4,500.0	0.00	0.00	4,500.0	0.0	0.0	0.0	0.00	0.00	0.00
4,600.0	0.00	0.00	4,600.0	0.0	0.0	0.0	0.00	0.00	0.00
4,634.0	0.00	0.00	4,634.0	0.0	0.0	0.0	0.00	0.00	0.00
Greenhorn									

Microsoft
Planning Report

Database: EDM 5000.1.9.0 Production DB
Company: Oasis
Project: Indian Hills
Site: 153N-101W-13/24
Well: Jefferies 5301 43-12B
Wellbore: Jefferies 5301 43-12B
Design: Plan #1

Local Co-ordinate Reference: Well Jefferies 5301 43-12B
TVD Reference: WELL @ 2118.0ft (Original Well Elev)
MD Reference: WELL @ 2118.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
4,700.0	0.00	0.00	4,700.0	0.0	0.0	0.0	0.00	0.00	0.00
4,800.0	0.00	0.00	4,800.0	0.0	0.0	0.0	0.00	0.00	0.00
4,900.0	0.00	0.00	4,900.0	0.0	0.0	0.0	0.00	0.00	0.00
5,000.0	0.00	0.00	5,000.0	0.0	0.0	0.0	0.00	0.00	0.00
5,030.0	0.00	0.00	5,030.0	0.0	0.0	0.0	0.00	0.00	0.00
Mowry									
5,100.0	0.00	0.00	5,100.0	0.0	0.0	0.0	0.00	0.00	0.00
5,200.0	0.00	0.00	5,200.0	0.0	0.0	0.0	0.00	0.00	0.00
5,300.0	0.00	0.00	5,300.0	0.0	0.0	0.0	0.00	0.00	0.00
5,400.0	0.00	0.00	5,400.0	0.0	0.0	0.0	0.00	0.00	0.00
5,461.0	0.00	0.00	5,461.0	0.0	0.0	0.0	0.00	0.00	0.00
Dakota									
5,500.0	0.00	0.00	5,500.0	0.0	0.0	0.0	0.00	0.00	0.00
5,600.0	0.00	0.00	5,600.0	0.0	0.0	0.0	0.00	0.00	0.00
5,700.0	0.00	0.00	5,700.0	0.0	0.0	0.0	0.00	0.00	0.00
5,800.0	0.00	0.00	5,800.0	0.0	0.0	0.0	0.00	0.00	0.00
5,900.0	0.00	0.00	5,900.0	0.0	0.0	0.0	0.00	0.00	0.00
6,000.0	0.00	0.00	6,000.0	0.0	0.0	0.0	0.00	0.00	0.00
6,100.0	0.00	0.00	6,100.0	0.0	0.0	0.0	0.00	0.00	0.00
6,200.0	0.00	0.00	6,200.0	0.0	0.0	0.0	0.00	0.00	0.00
6,300.0	0.00	0.00	6,300.0	0.0	0.0	0.0	0.00	0.00	0.00
6,377.0	0.00	0.00	6,377.0	0.0	0.0	0.0	0.00	0.00	0.00
Rierdon									
6,400.0	0.00	0.00	6,400.0	0.0	0.0	0.0	0.00	0.00	0.00
6,500.0	0.00	0.00	6,500.0	0.0	0.0	0.0	0.00	0.00	0.00
6,600.0	0.00	0.00	6,600.0	0.0	0.0	0.0	0.00	0.00	0.00
6,700.0	0.00	0.00	6,700.0	0.0	0.0	0.0	0.00	0.00	0.00
6,800.0	0.00	0.00	6,800.0	0.0	0.0	0.0	0.00	0.00	0.00
6,896.0	0.00	0.00	6,896.0	0.0	0.0	0.0	0.00	0.00	0.00
Dunham Salt									
6,900.0	0.00	0.00	6,900.0	0.0	0.0	0.0	0.00	0.00	0.00
6,963.0	0.00	0.00	6,963.0	0.0	0.0	0.0	0.00	0.00	0.00
Dunham Salt Base									
6,968.0	0.00	0.00	6,968.0	0.0	0.0	0.0	0.00	0.00	0.00
Spearfish									
7,000.0	0.00	0.00	7,000.0	0.0	0.0	0.0	0.00	0.00	0.00
7,100.0	0.00	0.00	7,100.0	0.0	0.0	0.0	0.00	0.00	0.00
7,200.0	0.00	0.00	7,200.0	0.0	0.0	0.0	0.00	0.00	0.00
7,212.0	0.00	0.00	7,212.0	0.0	0.0	0.0	0.00	0.00	0.00
Pine Salt									
7,300.0	0.00	0.00	7,300.0	0.0	0.0	0.0	0.00	0.00	0.00
7,337.0	0.00	0.00	7,337.0	0.0	0.0	0.0	0.00	0.00	0.00
Pine Salt Base									
7,365.0	0.00	0.00	7,365.0	0.0	0.0	0.0	0.00	0.00	0.00
Opeche Salt									
7,400.0	0.00	0.00	7,400.0	0.0	0.0	0.0	0.00	0.00	0.00
7,444.0	0.00	0.00	7,444.0	0.0	0.0	0.0	0.00	0.00	0.00
Opeche Salt Base									
7,500.0	0.00	0.00	7,500.0	0.0	0.0	0.0	0.00	0.00	0.00
7,600.0	0.00	0.00	7,600.0	0.0	0.0	0.0	0.00	0.00	0.00
7,625.0	0.00	0.00	7,625.0	0.0	0.0	0.0	0.00	0.00	0.00
Broom Creek (Top of Minnelusa Gp.)									
7,668.0	0.00	0.00	7,668.0	0.0	0.0	0.0	0.00	0.00	0.00

Microsoft
Planning Report

Database: EDM 5000.1.9.0 Production DB
Company: Oasis
Project: Indian Hills
Site: 153N-101W-13/24
Well: Jefferies 5301 43-12B
Wellbore: Jefferies 5301 43-12B
Design: Plan #1

Local Co-ordinate Reference: Well Jefferies 5301 43-12B
TVD Reference: WELL @ 2118.0ft (Original Well Elev)
MD Reference: WELL @ 2118.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
Amsden									
7,700.0	0.00	0.00	7,700.0	0.0	0.0	0.0	0.00	0.00	0.00
7,800.0	0.00	0.00	7,800.0	0.0	0.0	0.0	0.00	0.00	0.00
7,844.0	0.00	0.00	7,844.0	0.0	0.0	0.0	0.00	0.00	0.00
Tyler									
7,900.0	0.00	0.00	7,900.0	0.0	0.0	0.0	0.00	0.00	0.00
8,000.0	0.00	0.00	8,000.0	0.0	0.0	0.0	0.00	0.00	0.00
8,031.0	0.00	0.00	8,031.0	0.0	0.0	0.0	0.00	0.00	0.00
Otter (Base of Minnelusa Gp.)									
8,100.0	0.00	0.00	8,100.0	0.0	0.0	0.0	0.00	0.00	0.00
8,200.0	0.00	0.00	8,200.0	0.0	0.0	0.0	0.00	0.00	0.00
8,300.0	0.00	0.00	8,300.0	0.0	0.0	0.0	0.00	0.00	0.00
8,380.0	0.00	0.00	8,380.0	0.0	0.0	0.0	0.00	0.00	0.00
Kibbey									
8,400.0	0.00	0.00	8,400.0	0.0	0.0	0.0	0.00	0.00	0.00
8,500.0	0.00	0.00	8,500.0	0.0	0.0	0.0	0.00	0.00	0.00
8,527.0	0.00	0.00	8,527.0	0.0	0.0	0.0	0.00	0.00	0.00
Charles Salt									
8,600.0	0.00	0.00	8,600.0	0.0	0.0	0.0	0.00	0.00	0.00
8,700.0	0.00	0.00	8,700.0	0.0	0.0	0.0	0.00	0.00	0.00
8,800.0	0.00	0.00	8,800.0	0.0	0.0	0.0	0.00	0.00	0.00
8,900.0	0.00	0.00	8,900.0	0.0	0.0	0.0	0.00	0.00	0.00
9,000.0	0.00	0.00	9,000.0	0.0	0.0	0.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,100.0	0.0	0.0	0.0	0.00	0.00	0.00
9,150.0	0.00	0.00	9,150.0	0.0	0.0	0.0	0.00	0.00	0.00
UB									
9,200.0	0.00	0.00	9,200.0	0.0	0.0	0.0	0.00	0.00	0.00
9,227.0	0.00	0.00	9,227.0	0.0	0.0	0.0	0.00	0.00	0.00
Base Last Salt									
9,275.0	0.00	0.00	9,275.0	0.0	0.0	0.0	0.00	0.00	0.00
Ratcliffe									
9,300.0	0.00	0.00	9,300.0	0.0	0.0	0.0	0.00	0.00	0.00
9,400.0	0.00	0.00	9,400.0	0.0	0.0	0.0	0.00	0.00	0.00
9,451.0	0.00	0.00	9,451.0	0.0	0.0	0.0	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,500.0	0.0	0.0	0.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,600.0	0.0	0.0	0.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,700.0	0.0	0.0	0.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,800.0	0.0	0.0	0.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,900.0	0.0	0.0	0.0	0.00	0.00	0.00
10,000.0	0.00	0.00	10,000.0	0.0	0.0	0.0	0.00	0.00	0.00
10,025.0	0.00	0.00	10,025.0	0.0	0.0	0.0	0.00	0.00	0.00
Lodgepole									
10,100.0	0.00	0.00	10,100.0	0.0	0.0	0.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,200.0	0.0	0.0	0.0	0.00	0.00	0.00
10,300.0	0.00	0.00	10,300.0	0.0	0.0	0.0	0.00	0.00	0.00
10,321.3	0.00	0.00	10,321.3	0.0	0.0	0.0	0.00	0.00	0.00
KOP - Start Build 13.00									
10,325.0	0.49	209.94	10,325.0	0.0	0.0	0.0	13.00	13.00	0.00
10,350.0	3.74	209.94	10,350.0	-0.8	-0.5	0.8	13.00	13.00	0.00
10,375.0	6.99	209.94	10,374.9	-2.8	-1.6	2.9	13.00	13.00	0.00
10,400.0	10.24	209.94	10,399.6	-6.1	-3.5	6.3	13.00	13.00	0.00
10,425.0	13.49	209.94	10,424.0	-10.5	-6.1	10.9	13.00	13.00	0.00

Microsoft
Planning Report

Database: EDM 5000.1.9.0 Production DB
Company: Oasis
Project: Indian Hills
Site: 153N-101W-13/24
Well: Jefferies 5301 43-12B
Wellbore: Jefferies 5301 43-12B
Design: Plan #1

Local Co-ordinate Reference: Well Jefferies 5301 43-12B
TVD Reference: WELL @ 2118.0ft (Original Well Elev)
MD Reference: WELL @ 2118.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,450.0	16.74	209.94	10,448.2	-16.2	-9.3	16.8	13.00	13.00	0.00
10,475.0	19.99	209.94	10,471.9	-23.0	-13.2	23.9	13.00	13.00	0.00
10,500.0	23.24	209.94	10,495.1	-31.0	-17.8	32.2	13.00	13.00	0.00
10,525.0	26.49	209.94	10,517.8	-40.1	-23.1	41.6	13.00	13.00	0.00
10,550.0	29.74	209.94	10,539.9	-50.3	-29.0	52.2	13.00	13.00	0.00
10,575.0	32.99	209.94	10,561.2	-61.6	-35.5	64.0	13.00	13.00	0.00
10,600.0	36.24	209.94	10,581.8	-73.9	-42.5	76.7	13.00	13.00	0.00
10,625.0	39.49	209.94	10,601.5	-87.2	-50.2	90.5	13.00	13.00	0.00
10,650.0	42.74	209.94	10,620.4	-101.4	-58.4	105.3	13.00	13.00	0.00
10,675.0	45.99	209.94	10,638.2	-116.6	-67.1	121.1	13.00	13.00	0.00
10,700.0	49.24	209.94	10,655.1	-132.5	-76.3	137.7	13.00	13.00	0.00
10,725.0	52.49	209.94	10,670.9	-149.3	-86.0	155.1	13.00	13.00	0.00
10,750.0	55.74	209.94	10,685.5	-166.9	-96.1	173.4	13.00	13.00	0.00
10,775.0	58.99	209.94	10,699.0	-185.1	-106.6	192.3	13.00	13.00	0.00
10,800.0	62.24	209.94	10,711.3	-204.0	-117.5	211.9	13.00	13.00	0.00
10,825.0	65.49	209.94	10,722.3	-223.5	-128.7	232.1	13.00	13.00	0.00
10,850.0	68.74	209.94	10,732.0	-243.4	-140.2	252.9	13.00	13.00	0.00
10,873.7	71.82	209.94	10,740.0	-262.8	-151.3	273.0	13.00	13.00	0.00
False Bakken									
10,875.0	71.99	209.94	10,740.4	-263.8	-151.9	274.1	13.00	13.00	0.00
10,900.0	75.24	209.94	10,747.4	-284.6	-163.9	295.6	13.00	13.00	0.00
10,925.0	78.49	209.94	10,753.1	-305.7	-176.0	317.6	13.00	13.00	0.00
10,934.9	79.77	209.94	10,755.0	-314.1	-180.9	326.3	13.00	13.00	0.00
Upper Bakken									
10,950.0	81.74	209.94	10,757.4	-327.0	-188.3	339.7	13.00	13.00	0.00
10,975.0	84.99	209.94	10,760.3	-348.5	-200.7	362.1	13.00	13.00	0.00
11,000.0	88.24	209.94	10,761.8	-370.2	-213.2	384.5	13.00	13.00	0.00
11,013.5	90.00	209.94	10,762.0	-381.9	-219.9	396.7	13.00	13.00	0.00
EOC - Start 67.0 hold at 11013.6 MD - Middle Bakken Sand Target - Middle Bakken									
11,080.6	90.00	209.94	10,762.0	-440.0	-253.4	457.1	0.01	0.01	0.00
Start DLS 1.50 TFO -90.00									
11,100.0	90.00	209.65	10,762.0	-456.8	-263.0	474.6	1.50	0.00	-1.50
7"									
11,200.0	90.00	208.15	10,762.0	-544.4	-311.4	565.4	1.50	0.00	-1.50
11,300.0	90.00	206.65	10,762.0	-633.2	-357.4	657.2	1.50	0.00	-1.50
11,400.0	90.00	205.15	10,762.0	-723.1	-401.1	750.1	1.50	0.00	-1.50
11,500.0	90.00	203.65	10,762.0	-814.2	-442.4	843.9	1.50	0.00	-1.50
11,600.0	90.00	202.15	10,762.0	-906.3	-481.3	938.6	1.50	0.00	-1.50
11,700.0	90.00	200.65	10,762.0	-999.4	-517.7	1,034.1	1.50	0.00	-1.50
11,800.0	90.00	199.15	10,762.0	-1,093.5	-551.8	1,130.3	1.50	0.00	-1.50
11,900.0	90.00	197.65	10,762.0	-1,188.3	-583.3	1,227.2	1.50	0.00	-1.50
12,000.0	90.00	196.15	10,762.0	-1,284.0	-612.4	1,324.7	1.50	0.00	-1.50
12,100.0	90.00	194.65	10,762.0	-1,380.4	-639.0	1,422.8	1.50	0.00	-1.50
12,200.0	90.00	193.15	10,762.0	-1,477.5	-663.0	1,521.4	1.50	0.00	-1.50
12,300.0	90.00	191.65	10,762.0	-1,575.2	-684.4	1,620.3	1.50	0.00	-1.50
12,400.0	90.00	190.15	10,762.0	-1,673.4	-703.3	1,719.6	1.50	0.00	-1.50
12,500.0	90.00	188.65	10,762.0	-1,772.0	-719.7	1,819.2	1.50	0.00	-1.50
12,600.0	90.00	187.15	10,762.0	-1,871.1	-733.4	1,919.0	1.50	0.00	-1.50
12,700.0	90.00	185.65	10,762.0	-1,970.4	-744.5	2,018.9	1.50	0.00	-1.50
12,800.0	90.00	184.15	10,762.0	-2,070.1	-753.1	2,118.9	1.50	0.00	-1.50
12,900.0	90.00	182.65	10,762.0	-2,169.9	-759.0	2,218.9	1.50	0.00	-1.50
13,000.0	90.00	181.15	10,762.0	-2,269.8	-762.3	2,318.8	1.50	0.00	-1.50
13,076.5	90.00	180.00	10,762.0	-2,346.3	-763.1	2,395.1	1.50	0.00	-1.50

Microsoft
Planning Report

Database: EDM 5000.1.9.0 Production DB
Company: Oasis
Project: Indian Hills
Site: 153N-101W-13/24
Well: Jefferies 5301 43-12B
Wellbore: Jefferies 5301 43-12B
Design: Plan #1

Local Co-ordinate Reference:
TVD Reference:
MD Reference:
North Reference:
Survey Calculation Method:

Well Jefferies 5301 43-12B
WELL @ 2118.0ft (Original Well Elev)
WELL @ 2118.0ft (Original Well Elev)
True
Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
Start 8233.7 hold at 13076.5 MD									
13,100.0	90.00	180.00	10,762.0	-2,369.8	-763.1	2,418.6	0.00	0.00	0.00
13,200.0	90.00	180.00	10,762.0	-2,469.8	-763.1	2,518.3	0.00	0.00	0.00
13,300.0	90.00	180.00	10,762.0	-2,569.8	-763.1	2,618.1	0.00	0.00	0.00
13,400.0	90.00	180.00	10,762.0	-2,669.8	-763.1	2,717.8	0.00	0.00	0.00
13,500.0	90.00	180.00	10,762.0	-2,769.8	-763.1	2,817.5	0.00	0.00	0.00
13,600.0	90.00	180.00	10,762.0	-2,869.8	-763.1	2,917.3	0.00	0.00	0.00
13,700.0	90.00	180.00	10,762.0	-2,969.8	-763.1	3,017.0	0.00	0.00	0.00
13,800.0	90.00	180.00	10,762.0	-3,069.8	-763.1	3,116.8	0.00	0.00	0.00
13,900.0	90.00	180.00	10,762.0	-3,169.8	-763.1	3,216.5	0.00	0.00	0.00
14,000.0	90.00	180.00	10,762.0	-3,269.8	-763.1	3,316.2	0.00	0.00	0.00
14,100.0	90.00	180.00	10,762.0	-3,369.8	-763.1	3,416.0	0.00	0.00	0.00
14,200.0	90.00	180.00	10,762.0	-3,469.8	-763.1	3,515.7	0.00	0.00	0.00
14,300.0	90.00	180.00	10,762.0	-3,569.8	-763.1	3,615.5	0.00	0.00	0.00
14,400.0	90.00	180.00	10,762.0	-3,669.8	-763.1	3,715.2	0.00	0.00	0.00
14,500.0	90.00	180.00	10,762.0	-3,769.8	-763.1	3,814.9	0.00	0.00	0.00
14,600.0	90.00	180.00	10,762.0	-3,869.8	-763.1	3,914.7	0.00	0.00	0.00
14,700.0	90.00	180.00	10,762.0	-3,969.8	-763.1	4,014.4	0.00	0.00	0.00
14,800.0	90.00	180.00	10,762.0	-4,069.8	-763.1	4,114.2	0.00	0.00	0.00
14,900.0	90.00	180.00	10,762.0	-4,169.8	-763.1	4,213.9	0.00	0.00	0.00
15,000.0	90.00	180.00	10,762.0	-4,269.8	-763.1	4,313.7	0.00	0.00	0.00
15,100.0	90.00	180.00	10,762.0	-4,369.8	-763.1	4,413.4	0.00	0.00	0.00
15,200.0	90.00	180.00	10,762.0	-4,469.8	-763.1	4,513.1	0.00	0.00	0.00
15,300.0	90.00	180.00	10,762.0	-4,569.8	-763.1	4,612.9	0.00	0.00	0.00
15,400.0	90.00	180.00	10,762.0	-4,669.8	-763.1	4,712.6	0.00	0.00	0.00
15,500.0	90.00	180.00	10,762.0	-4,769.8	-763.1	4,812.4	0.00	0.00	0.00
15,600.0	90.00	180.00	10,762.0	-4,869.8	-763.1	4,912.1	0.00	0.00	0.00
15,700.0	90.00	180.00	10,762.0	-4,969.8	-763.1	5,011.8	0.00	0.00	0.00
15,800.0	90.00	180.00	10,762.0	-5,069.8	-763.1	5,111.6	0.00	0.00	0.00
15,900.0	90.00	180.00	10,762.0	-5,169.8	-763.1	5,211.3	0.00	0.00	0.00
16,000.0	90.00	180.00	10,762.0	-5,269.8	-763.1	5,311.1	0.00	0.00	0.00
16,100.0	90.00	180.00	10,762.0	-5,369.8	-763.1	5,410.8	0.00	0.00	0.00
16,200.0	90.00	180.00	10,762.0	-5,469.8	-763.1	5,510.5	0.00	0.00	0.00
16,300.0	90.00	180.00	10,762.0	-5,569.8	-763.0	5,610.3	0.00	0.00	0.00
16,400.0	90.00	180.00	10,762.0	-5,669.8	-763.0	5,710.0	0.00	0.00	0.00
16,500.0	90.00	180.00	10,762.0	-5,769.8	-763.0	5,809.8	0.00	0.00	0.00
16,600.0	90.00	180.00	10,762.0	-5,869.8	-763.0	5,909.5	0.00	0.00	0.00
16,700.0	90.00	180.00	10,762.0	-5,969.8	-763.0	6,009.2	0.00	0.00	0.00
16,800.0	90.00	180.00	10,762.0	-6,069.8	-763.0	6,109.0	0.00	0.00	0.00
16,900.0	90.00	180.00	10,762.0	-6,169.8	-763.0	6,208.7	0.00	0.00	0.00
17,000.0	90.00	180.00	10,762.0	-6,269.8	-763.0	6,308.5	0.00	0.00	0.00
17,100.0	90.00	180.00	10,762.0	-6,369.8	-763.0	6,408.2	0.00	0.00	0.00
17,200.0	90.00	180.00	10,762.0	-6,469.8	-763.0	6,508.0	0.00	0.00	0.00
17,300.0	90.00	180.00	10,762.0	-6,569.8	-763.0	6,607.7	0.00	0.00	0.00
17,400.0	90.00	180.00	10,762.0	-6,669.8	-763.0	6,707.4	0.00	0.00	0.00
17,500.0	90.00	180.00	10,762.0	-6,769.8	-763.0	6,807.2	0.00	0.00	0.00
17,600.0	90.00	180.00	10,762.0	-6,869.8	-763.0	6,906.9	0.00	0.00	0.00
17,700.0	90.00	180.00	10,762.0	-6,969.8	-763.0	7,006.7	0.00	0.00	0.00
17,800.0	90.00	180.00	10,762.0	-7,069.8	-763.0	7,106.4	0.00	0.00	0.00
17,900.0	90.00	180.00	10,762.0	-7,169.8	-763.0	7,206.1	0.00	0.00	0.00
18,000.0	90.00	180.00	10,762.0	-7,269.8	-763.0	7,305.9	0.00	0.00	0.00
18,100.0	90.00	180.00	10,762.0	-7,369.8	-763.0	7,405.6	0.00	0.00	0.00
18,200.0	90.00	180.00	10,762.0	-7,469.8	-763.0	7,505.4	0.00	0.00	0.00
18,300.0	90.00	180.00	10,762.0	-7,569.8	-763.0	7,605.1	0.00	0.00	0.00

Microsoft
Planning Report

Database: EDM 5000.1.9.0 Production DB
Company: Oasis
Project: Indian Hills
Site: 153N-101W-13/24
Well: Jefferies 5301 43-12B
Wellbore: Jefferies 5301 43-12B
Design: Plan #1

Local Co-ordinate Reference: Well Jefferies 5301 43-12B
TVD Reference: WELL @ 2118.0ft (Original Well Elev)
MD Reference: WELL @ 2118.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
18,400.0	90.00	180.00	10,762.0	-7,669.8	-763.0	7,704.8	0.00	0.00	0.00
18,500.0	90.00	180.00	10,762.0	-7,769.8	-763.0	7,804.6	0.00	0.00	0.00
18,600.0	90.00	180.00	10,762.0	-7,869.8	-763.0	7,904.3	0.00	0.00	0.00
18,700.0	90.00	180.00	10,762.0	-7,969.8	-763.0	8,004.1	0.00	0.00	0.00
18,800.0	90.00	180.00	10,762.0	-8,069.8	-763.0	8,103.8	0.00	0.00	0.00
18,900.0	90.00	180.00	10,762.0	-8,169.8	-763.0	8,203.5	0.00	0.00	0.00
19,000.0	90.00	180.00	10,762.0	-8,269.8	-763.0	8,303.3	0.00	0.00	0.00
19,100.0	90.00	180.00	10,762.0	-8,369.8	-763.0	8,403.0	0.00	0.00	0.00
19,200.0	90.00	180.00	10,762.0	-8,469.8	-763.0	8,502.8	0.00	0.00	0.00
19,300.0	90.00	180.00	10,762.0	-8,569.8	-763.0	8,602.5	0.00	0.00	0.00
19,400.0	90.00	180.00	10,762.0	-8,669.8	-763.0	8,702.3	0.00	0.00	0.00
19,500.0	90.00	180.00	10,762.0	-8,769.8	-763.0	8,802.0	0.00	0.00	0.00
19,600.0	90.00	180.00	10,762.0	-8,869.8	-763.0	8,901.7	0.00	0.00	0.00
19,700.0	90.00	180.00	10,762.0	-8,969.8	-763.0	9,001.5	0.00	0.00	0.00
19,800.0	90.00	180.00	10,762.0	-9,069.8	-763.0	9,101.2	0.00	0.00	0.00
19,900.0	90.00	180.00	10,762.0	-9,169.8	-763.0	9,201.0	0.00	0.00	0.00
20,000.0	90.00	180.00	10,762.0	-9,269.8	-763.0	9,300.7	0.00	0.00	0.00
20,100.0	90.00	180.00	10,762.0	-9,369.8	-763.0	9,400.4	0.00	0.00	0.00
20,200.0	90.00	180.00	10,762.0	-9,469.8	-763.0	9,500.2	0.00	0.00	0.00
20,300.0	90.00	180.00	10,762.0	-9,569.8	-763.0	9,599.9	0.00	0.00	0.00
20,400.0	90.00	180.00	10,762.0	-9,669.8	-763.0	9,699.7	0.00	0.00	0.00
20,500.0	90.00	180.00	10,762.0	-9,769.8	-763.0	9,799.4	0.00	0.00	0.00
20,600.0	90.00	180.00	10,762.0	-9,869.8	-763.0	9,899.1	0.00	0.00	0.00
20,700.0	90.00	180.00	10,762.0	-9,969.8	-763.0	9,998.9	0.00	0.00	0.00
20,800.0	90.00	180.00	10,762.0	-10,069.8	-763.0	10,098.6	0.00	0.00	0.00
20,900.0	90.00	180.00	10,762.0	-10,169.8	-763.0	10,198.4	0.00	0.00	0.00
21,000.0	90.00	180.00	10,762.0	-10,269.8	-763.0	10,298.1	0.00	0.00	0.00
21,100.0	90.00	180.00	10,762.0	-10,369.8	-763.0	10,397.8	0.00	0.00	0.00
21,200.0	90.00	180.00	10,762.0	-10,469.8	-763.0	10,497.6	0.00	0.00	0.00
21,300.0	90.00	180.00	10,762.0	-10,569.8	-763.0	10,597.3	0.00	0.00	0.00
21,310.2	90.00	180.00	10,762.0	-10,580.0	-763.0	10,607.5	0.00	0.00	0.00
TD at 21310.2									

Design Targets

Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (ft)	+N/-S (ft)	+E/-W (ft)	Northing (m)	Easting (m)	Latitude	Longitude
- hit/miss target									
- Shape									
Jefferies 5301 43-12B P	0.00	0.00	10,762.0	-10,580.0	-763.0	121,865.49	367,674.51	48° 3' 13.656 N	103° 37' 3.06' W
- plan hits target center									
- Point									

Casing Points

Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)
2,170.0	2,170.0	9 5/8"	9.625	13.500
11,100.0	10,762.0	7"	7.000	8.750

Microsoft
Planning Report

Database: EDM 5000.1 9.0 Production DB
Company: Oasis
Project: Indian Hills
Site: 153N-101W-13/24
Well: Jefferies 5301 43-12B
Wellbore: Jefferies 5301 43-12B
Design: Plan #1

Local Co-ordinate Reference: Well Jefferies 5301 43-12B
TVD Reference: WELL @ 2118.0ft (Original Well Elev)
MD Reference: WELL @ 2118.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Formations

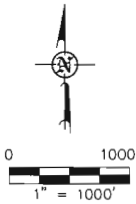
Measured Depth (ft)	Vertical Depth (ft)	Name	Lithology	Dip (°)	Dip Direction (°)
1,968.0	1,968.0	Pierre			
4,634.0	4,634.0	Greenhorn			
5,030.0	5,030.0	Mowry			
5,461.0	5,461.0	Dakota			
6,377.0	6,377.0	Rierdon			
6,896.0	6,896.0	Dunham Salt			
6,963.0	6,963.0	Dunham Salt Base			
6,968.0	6,968.0	Spearfish			
7,212.0	7,212.0	Pine Salt			
7,337.0	7,337.0	Pine Salt Base			
7,365.0	7,365.0	Opeche Salt			
7,444.0	7,444.0	Opeche Salt Base			
7,625.0	7,625.0	Broom Creek (Top of Minnelusa Gp.)			
7,668.0	7,668.0	Amsden			
7,844.0	7,844.0	Tyler			
8,031.0	8,031.0	Otter (Base of Minnelusa Gp.)			
8,380.0	8,380.0	Kibbey			
8,527.0	8,527.0	Charles Salt			
9,150.0	9,150.0	UB			
9,227.0	9,227.0	Base Last Salt			
9,275.0	9,275.0	Ratcliffe			
9,451.0	9,451.0	Mission Canyon			
10,025.0	10,025.0	Lodgepole			
10,873.7	10,740.0	False Bakken			
10,934.9	10,755.0	Upper Bakken			
11,013.5	10,762.0	Middle Bakken Sand Target			
11,013.5	10,762.0	Middle Bakken			

Plan Annotations

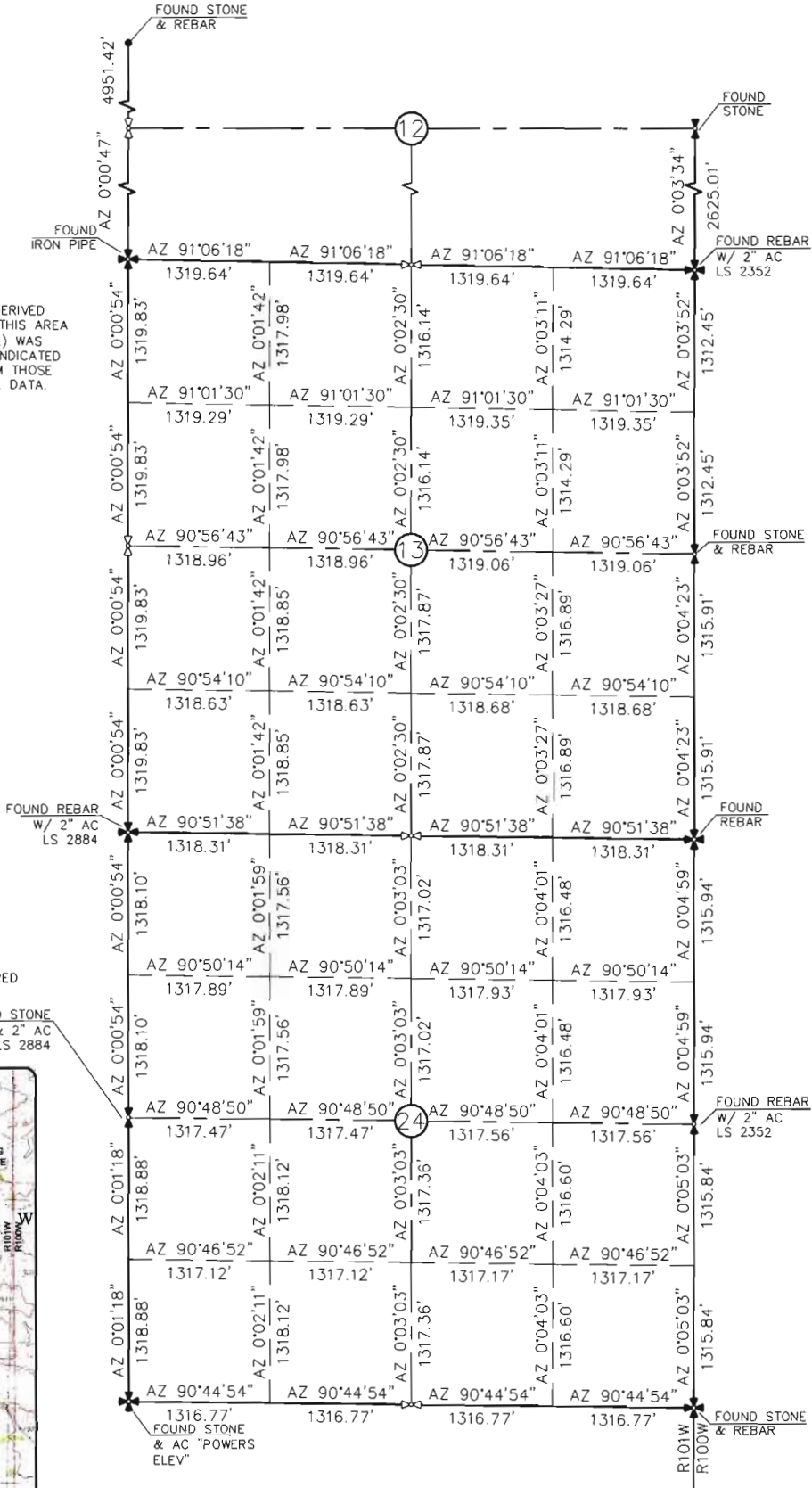
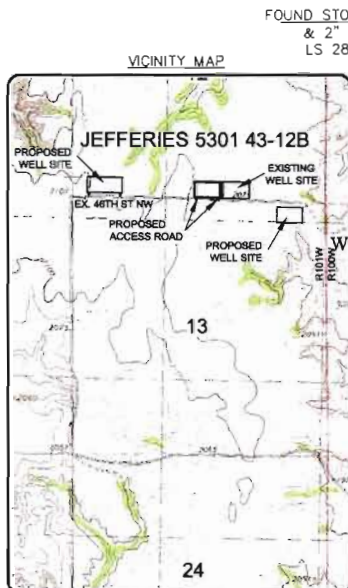
Measured Depth (ft)	Vertical Depth (ft)	Local Coordinates		Comment
		+N/-S (ft)	+E/-W (ft)	
10,321.3	10,321.3	0.0	0.0	KOP - Start Build 13.00
11,013.6	10,762.0	-381.9	-220.0	EOC - Start 67.0 hold at 11013.6 MD
11,080.6	10,762.0	-440.0	-253.4	Start DLS 1.50 TFO -90.00
13,076.5	10,762.0	-2,346.3	-763.1	Start 8233.7 hold at 13076.5 MD
21,310.2	10,762.0	-10,580.0	-763.0	TD at 21310.2

SECTION BREAKDOWN
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
 "JEFFERIES 5301 43-12B"
 250 FEET FROM SOUTH LINE AND 2510 FEET FROM EAST LINE
 SECTIONS 12, 13, & 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

ALL BEARINGS ARE BASED ON G.P.S. DERIVED BEARINGS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1900. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA.



⊕ - MONUMENT - RECOVERED
 ⊗ - MONUMENT - NOT RECOVERED

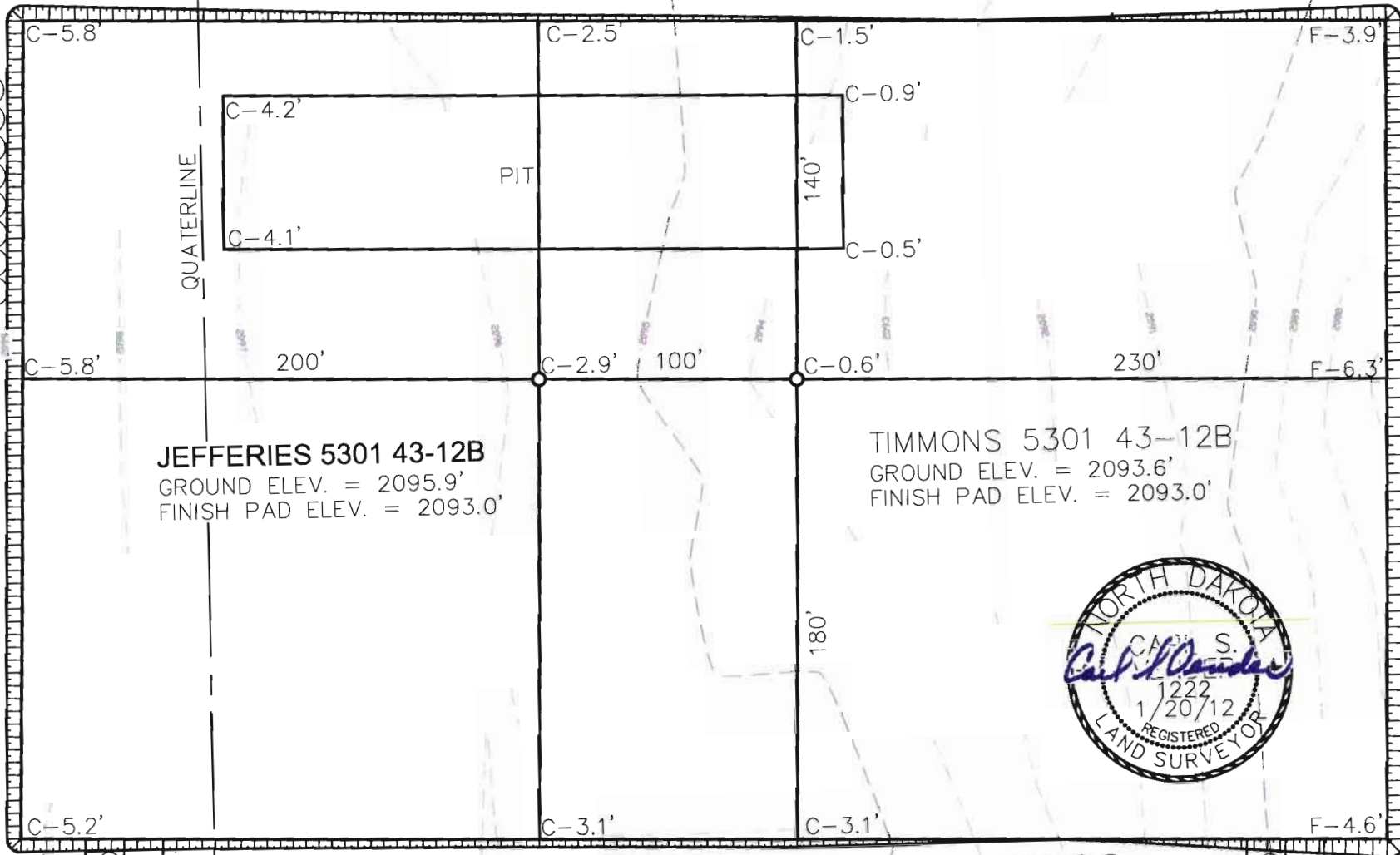


© 2011, INTERSTATE ENGINEERING, INC.

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"

250 FEET FROM SOUTH LINE AND 2510 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

Revision No.	Date	By	Description
REV 1	1/19/12	LS	SHARPOW WELL LOCATIONS

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: J.J.S. Project No.: S11-9-361
Checked By: C.S.V. Date: DEC 2011

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sully, ND 58270
Ph: (406) 433-5670
Fax: (406) 433-5618
www.inteng.com
Other offices in Minnesota, North Dakota and South Dakota

© 2011, INTERSTATE ENGINEERING, INC.
INTERSTATE ENGINEERING
Professionals you need, people you trust

3/8
SHEET NO.

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"

LANDOWNER:
LARRY HEEN
SE1/4SW1/4 & SW1/4SE1/4 SEC 12
140.1 FT = 9 RODS

AZ 181°06'
69.9'

-2225.2'-

EX. 33' R/W AZ 181°06'
70.2'

LEX. 33' R/W

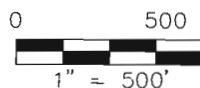
75.2'

TESS 5301 14-13B

EX. 33' R/W

EX. 33' R/W

FOUND REBAR
W/ 2" AC
LS 2352



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.



© 2011, INTERSTATE ENGINEERING, INC.



4/8

Deafness is not a disease, it is a condition.

COASTS PETROLEUM IN NORTH AMERICA LLC

ROLEUM NORTH AMERICA ACCESS APPROACH

ACCESS APPROACH
SECTION 12. T153N. R101W

Order No:	J.J.S.	Product No:	S11-09-381
-----------	--------	-------------	------------

Drummen Bv.	J.J.S.	Product No.:
-------------	--------	--------------

Checked By: C.S.V. Date: DEC 2011

Checked By: _____ C.S.V. _____ Date: _____

Revision No.	Date	By	Description
--------------	------	----	-------------

10/10/2019

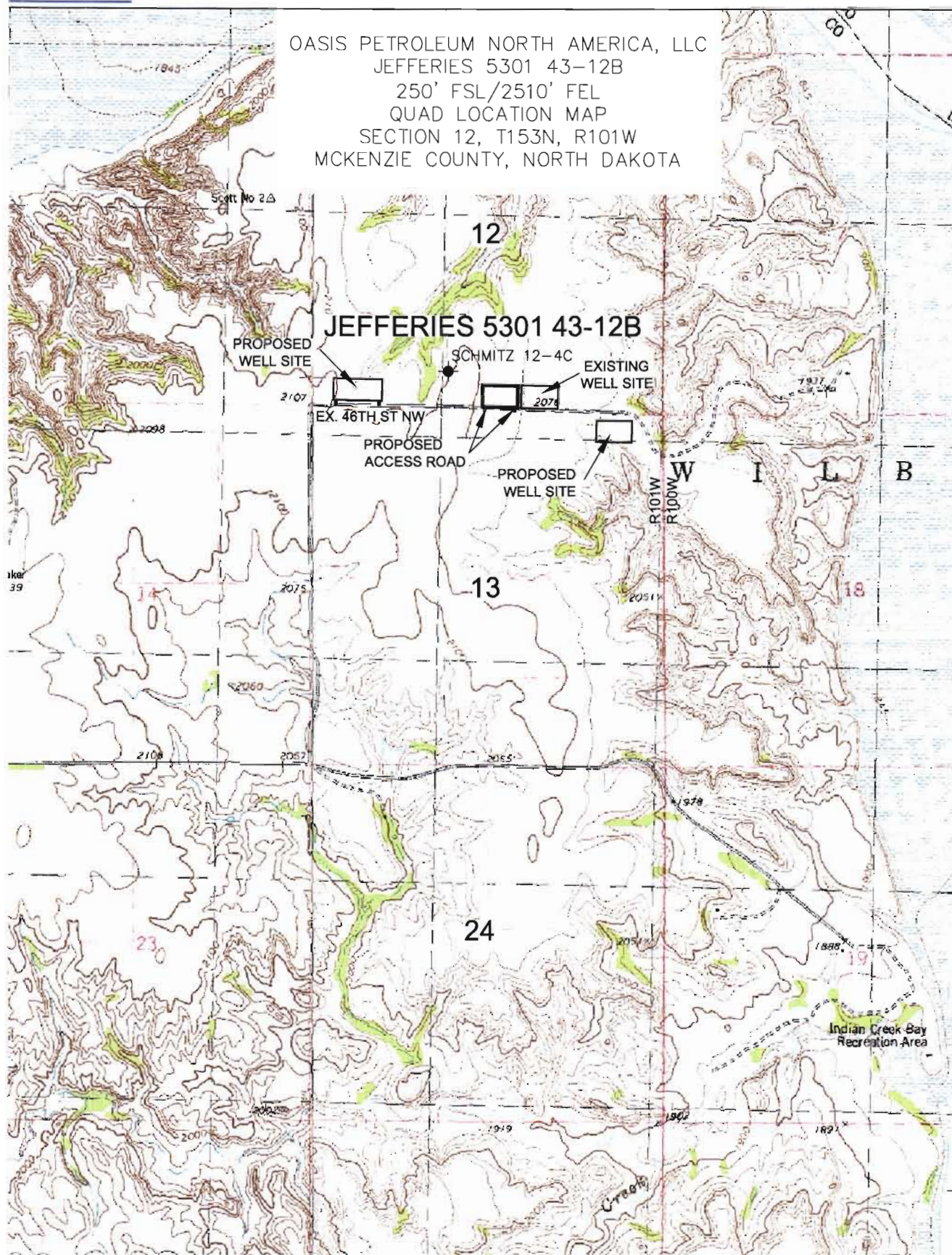
Revision No.	Date	By	Description
--------------	------	----	-------------

Revision No.	Date	By	Description
--------------	------	----	-------------

Revision No.	Date	By	Description
--------------	------	----	-------------

Revision No.	Date	By	Description
--------------	------	----	-------------

OASIS PETROLEUM NORTH AMERICA, LLC
JEFFERIES 5301 43-12B
250' FSL/2510' FEL
QUAD LOCATION MAP
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA



© 2011, INTERSTATE ENGINEERING, INC.

5/8

SHEET NO.



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengi.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

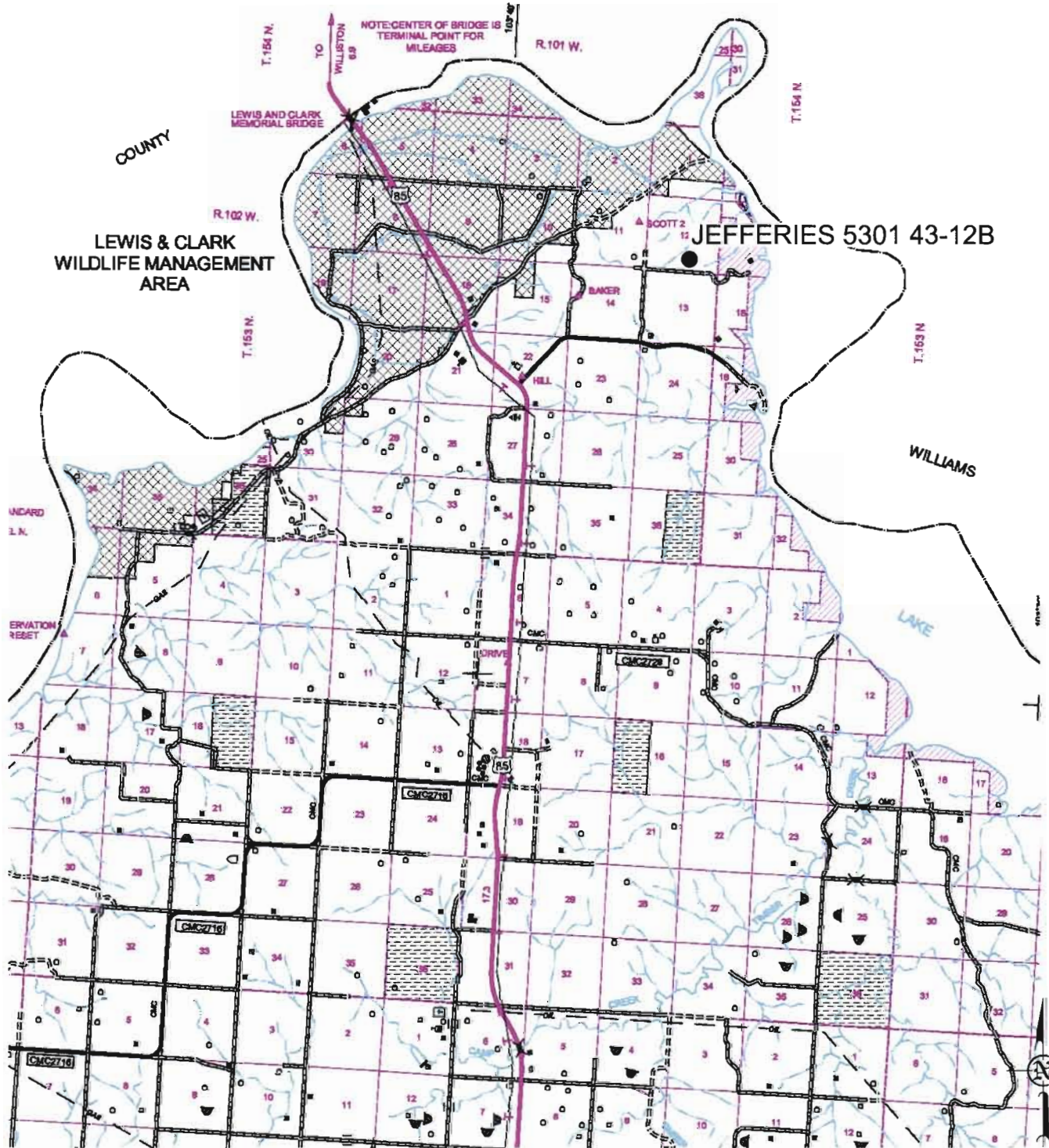
Drawn By: J.J.S. Project No.: S11-09-361
Checked By: C.S.V. Date: DEC 2011

Revision No.	Date	By	Description
REV 1	1/16/12	JJS	SWAPPED WELL LOCATIONS

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"

250 FEET FROM SOUTH LINE AND 2510 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2011, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

6/8

SHEET NO.



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengi.com
Other offices in Minnesota, North Dakota and South Dakota

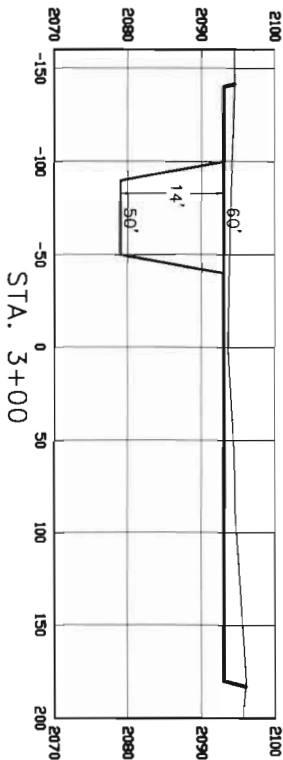
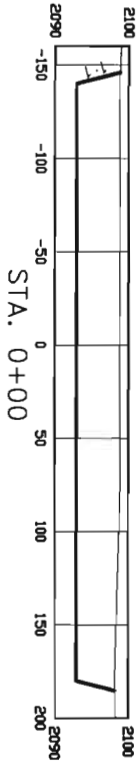
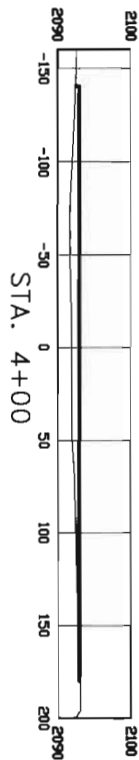
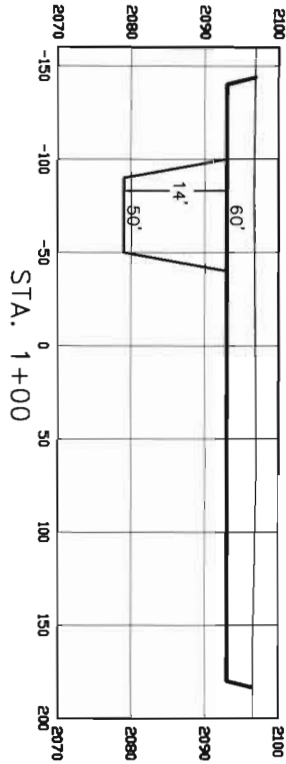
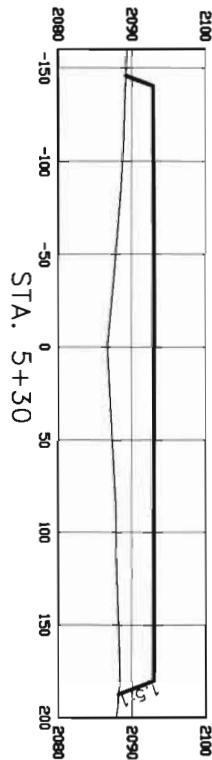
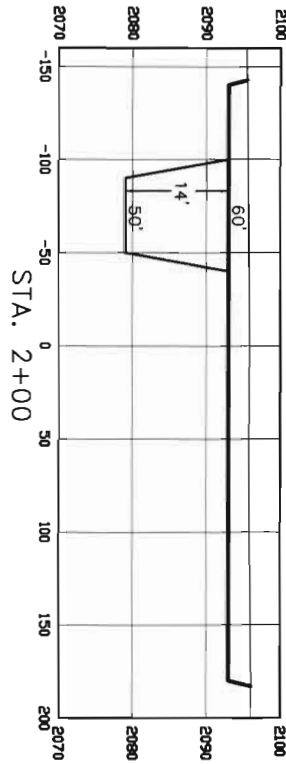
OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S11-09-361
Checked By: C.S.V. Date: DEC 2011

Revision No.	Date	By	Description
REV 1	1/19/12	JJS	SHARPER WELL LOCATIONS

CROSS SECTIONS

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"
250 FEET FROM SOUTH LINE AND 2510 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



SCALE
HORIZ 1"=100'
VERT 1"=20'

© 2011, INTERSTATE ENGINEERING, INC.

7/8
SHEET NO.



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengi.com
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
PAD CROSS SECTIONS
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: J.J.S. Project No.: S11-9-361
Checked By: C.S.V. Date: DEC 2011

Revision No.	Date	By	Description
REV 1	1/18/12	JJS	SWAPPED WELL LOCATIONS

WELL LOCATION SITE QUANTITIES

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"

250 FEET FROM SOUTH LINE AND 2510 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

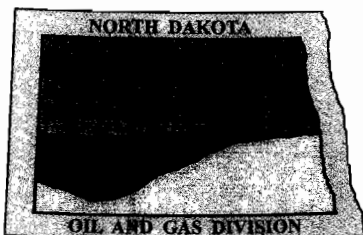
WELL SITE ELEVATION	2095.9
WELL PAD ELEVATION	2093.0

EXCAVATION	12,909
PLUS PIT	<u>3,150</u>
	16,059
EMBANKMENT	4,705
PLUS SHRINKAGE (30%)	<u>1,412</u>
	6,117
STOCKPILE PIT	3,150
STOCKPILE TOP SOIL (6")	3,284
STOCKPILE FROM PAD	3,508
DISTURBED AREA FROM PAD	4.07 ACRES

NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
CUT END SLOPES AT 1:1
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION

2510' FEL
250' FSL



Oil and Gas Division

22220

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

ROBIN E. HESKETH
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN, SUITE 1500
HOUSTON, TX 77002 USA

Date: 1/19/2012

RE: CORES AND SAMPLES

Well Name: **JEFFERIES 5301 43-12B** Well File No.: **22220**
Location: **SWSE 12-153-101** County: **MCKENZIE**
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: **BAKKEN**

Dear ROBIN E. HESKETH:

North Dakota Century Code (NDCC) Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for the NDCC Section 38-08-04 and North Dakota Administrative Code 43-02-03-38.1.
- 2) Samples shall include all cuttings from:

Base of the Last Charles Salt

Samples of cuttings shall be taken at 30' maximum intervals through all vertical, build and horizontal sections. Samples must be washed, dried, packed in sample envelopes in correct order with labels showing operator, well name, location and depth, and forwarded in standard boxes to the State Geologist within 30 days of the completion of drilling operations.

- 3) Cores: ALL CORES cut shall be preserved in correct order, properly boxed, and forwarded to the State Geologist within 90 days of completion of drilling operations. Any extension of time must have written approval from the State Geologist.
- 4) All cores, core chips, and samples must be shipped, prepaid, to the State Geologist at the following address:

**ND Geological Survey Core Library
Campus Road and Cornell
Grand Forks, ND 58202**
- 5) NDCC Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely

Richard A. Suggs
Geologist

SURFACE DAMAGE SETTLEMENT AND RELEASE



In consideration for the sum of _____ Dollars

(\$ _____) paid by Oasis Petroleum North America LLC ("Oasis") to the undersigned surface owners, Larry P. Heen, a married man dealing in his sole & separate property ("Owners," and together with Oasis, the "Parties") for themselves and their heirs, successors, administrators and assigns, hereby acknowledge the receipt and sufficiency of said payment as a full and complete settlement for and as a release of all claims for loss, damage or injury to the Subject Lands (as defined herein) arising out of the Operations (as defined herein) of the Jefferies 5301 43-12B & Timmons 5301 43-12B the "Well(s)" located on the approximately (6) six acre tract of land identified on the plat attached hereto as Exhibit "A" (the "Subject Lands") and which is situated on the following described real property located in McKenzie County, State of North Dakota, to wit:

22220

Township 153 North, Range 101 West, 5th P.M.
Section 12: SE4SW4, SW4SE4

This pad shall accommodate the drilling of the Jefferies 5301 43-12B well and the Timmons 5301 43-12B well on the same location. The undersigned is fully aware that the cuttings generated from the drilling of the above described wells will be buried on site on the above described location.

The Parties agree that the settlement and release described herein does not include any claims by any third party against the Owners for personal injury or property damage arising directly out of Oasis's Operations, and Oasis agrees to indemnify, defend and hold harmless Owners against all liabilities arising from such claim (except as such claim arises from the gross negligence or willful misconduct of the Owners).

In further consideration of the payments specified herein, Oasis is hereby specifically granted the right to construct, install and operate, replace or remove pads, pits, pumps, compressors, tanks, roads, pipelines, equipment or other facilities on the above described tract of land necessary for its drilling, completion, operation and/or plugging and abandonment of the Well(s) (the "Operations"), and to the extent such facilities are maintained by Oasis for use on the Subject Lands, this agreement shall permit Oasis's use of such facilities for the Operations on the Subject Lands.

Should commercial production be established from the Well(s), Oasis agrees to pay Owners an annual amount of _____ per year beginning one year after the completion of the Wells and to be paid annually until the Wells is plugged and abandoned.

The Parties expressly agree and acknowledge that the payments described herein to be made by Oasis to the Owners constitute full satisfaction of the requirements of Chapter 38.11.1 of the North Dakota Century Code and, once in effect, the amended Chapter 38.11.1 of the North Dakota Century Code enacted by House Bill 1241. The Parties further expressly agree and acknowledge that the \$ _____ payment set forth above constitutes full and adequate consideration for damage and disruption required under Section 38.11.1-04 of the North Dakota Century Code, and that the _____ payment set forth above constitutes full and adequate consideration for loss of production payments under Section 38.11.1-08.1 of the North Dakota Century Code.

Oasis shall keep the Site free of noxious weeds, and shall take reasonable steps to control erosion and washouts on the Site. Oasis shall restore the Site to a condition as near to the original condition of the Site as is reasonably possible, including the re-contouring, replacing of topsoil and re-seeding of the Site (such actions, the "Restoration").

The surface owners grant Oasis access to the Wells in the location(s) shown on the plats attached hereto as Exhibit "A".

Upon written request and the granting of a full release by the Owners of further Restoration by Oasis with respect to the affected area described in this paragraph, Oasis shall leave in place any road built by it in its Operations for the benefit of the Owners after abandoning its Operations, and shall have no further maintenance obligations with respect to any such road.

This agreement shall apply to the Parties and their respective successors, assigns, parent and subsidiary companies, affiliates and related companies, trusts and partnerships, as well as their contractors, subcontractors, officers, directors, agents and employees.

This agreement may be executed in multiple counterparts, each of which shall be an original, but all of which shall constitute one instrument.

[Signature Page Follows.]

DATED this 13th day of December

2011

SURFACE OWNERS

Larry P. Heen
Larry P. Heen, a married man dealing in his sole &
separate property

Address: 14033 45th Street NW

Williston, ND 58801

Phone: 701-572-6991

STATE OF North Dakota

ACKNOWLEDGMENT INDIVIDUAL

COUNTY OF McKenzie

} SS.

BE IT REMEMBERED, That on this 13th day of December, 2011 before me, a Notary Public,
in and for said County and State, personally appeared Larry P. Heen, a married man dealing in his sole &
separate property, to me known to be the identical person described in
and who executed the within and foregoing instrument and acknowledged to me that he executed the
same as his free and voluntary act and deed for the uses and purposes therein set forth.

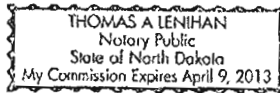
IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal,
the day and year first above written.

My commission expires: April 9, 2013

Thomas A. Lenihan

Notary Public

NOTARY STAMP



STATE OF _____

ACKNOWLEDGMENT CORPORATION

COUNTY OF _____

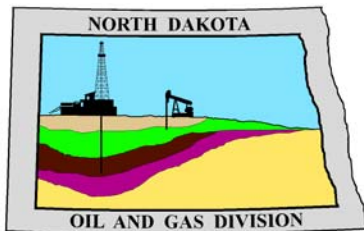
Before me the undersigned, a Notary Public, in and for said County and State, on this _____ day of _____,
2011, personally appeared _____, to me known to be the identical person who
subscribed the name of the maker thereof to the foregoing instrument as its _____ and
acknowledged to me that _____ executed the same as _____ free and voluntary act and deed and as the free and
voluntary act and deed of such corporation, for the uses and purposes therein set forth.

Given under my hand and seal of office the day and year last above written.

My commission expires: _____

Notary Public

NOTARY STAMP



Oil and Gas Division

Lynn D. Helms - Director Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

January 10, 2012

Kaitlin Bass
Operations Assistant
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 1500
Houston, TX 77002

**RE: HORIZONTAL WELL
JEFFERIES 5301 43-12B
SWSE Section 12-153N-101W
McKenzie County
Well File # 22220**

Dear Kaitlin:

Pursuant to Commission Order No. 18012, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **200' setback** from the north & south boundaries and **500' setback** from the east & west boundaries within the 1280 acre spacing unit consisting of Sections 13 & 24-T153N-R101W.

PERMIT STIPULATIONS: A Closed Mud System is required on multi-well pads, although the disposal of drill cuttings is contingent upon site specific conditions to be determined by an NDIC Field Inspector. In cases where a spacing unit is accessed from an off-site drill pad, an affidavit must be provided affirming that the surface owner of the multi-well pad agrees to accept burial on their property of the cuttings generated from drilling the well(s) into an offsite spacing/drilling unit. OASIS PETRO NO AMER must contact NDIC Field Inspector Marc Binns at 701-220-5989 prior to location construction.

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. The minimum legal coordinates from the well head at casing point is: 450' S & 1910' E. Also, based on the azimuth of the proposed lateral the maximum legal coordinates from the well head is: 1910' E & 10586' S.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card. The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the

Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Kaitlin Bass
January 10, 2012
Page 2

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Reserve pit

Please be advised that conditions may be imposed on the use and reclamation of a drilling reserve pit on this site if specific site conditions warrant.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of a Cement Bond Log from which the presence of cement can be determined in every well in which production or intermediate casing has been set and a Gamma Ray Log must be run from total depth to ground level elevation of the well bore. All logs must be submitted as one paper copy and one digital copy in LAS (Log ASCII) format, or a format approved by the Director. Image logs that include, but are not limited to, Mud Logs, Cement Bond Logs, and Cyberlook Logs, cannot be produced in their entirety as LAS (Log ASCII) files. To create a solution and establish a standard format for industry to follow when submitting image logs, the Director has given approval for the operator to submit an image log as a TIFF (*.tif) formatted file. The TIFF (*.tif) format will be accepted only when the log cannot be produced in its entirety as a LAS (Log ASCII) file format. The digital copy may be submitted on a 3.5" floppy diskette, a standard CD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

David Tabor
Engineering Technician IV



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 12 / 30 / 2011	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC			Telephone Number 281-404-9461
Address 1001 Fannin Suite 1500		City Houston	State TX Zip Code 77002

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name JEFFERIES				Well Number 5301 43-12B			
Surface Footages 250 F S L 2410 F E L		Qtr-Qtr SWSE	Section 12	Township 153 N	Range 101 W	County McKenzie	
Longstring Casing Point Footages 206 F N L 2644 F E L		Qtr-Qtr NENW	Section 13	Township 153 N	Range 101 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 456 S From WH 234 W From WH		Azimuth 207 °	Longstring Total Depth 11085 Feet MD 10762 Feet TVD				
Bottom Hole Footages From Nearest Section Line 200 F S L 2000 F W L		Qtr-Qtr SESW	Section 24	Township 153 N	Range 101 W	County McKenzie	
Bottom Hole Coordinates From Well Head 10586 S From WH 857 W From WH		KOP Lateral 1 10321 Feet MD	Azimuth Lateral 1 180 °		Estimated Total Depth Lateral 1 21314 Feet MD 10762 Feet TVD		
Latitude of Well Head 48 ° 04 ' 58.05 "		Longitude of Well Head -103 ° 36 ' 50.36 "		NAD Reference NAD83		Description of (Subject to NDIC Approval) SPACING UNIT: Sections 13 & 24-T153N-R101W	
Ground Elevation 2094 Feet Above S.L.		Acres in Spacing/Drilling Unit 1280		Spacing/Drilling Unit Setback Requirement 200 Feet N/S 500 Feet E/W		Industrial Commission Order 18012	
North Line of Spacing/Drilling Unit 5278 Feet		South Line of Spacing/Drilling Unit 5267 Feet		East Line of Spacing/Drilling Unit 10520 Feet		West Line of Spacing/Drilling Unit 10553 Feet	
Objective Horizons BAKKEN						Pierre Shale Top 1968	
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 2170 Feet	Cement Volume 643 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 29&32 Lb./Ft.	Longstring Total Depth 11085 Feet MD 10762 Feet TVD		Cement Volume 741 Sacks	Cement Top 4961 Feet	Top Dakota Sand 5461 Feet
Base Last Charles Salt (If Applicable) 9227 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs CBL/GR-TOC/GR-BSC							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Salt Water Gel			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor Ryan	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Email Attachments: Drill plan with drilling fluid data, Well Summary with Csg/Cmt design, Direct Plan/Plot, and Plats

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.			Date 12 / 9 / 2011
ePermit	Printed Name Kaitlin Bass	Title Operations Assistant	

FOR STATE USE ONLY

Permit and File Number 22220	API Number 33 - 053 - 03936
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

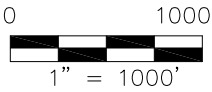
FOR STATE USE ONLY

Date Approved 1 / 10 / 2012
By David Tabor
Title Engineering Technician IV

WELL LOCATION PLAT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"
250 FEET FROM SOUTH LINE AND 2410 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

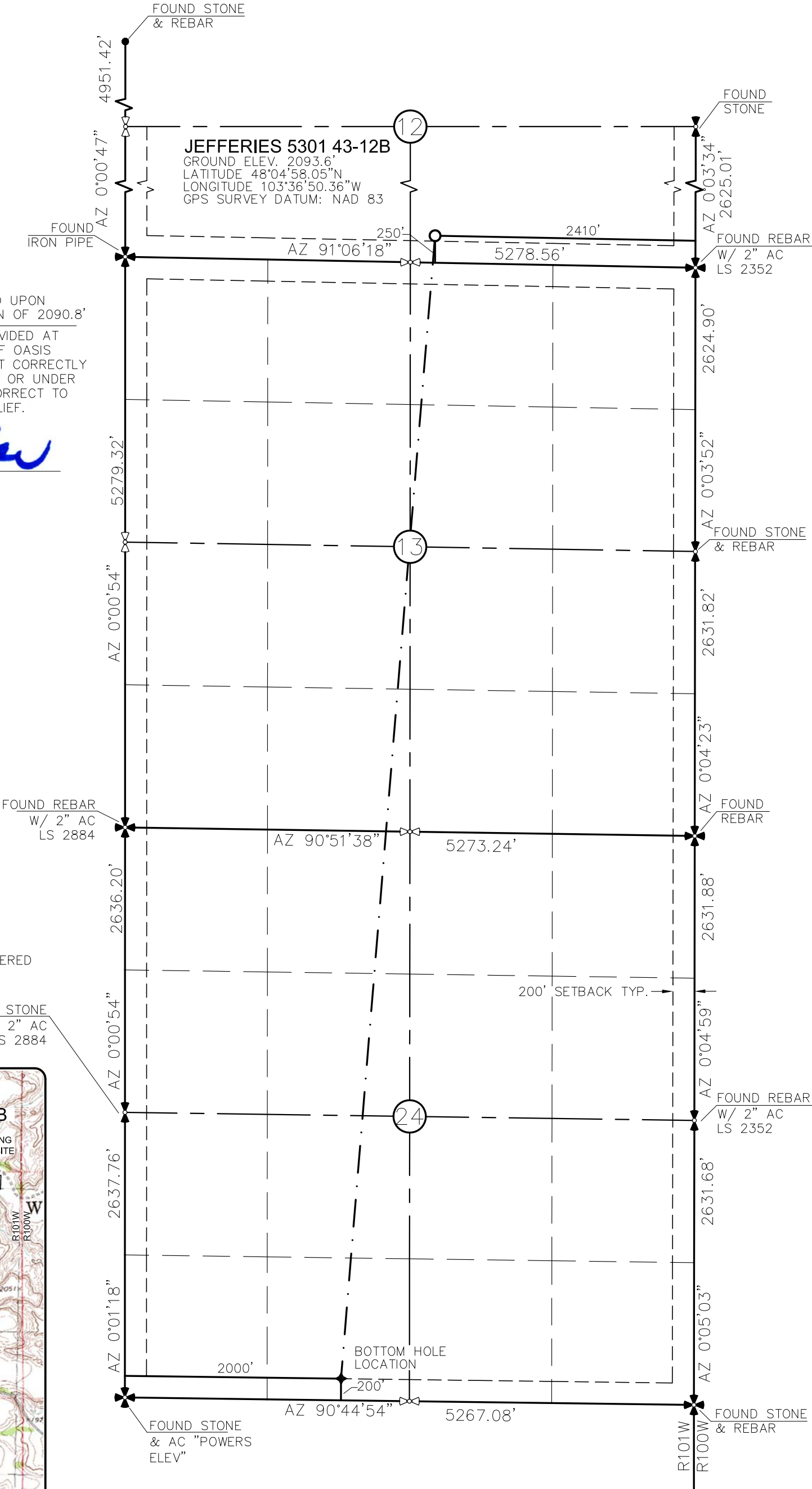
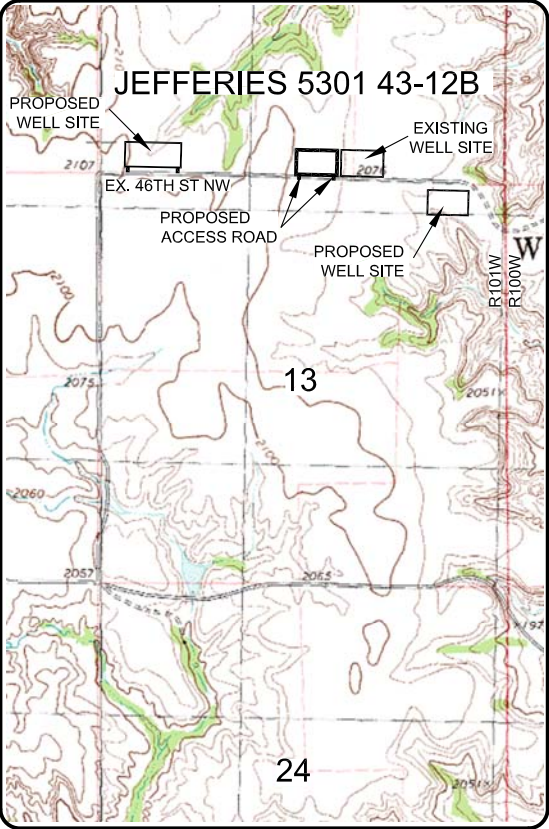
STAKED ON 12/6/11
VERTICAL CONTROL DATUM WAS BASED UPON
CONTROL POINT 13 WITH AN ELEVATION OF 2090.8'
THIS SURVEY AND PLAT IS BEING PROVIDED AT
THE REQUEST OF FABIAN KJORSTAD OF OASIS
PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY
REPRESENTS WORK PERFORMED BY ME OR UNDER
MY SUPERVISION AND IS TRUE AND CORRECT TO
THE BEST OF MY KNOWLEDGE AND BELIEF.

Carl S. Vender
CARL S. VENDER LS 1222



- ⊕ — MONUMENT — RECOVERED
⊗ — MONUMENT — NOT RECOVERED

VICINITY MAP



© 2011, INTERSTATE ENGINEERING, INC.

1/8

SHEET NO.



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengi.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S11-09-361
Checked By: C.S.V. Date: DEC 2011

Revision No.	Date	By	Description

C:\2011\S11-09-361\Oasis Petroleum 2 Wells Sec 12 13 & 24 T153N R101W\CAD\JEFFERIES WELL.dwg - 12/7/2011 1:22 PM jsh schmierer

**Oasis Petroleum
Well Summary
Jefferies 5301 43-12B
Section 12 T153N R101W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
9-5/8"	0' to 2,170'	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
0' to 2,170'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 1.98	3520 / 3.46	453 / 2.70	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.0 ppg fluid on backside.
- b) Burst pressure based on 9 ppg fluid with no fluid on backside.
- c) Based on string weight in 9.0 ppg fluid at 2,170' TVD plus 100k# overpull.
(Buoyed weight equals 67k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2" hole with 55% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **443 sks** (234 bbls) 11.5 lb/gal VARICEM CEMENT with 0.25 lb/sk Poly-E-Flake (lost circulation additive)

Tail Slurry: **200 sks** (72 bbls) 13 lb/gal VARICEM CEMENT with 0.25 lb/sk Poly-E-Flake (lost circulation additive)

**Oasis Petroleum
Well Summary
Jefferies 5301 43-12B
Section 12 T153N R101W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Intermediate Casing Design

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' – 6,750'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770
7"	6,750' – 10,321' (KOP)	32	HCP-110	LTC	6.094"	6.000"	6,730	8,970	9,870
7"	10,321' – 11,085'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770

**Special Drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 6,750'	6,750'	7", 29#, P-110, LTC, 8rd	8,530 / 2.43*	11,220 / 1.19	797 / 2.09
6,750' – 10,321'	3,571'	7", 32#, HCP-110, LTC, 8rd	11,820 / 2.09*	12,460 / 1.29	
6,750' – 10,321'	3,571'	7", 32#, HCP-110, LTC, 8rd	11,820 / 1.05**	12,460 / 1.29	
10,321' – 11,085'	764'	7", 29 lb, P-110, LTC, 8rd	8,530 / 1.52*	11,220 / 1.16	

API Rating & Safety Factor

- a. *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b. Burst pressure based on 9,000 psig maxpress for stimulation plus 10.2 ppg fluid in casing and 9.0 ppg fluid on backside—to 10,762' TVD.
- c. Based on string weight in 10 ppg fluid, (280k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **100 bbls** Saltwater
70 sks Pozmix A
20 bbls Fresh Water

Lead Slurry: **110 sks** (50 bbls) 11.8 lb/gal ECONOCHEM SYSTEM with 0.3% Fe-2 (additive material) and 0.25 lb/sk Poly-E-Flake (lost circulation additive)

Primary Slurry: **349 sks** (86 bbls) 14 lb/gal EXTENDACHEM SYSTEM with 0.6% HR-5 (retarder) and 0.25 lb/sk Poly-E-Flake (lost circulation additive)

Tail Slurry: **282 sks** (78 bbls) 15.6 lb/gal HALCEM SYSTEM with 0.2% HR-5 (retarder), 0.25 lb/sk Poly-E-Flake (lost circulation additive) and 35% SSA-1 (additive material)

**Oasis Petroleum
Well Summary
Jefferies 5301 43-12B
Section 12 T153N R101W
McKenzie County, ND**

PRODUCTION LINER

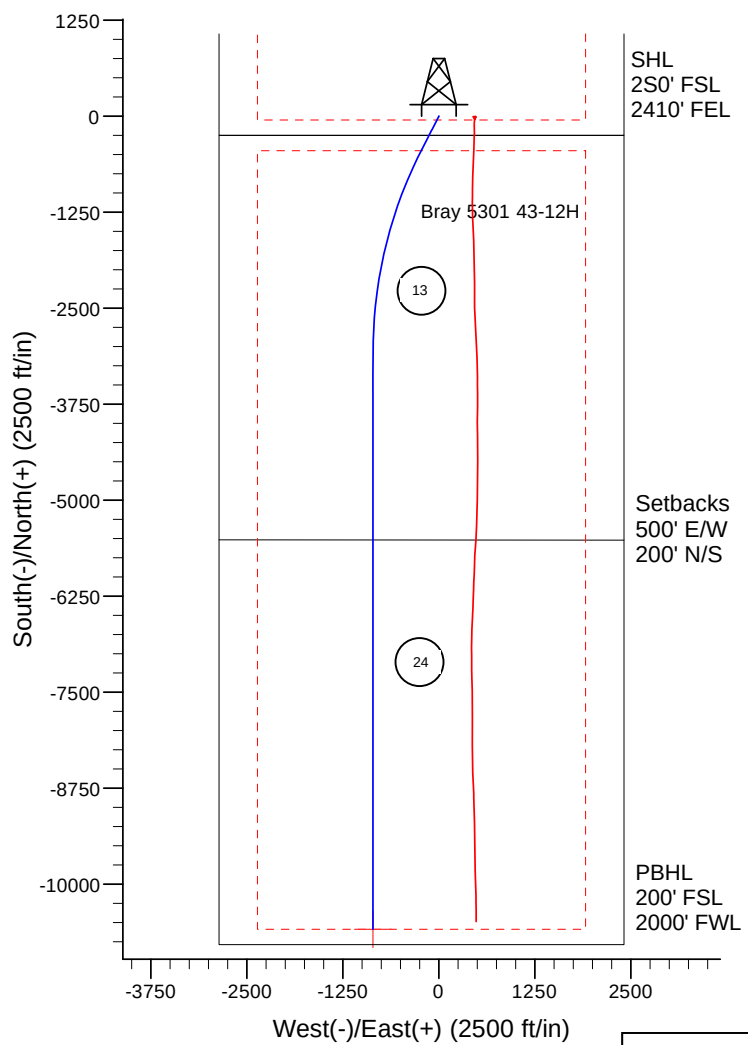
							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
4-1/2"	10,270' to 21,313'	11.6	P-110	BTC	4.000"	3.875"			

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
10,270' to 21,313'	4-1/2", 11.6 lb, P-110, BTC	7,560 / 1.42	10,690 / 1.105	279 / 1.33	\$13.25

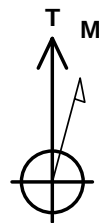
API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,762' TVD.
- b) Burst pressure based on 9,000 psi Stimulation pressure with 10.2 ppg internal fluid gradient with 9.0 ppg gradient on backside @ 10,762' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 110k lbs.) plus 100k lbs overpull.

DRILLING PLAN									
PROSPECT/FIELD	Indian Hills		Horizontal Middle Bakken		COUNTY/STATE	McKenzie Co., ND			
OPERATOR	Oasis Operating				RIG	Nabors 149			
WELL NO.	5301 43-12B				LEASE	Jefferies			
LOCATION	SWSE 12-153N-101W		Surface Location (survey plat): 250' fsl			2410' fel			
EST. T.D.	21,313'				GROUND ELEV:		2093		Finished Pad Elev.
TOTAL LATERAL:		10,228' (est)			KB ELEV:		2118		Sub Hieght: 25
PROGNOSIS: Based on 2,118' KB(est)					LOGS: Type Interval				
OH Logs: File to omit									
CBL/GR: Above top of cement/GR to base of casing									
MWD GR: KOP to lateral TD									
DEVIATION:									
Surf:					3 deg. max., 1 deg / 100'; srvy every 500'				
Prod:					5 deg. max., 1 deg / 100'; srvy every 100'				
DST'S:					None planned				
CORES:					None planned				
MUDLOGGING:					Two-Man: 8,327'				
					~200' above the Charles (Kibbey) to Casing point; Casing point to TD				
					30' samples at direction of wellsite geologist; 10' through target @ curve land				
BOP:					11" 5000 psi blind, pipe & annular				
Dip Rate: -0.25° or .63' /100' DOWN first 4000' then +0.55° or .93'/100' UP									
Max. Anticipated BHP: 4676					Surface Formation: Glacial till				
MUD:	Interval		Type		WT	Vis	WL	Remarks	
Surface	0' -	2,170'	FW/Gel - Lime Sweeps		8.6 - 8.9	28-34	NC	Circ Mud Tanks	
Intermediate	2,170' -	11,085'	Invert		9.6-10.4	40-60	30+(HpHt)	Circ Mud Tanks	
Liner	11,085' -	21,313'	Salt Water		9.3-10.4	28-34	NC	Circ Mud Tanks	
CASING:	Size	Wt ppf	Hole	Depth	Cement	WOC	Remarks		
Surface:	9-5/8"	36#	13-1/2"	2,170'	To Surface	12	100' into Pierre		
Intermediate:	7"	29/32#	8-3/4"	11,085'	4,961'	24	500' above Dakota		
Production:	4.5"	11.6#	6"	21,313'	TOL @ 10,270'		50' above KOP		
Production Liner:									
PROBABLE PLUGS, IF REQ'D:									
OTHER:	MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI			
Surface:	2,170'	2,170'	250' FSL	2410' FEL	12-T153N-R101W	Survey Company:			
KOP:	10,321'	10,321'	250' FSL	2410' FEL	12-T153N-R101W	Build Rate:	13 deg /100'		
EOC	11,013'	10,762'	190' FNL	2635' FEL	13-T153N-R101W				
Casing Point:	11,085'	10,762'	206' FNL	2644' FEL	13-T153N-R101W				
Middle Bakken Lateral TD:	21,313'	10,762'	200' FSL	2000' FWL	24-T153N-R101W				
Comments:									
DRILL TO KOP.									
DRILL CURVE TO 90 DEG AND 7" CASING POINT									
SET 7" CASING. DRILL MIDDLE BAKKEN LATERAL.									
MWD Surveys will be taken every 100' in vertical hole, and a minimum of every 30' while building curve and every 90' while drilling lateral.									
MWD GR to be run from KOP to Lateral TD.									
GR must be run to ground surface.									
Geology: MRB 12-9-2011									
Prepared by:									
Engineering: L. Strong 12/9/2011									



Project: Indian Hills
 Site: 153N-101W-13/24
 Well: Jefferies 5301 43-12B
 Wellbore: OH
 Design: Plan #1



Azimuths to True North
 Magnetic North: 8.56°
 Magnetic Field
 Strength: 56724.3snT
 Dip Angle: 73.09°
 Date: 12/9/2011
 Model: IGRF200510

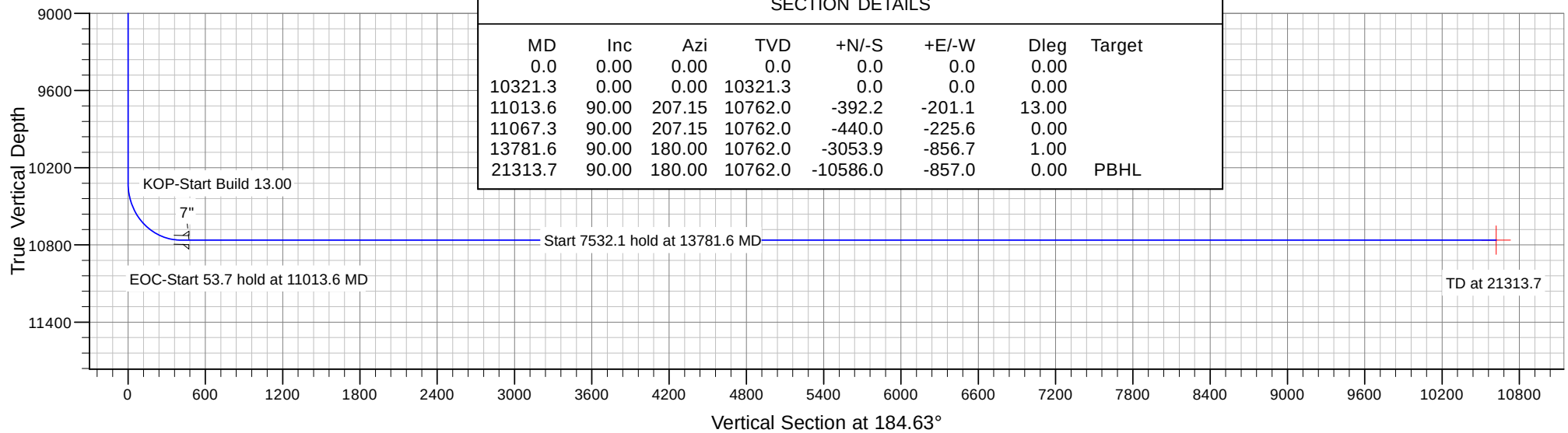
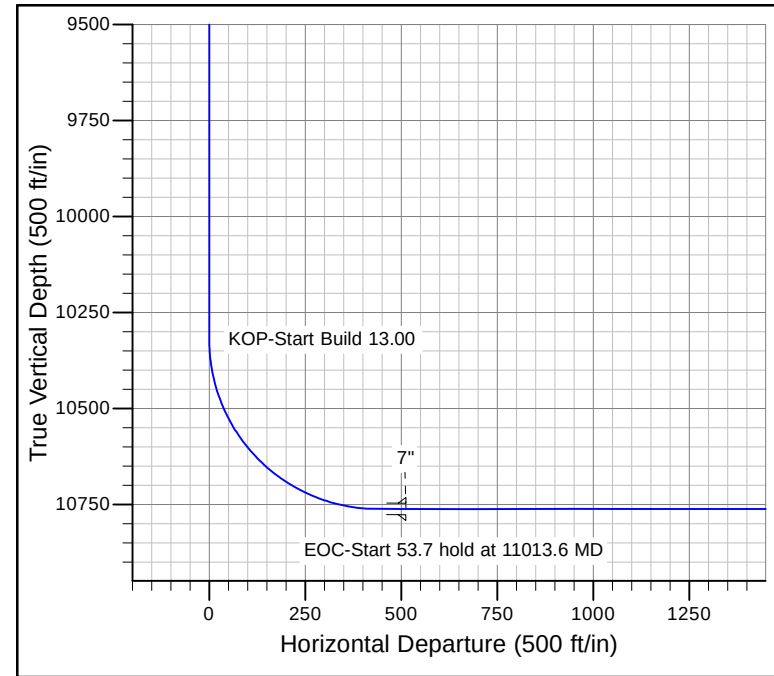
CASING DETAILS

TVD	MD	Name	Size
2170.0	2170.0	9 5/8"	9.625
10762.0	11085.0	7"	7.000

SITE DETAILS: 153N-101W-13/24

Site Centre Latitude: 48° 4' 57.960 N
 Longitude: 103° 36' 43.250 W

Positional Uncertainty: 0.0
 Convergence: -2.32
 Local North: True



SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
10321.3	0.00	0.00	10321.3	0.0	0.0	0.00	
11013.6	90.00	207.15	10762.0	-392.2	-201.1	13.00	
11067.3	90.00	207.15	10762.0	-440.0	-225.6	0.00	
13781.6	90.00	180.00	10762.0	-3053.9	-856.7	1.00	
21313.7	90.00	180.00	10762.0	-10586.0	-857.0	0.00	PBHL

Oasis

Indian Hills

153N-101W-13/24

Jefferies 5301 43-12B

OH

Plan: Plan #1

Standard Planning Report

09 December, 2011

Oasis Petroleum

Planning Report

Database:	Compass	Local Co-ordinate Reference:	Well Jefferies 5301 43-12B
Company:	Oasis	TVD Reference:	WELL @ 2118.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2118.0ft (Original Well Elev)
Site:	153N-101W-13/24	North Reference:	True
Well:	Jefferies 5301 43-12B	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-101W-13/24			
Site Position:		Northing:	125,067.66 m	Latitude: 48° 4' 57.960 N
From:	Lat/Long	Easting:	368,214.56 m	Longitude: 103° 36' 43.250 W
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in	Grid Convergence: -2.32 °

Well	Jefferies 5301 43-12B			
Well Position	+N/-S	9.1 ft	Northing:	125,076.39 m
	+E/-W	-482.7 ft	Easting:	368,067.66 m
Position Uncertainty		0.0 ft	Wellhead Elevation:	
			Ground Level:	2,093.0 ft

Wellbore	OH				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	12/9/2011	8.56	73.09	56,724

Design	Plan #1			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD)	+N/-S (ft)	+E/-W (ft)	Direction (°)
	0.0	0.0	0.0	184.63

Plan Sections										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
10,321.3	0.00	0.00	10,321.3	0.0	0.0	0.00	0.00	0.00	0.00	
11,013.6	90.00	207.15	10,762.0	-392.2	-201.1	13.00	13.00	0.00	207.15	
11,067.3	90.00	207.15	10,762.0	-440.0	-225.6	0.00	0.00	0.00	0.00	
13,781.6	90.00	180.00	10,762.0	-3,053.9	-856.7	1.00	0.00	-1.00	-90.00	
21,313.7	90.00	180.00	10,762.0	-10,586.0	-857.0	0.00	0.00	0.00	0.00	Jefferies 5301 43-12E

Oasis Petroleum

Planning Report

Database:	Compass	Local Co-ordinate Reference:	Well Jefferies 5301 43-12B
Company:	Oasis	TVD Reference:	WELL @ 2118.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2118.0ft (Original Well Elev)
Site:	153N-101W-13/24	North Reference:	True
Well:	Jefferies 5301 43-12B	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	0.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.00	0.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.00	0.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.00	0.00	400.0	0.0	0.0	0.0	0.00	0.00	0.00
500.0	0.00	0.00	500.0	0.0	0.0	0.0	0.00	0.00	0.00
600.0	0.00	0.00	600.0	0.0	0.0	0.0	0.00	0.00	0.00
700.0	0.00	0.00	700.0	0.0	0.0	0.0	0.00	0.00	0.00
800.0	0.00	0.00	800.0	0.0	0.0	0.0	0.00	0.00	0.00
900.0	0.00	0.00	900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,000.0	0.00	0.00	1,000.0	0.0	0.0	0.0	0.00	0.00	0.00
1,100.0	0.00	0.00	1,100.0	0.0	0.0	0.0	0.00	0.00	0.00
1,200.0	0.00	0.00	1,200.0	0.0	0.0	0.0	0.00	0.00	0.00
1,300.0	0.00	0.00	1,300.0	0.0	0.0	0.0	0.00	0.00	0.00
1,400.0	0.00	0.00	1,400.0	0.0	0.0	0.0	0.00	0.00	0.00
1,500.0	0.00	0.00	1,500.0	0.0	0.0	0.0	0.00	0.00	0.00
1,600.0	0.00	0.00	1,600.0	0.0	0.0	0.0	0.00	0.00	0.00
1,700.0	0.00	0.00	1,700.0	0.0	0.0	0.0	0.00	0.00	0.00
1,800.0	0.00	0.00	1,800.0	0.0	0.0	0.0	0.00	0.00	0.00
1,900.0	0.00	0.00	1,900.0	0.0	0.0	0.0	0.00	0.00	0.00
1,968.0	0.00	0.00	1,968.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,000.0	0.00	0.00	2,000.0	0.0	0.0	0.0	0.00	0.00	0.00
2,100.0	0.00	0.00	2,100.0	0.0	0.0	0.0	0.00	0.00	0.00
2,170.0	0.00	0.00	2,170.0	0.0	0.0	0.0	0.00	0.00	0.00
9 5/8"									
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.0	0.00	0.00	0.00
2,300.0	0.00	0.00	2,300.0	0.0	0.0	0.0	0.00	0.00	0.00
2,400.0	0.00	0.00	2,400.0	0.0	0.0	0.0	0.00	0.00	0.00
2,500.0	0.00	0.00	2,500.0	0.0	0.0	0.0	0.00	0.00	0.00
2,600.0	0.00	0.00	2,600.0	0.0	0.0	0.0	0.00	0.00	0.00
2,700.0	0.00	0.00	2,700.0	0.0	0.0	0.0	0.00	0.00	0.00
2,800.0	0.00	0.00	2,800.0	0.0	0.0	0.0	0.00	0.00	0.00
2,900.0	0.00	0.00	2,900.0	0.0	0.0	0.0	0.00	0.00	0.00
3,000.0	0.00	0.00	3,000.0	0.0	0.0	0.0	0.00	0.00	0.00
3,100.0	0.00	0.00	3,100.0	0.0	0.0	0.0	0.00	0.00	0.00
3,200.0	0.00	0.00	3,200.0	0.0	0.0	0.0	0.00	0.00	0.00
3,300.0	0.00	0.00	3,300.0	0.0	0.0	0.0	0.00	0.00	0.00
3,400.0	0.00	0.00	3,400.0	0.0	0.0	0.0	0.00	0.00	0.00
3,500.0	0.00	0.00	3,500.0	0.0	0.0	0.0	0.00	0.00	0.00
3,600.0	0.00	0.00	3,600.0	0.0	0.0	0.0	0.00	0.00	0.00
3,700.0	0.00	0.00	3,700.0	0.0	0.0	0.0	0.00	0.00	0.00
3,800.0	0.00	0.00	3,800.0	0.0	0.0	0.0	0.00	0.00	0.00
3,900.0	0.00	0.00	3,900.0	0.0	0.0	0.0	0.00	0.00	0.00
4,000.0	0.00	0.00	4,000.0	0.0	0.0	0.0	0.00	0.00	0.00
4,100.0	0.00	0.00	4,100.0	0.0	0.0	0.0	0.00	0.00	0.00
4,200.0	0.00	0.00	4,200.0	0.0	0.0	0.0	0.00	0.00	0.00
4,300.0	0.00	0.00	4,300.0	0.0	0.0	0.0	0.00	0.00	0.00
4,400.0	0.00	0.00	4,400.0	0.0	0.0	0.0	0.00	0.00	0.00
4,500.0	0.00	0.00	4,500.0	0.0	0.0	0.0	0.00	0.00	0.00
4,600.0	0.00	0.00	4,600.0	0.0	0.0	0.0	0.00	0.00	0.00
4,634.0	0.00	0.00	4,634.0	0.0	0.0	0.0	0.00	0.00	0.00
Greenhorn									

Oasis Petroleum

Planning Report

Database:	Compass	Local Co-ordinate Reference:	Well Jefferies 5301 43-12B
Company:	Oasis	TVD Reference:	WELL @ 2118.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2118.0ft (Original Well Elev)
Site:	153N-101W-13/24	North Reference:	True
Well:	Jefferies 5301 43-12B	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
4,700.0	0.00	0.00	4,700.0	0.0	0.0	0.0	0.00	0.00	0.00
4,800.0	0.00	0.00	4,800.0	0.0	0.0	0.0	0.00	0.00	0.00
4,900.0	0.00	0.00	4,900.0	0.0	0.0	0.0	0.00	0.00	0.00
5,000.0	0.00	0.00	5,000.0	0.0	0.0	0.0	0.00	0.00	0.00
5,030.0	0.00	0.00	5,030.0	0.0	0.0	0.0	0.00	0.00	0.00
Mowry									
5,100.0	0.00	0.00	5,100.0	0.0	0.0	0.0	0.00	0.00	0.00
5,200.0	0.00	0.00	5,200.0	0.0	0.0	0.0	0.00	0.00	0.00
5,300.0	0.00	0.00	5,300.0	0.0	0.0	0.0	0.00	0.00	0.00
5,400.0	0.00	0.00	5,400.0	0.0	0.0	0.0	0.00	0.00	0.00
5,461.0	0.00	0.00	5,461.0	0.0	0.0	0.0	0.00	0.00	0.00
Dakota									
5,500.0	0.00	0.00	5,500.0	0.0	0.0	0.0	0.00	0.00	0.00
5,600.0	0.00	0.00	5,600.0	0.0	0.0	0.0	0.00	0.00	0.00
5,700.0	0.00	0.00	5,700.0	0.0	0.0	0.0	0.00	0.00	0.00
5,800.0	0.00	0.00	5,800.0	0.0	0.0	0.0	0.00	0.00	0.00
5,900.0	0.00	0.00	5,900.0	0.0	0.0	0.0	0.00	0.00	0.00
6,000.0	0.00	0.00	6,000.0	0.0	0.0	0.0	0.00	0.00	0.00
6,100.0	0.00	0.00	6,100.0	0.0	0.0	0.0	0.00	0.00	0.00
6,200.0	0.00	0.00	6,200.0	0.0	0.0	0.0	0.00	0.00	0.00
6,300.0	0.00	0.00	6,300.0	0.0	0.0	0.0	0.00	0.00	0.00
6,377.0	0.00	0.00	6,377.0	0.0	0.0	0.0	0.00	0.00	0.00
Rierdon									
6,400.0	0.00	0.00	6,400.0	0.0	0.0	0.0	0.00	0.00	0.00
6,500.0	0.00	0.00	6,500.0	0.0	0.0	0.0	0.00	0.00	0.00
6,600.0	0.00	0.00	6,600.0	0.0	0.0	0.0	0.00	0.00	0.00
6,700.0	0.00	0.00	6,700.0	0.0	0.0	0.0	0.00	0.00	0.00
6,800.0	0.00	0.00	6,800.0	0.0	0.0	0.0	0.00	0.00	0.00
6,896.0	0.00	0.00	6,896.0	0.0	0.0	0.0	0.00	0.00	0.00
Dunham Salt									
6,900.0	0.00	0.00	6,900.0	0.0	0.0	0.0	0.00	0.00	0.00
6,963.0	0.00	0.00	6,963.0	0.0	0.0	0.0	0.00	0.00	0.00
Dunham Salt Base									
6,968.0	0.00	0.00	6,968.0	0.0	0.0	0.0	0.00	0.00	0.00
Spearfish									
7,000.0	0.00	0.00	7,000.0	0.0	0.0	0.0	0.00	0.00	0.00
7,100.0	0.00	0.00	7,100.0	0.0	0.0	0.0	0.00	0.00	0.00
7,200.0	0.00	0.00	7,200.0	0.0	0.0	0.0	0.00	0.00	0.00
7,212.0	0.00	0.00	7,212.0	0.0	0.0	0.0	0.00	0.00	0.00
Pine Salt									
7,300.0	0.00	0.00	7,300.0	0.0	0.0	0.0	0.00	0.00	0.00
7,337.0	0.00	0.00	7,337.0	0.0	0.0	0.0	0.00	0.00	0.00
Pine Salt Base									
7,365.0	0.00	0.00	7,365.0	0.0	0.0	0.0	0.00	0.00	0.00
Opeche Salt									
7,400.0	0.00	0.00	7,400.0	0.0	0.0	0.0	0.00	0.00	0.00
7,444.0	0.00	0.00	7,444.0	0.0	0.0	0.0	0.00	0.00	0.00
Opeche Salt Base									
7,500.0	0.00	0.00	7,500.0	0.0	0.0	0.0	0.00	0.00	0.00
7,600.0	0.00	0.00	7,600.0	0.0	0.0	0.0	0.00	0.00	0.00
7,625.0	0.00	0.00	7,625.0	0.0	0.0	0.0	0.00	0.00	0.00
Broom Creek (Top of Minnelusa Gp.)									
7,668.0	0.00	0.00	7,668.0	0.0	0.0	0.0	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	Compass	Local Co-ordinate Reference:	Well Jefferies 5301 43-12B
Company:	Oasis	TVD Reference:	WELL @ 2118.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2118.0ft (Original Well Elev)
Site:	153N-101W-13/24	North Reference:	True
Well:	Jefferies 5301 43-12B	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
Amsden									
7,700.0	0.00	0.00	7,700.0	0.0	0.0	0.0	0.00	0.00	0.00
7,800.0	0.00	0.00	7,800.0	0.0	0.0	0.0	0.00	0.00	0.00
7,844.0	0.00	0.00	7,844.0	0.0	0.0	0.0	0.00	0.00	0.00
Tyler									
7,900.0	0.00	0.00	7,900.0	0.0	0.0	0.0	0.00	0.00	0.00
8,000.0	0.00	0.00	8,000.0	0.0	0.0	0.0	0.00	0.00	0.00
8,031.0	0.00	0.00	8,031.0	0.0	0.0	0.0	0.00	0.00	0.00
Otter (Base of Minnelusa Gp.)									
8,100.0	0.00	0.00	8,100.0	0.0	0.0	0.0	0.00	0.00	0.00
8,200.0	0.00	0.00	8,200.0	0.0	0.0	0.0	0.00	0.00	0.00
8,300.0	0.00	0.00	8,300.0	0.0	0.0	0.0	0.00	0.00	0.00
8,380.0	0.00	0.00	8,380.0	0.0	0.0	0.0	0.00	0.00	0.00
Kibbey									
8,400.0	0.00	0.00	8,400.0	0.0	0.0	0.0	0.00	0.00	0.00
8,500.0	0.00	0.00	8,500.0	0.0	0.0	0.0	0.00	0.00	0.00
8,527.0	0.00	0.00	8,527.0	0.0	0.0	0.0	0.00	0.00	0.00
Charles Salt									
8,600.0	0.00	0.00	8,600.0	0.0	0.0	0.0	0.00	0.00	0.00
8,700.0	0.00	0.00	8,700.0	0.0	0.0	0.0	0.00	0.00	0.00
8,800.0	0.00	0.00	8,800.0	0.0	0.0	0.0	0.00	0.00	0.00
8,900.0	0.00	0.00	8,900.0	0.0	0.0	0.0	0.00	0.00	0.00
9,000.0	0.00	0.00	9,000.0	0.0	0.0	0.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,100.0	0.0	0.0	0.0	0.00	0.00	0.00
9,150.0	0.00	0.00	9,150.0	0.0	0.0	0.0	0.00	0.00	0.00
UB									
9,200.0	0.00	0.00	9,200.0	0.0	0.0	0.0	0.00	0.00	0.00
9,227.0	0.00	0.00	9,227.0	0.0	0.0	0.0	0.00	0.00	0.00
Base Last Salt									
9,275.0	0.00	0.00	9,275.0	0.0	0.0	0.0	0.00	0.00	0.00
Ratcliffe									
9,300.0	0.00	0.00	9,300.0	0.0	0.0	0.0	0.00	0.00	0.00
9,400.0	0.00	0.00	9,400.0	0.0	0.0	0.0	0.00	0.00	0.00
9,451.0	0.00	0.00	9,451.0	0.0	0.0	0.0	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,500.0	0.0	0.0	0.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,600.0	0.0	0.0	0.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,700.0	0.0	0.0	0.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,800.0	0.0	0.0	0.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,900.0	0.0	0.0	0.0	0.00	0.00	0.00
10,000.0	0.00	0.00	10,000.0	0.0	0.0	0.0	0.00	0.00	0.00
10,025.0	0.00	0.00	10,025.0	0.0	0.0	0.0	0.00	0.00	0.00
Lodgepole									
10,100.0	0.00	0.00	10,100.0	0.0	0.0	0.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,200.0	0.0	0.0	0.0	0.00	0.00	0.00
10,300.0	0.00	0.00	10,300.0	0.0	0.0	0.0	0.00	0.00	0.00
10,321.3	0.00	0.00	10,321.3	0.0	0.0	0.0	0.00	0.00	0.00
KOP-Start Build 13.00									
10,325.0	0.49	207.15	10,325.0	0.0	0.0	0.0	13.00	13.00	0.00
10,350.0	3.74	207.15	10,350.0	-0.8	-0.4	0.9	13.00	13.00	0.00
10,375.0	6.99	207.15	10,374.9	-2.9	-1.5	3.0	13.00	13.00	0.00
10,400.0	10.24	207.15	10,399.6	-6.2	-3.2	6.5	13.00	13.00	0.00
10,425.0	13.49	207.15	10,424.0	-10.8	-5.5	11.2	13.00	13.00	0.00

Oasis Petroleum

Planning Report

Database:	Compass	Local Co-ordinate Reference:	Well Jefferies 5301 43-12B
Company:	Oasis	TVD Reference:	WELL @ 2118.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2118.0ft (Original Well Elev)
Site:	153N-101W-13/24	North Reference:	True
Well:	Jefferies 5301 43-12B	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,450.0	16.74	207.15	10,448.2	-16.6	-8.5	17.2	13.00	13.00	0.00
10,475.0	19.99	207.15	10,471.9	-23.6	-12.1	24.5	13.00	13.00	0.00
10,500.0	23.24	207.15	10,495.1	-31.8	-16.3	33.0	13.00	13.00	0.00
10,525.0	26.49	207.15	10,517.8	-41.2	-21.1	42.7	13.00	13.00	0.00
10,550.0	29.74	207.15	10,539.9	-51.6	-26.5	53.6	13.00	13.00	0.00
10,575.0	32.99	207.15	10,561.2	-63.2	-32.4	65.6	13.00	13.00	0.00
10,600.0	36.24	207.15	10,581.8	-75.9	-38.9	78.7	13.00	13.00	0.00
10,625.0	39.49	207.15	10,601.5	-89.5	-45.9	92.9	13.00	13.00	0.00
10,650.0	42.74	207.15	10,620.4	-104.1	-53.4	108.1	13.00	13.00	0.00
10,675.0	45.99	207.15	10,638.2	-119.7	-61.4	124.2	13.00	13.00	0.00
10,700.0	49.24	207.15	10,655.1	-136.1	-69.8	141.3	13.00	13.00	0.00
10,725.0	52.49	207.15	10,670.9	-153.4	-78.6	159.2	13.00	13.00	0.00
10,750.0	55.74	207.15	10,685.5	-171.4	-87.9	177.9	13.00	13.00	0.00
10,775.0	58.99	207.15	10,699.0	-190.1	-97.5	197.4	13.00	13.00	0.00
10,800.0	62.24	207.15	10,711.3	-209.5	-107.4	217.5	13.00	13.00	0.00
10,825.0	65.49	207.15	10,722.3	-229.5	-117.7	238.2	13.00	13.00	0.00
10,850.0	68.74	207.15	10,732.0	-250.0	-128.2	259.5	13.00	13.00	0.00
10,873.7	71.82	207.15	10,740.0	-269.8	-138.3	280.1	13.00	13.00	0.00
False Bakken									
10,875.0	71.99	207.15	10,740.4	-270.9	-138.9	281.2	13.00	13.00	0.00
10,900.0	75.24	207.15	10,747.4	-292.2	-149.8	303.4	13.00	13.00	0.00
10,925.0	78.49	207.15	10,753.1	-313.9	-160.9	325.9	13.00	13.00	0.00
10,934.9	79.77	207.15	10,755.0	-322.6	-165.4	334.9	13.00	13.00	0.00
Upper Bakken									
10,950.0	81.74	207.15	10,757.4	-335.8	-172.2	348.6	13.00	13.00	0.00
10,975.0	84.99	207.15	10,760.3	-357.9	-183.5	371.6	13.00	13.00	0.00
11,000.0	88.24	207.15	10,761.8	-380.1	-194.9	394.6	13.00	13.00	0.00
11,013.6	90.00	207.15	10,762.0	-392.2	-201.1	407.1	13.00	13.00	0.00
EOC-Start 53.7 hold at 11013.6 MD - Middle Bakken - Middle Bakken Sand Target									
11,067.3	90.00	207.15	10,762.0	-440.0	-225.6	456.8	0.00	0.00	0.00
Start DLS 1.00 TFO -90.00									
11,085.0	90.00	206.97	10,762.0	-455.8	-233.7	473.1	1.00	0.00	-1.00
7"									
11,100.0	90.00	206.82	10,762.0	-469.1	-240.4	487.0	1.00	0.00	-1.00
11,200.0	90.00	205.82	10,762.0	-558.8	-284.8	579.9	1.00	0.00	-1.00
11,300.0	90.00	204.82	10,762.0	-649.2	-327.5	673.5	1.00	0.00	-1.00
11,400.0	90.00	203.82	10,762.0	-740.3	-368.7	767.6	1.00	0.00	-1.00
11,500.0	90.00	202.82	10,762.0	-832.1	-408.3	862.4	1.00	0.00	-1.00
11,600.0	90.00	201.82	10,762.0	-924.6	-446.3	957.6	1.00	0.00	-1.00
11,700.0	90.00	200.82	10,762.0	-1,017.8	-482.6	1,053.4	1.00	0.00	-1.00
11,800.0	90.00	199.82	10,762.0	-1,111.6	-517.4	1,149.7	1.00	0.00	-1.00
11,900.0	90.00	198.82	10,762.0	-1,205.9	-550.4	1,246.4	1.00	0.00	-1.00
12,000.0	90.00	197.82	10,762.0	-1,300.9	-581.9	1,343.6	1.00	0.00	-1.00
12,100.0	90.00	196.82	10,762.0	-1,396.3	-611.6	1,441.1	1.00	0.00	-1.00
12,200.0	90.00	195.82	10,762.0	-1,492.3	-639.7	1,539.1	1.00	0.00	-1.00
12,300.0	90.00	194.82	10,762.0	-1,588.7	-666.1	1,637.3	1.00	0.00	-1.00
12,400.0	90.00	193.82	10,762.0	-1,685.6	-690.9	1,735.9	1.00	0.00	-1.00
12,500.0	90.00	192.82	10,762.0	-1,782.9	-713.9	1,834.7	1.00	0.00	-1.00
12,600.0	90.00	191.82	10,762.0	-1,880.6	-735.2	1,933.8	1.00	0.00	-1.00
12,700.0	90.00	190.82	10,762.0	-1,978.7	-754.9	2,033.2	1.00	0.00	-1.00
12,800.0	90.00	189.82	10,762.0	-2,077.1	-772.8	2,132.7	1.00	0.00	-1.00
12,900.0	90.00	188.82	10,762.0	-2,175.8	-789.0	2,232.3	1.00	0.00	-1.00
13,000.0	90.00	187.82	10,762.0	-2,274.7	-803.4	2,332.1	1.00	0.00	-1.00

Oasis Petroleum

Planning Report

Database:	Compass	Local Co-ordinate Reference:	Well Jefferies 5301 43-12B
Company:	Oasis	TVD Reference:	WELL @ 2118.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2118.0ft (Original Well Elev)
Site:	153N-101W-13/24	North Reference:	True
Well:	Jefferies 5301 43-12B	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
13,100.0	90.00	186.82	10,762.0	-2,373.9	-816.2	2,432.0	1.00	0.00	-1.00
13,200.0	90.00	185.82	10,762.0	-2,473.3	-827.2	2,532.0	1.00	0.00	-1.00
13,300.0	90.00	184.82	10,762.0	-2,572.8	-836.5	2,632.0	1.00	0.00	-1.00
13,400.0	90.00	183.82	10,762.0	-2,672.6	-844.0	2,732.0	1.00	0.00	-1.00
13,500.0	90.00	182.82	10,762.0	-2,772.4	-849.8	2,831.9	1.00	0.00	-1.00
13,600.0	90.00	181.82	10,762.0	-2,872.3	-853.8	2,931.8	1.00	0.00	-1.00
13,700.0	90.00	180.82	10,762.0	-2,972.3	-856.1	3,031.7	1.00	0.00	-1.00
13,781.6	90.00	180.00	10,762.0	-3,053.9	-856.7	3,113.1	1.00	0.00	-1.00
Start 7532.1 hold at 13781.6 MD									
13,800.0	90.00	180.00	10,762.0	-3,072.3	-856.7	3,131.4	0.00	0.00	0.00
13,900.0	90.00	180.00	10,762.0	-3,172.3	-856.7	3,231.1	0.00	0.00	0.00
14,000.0	90.00	180.00	10,762.0	-3,272.3	-856.7	3,330.7	0.00	0.00	0.00
14,100.0	90.00	180.00	10,762.0	-3,372.3	-856.7	3,430.4	0.00	0.00	0.00
14,200.0	90.00	180.00	10,762.0	-3,472.3	-856.7	3,530.1	0.00	0.00	0.00
14,300.0	90.00	180.00	10,762.0	-3,572.3	-856.7	3,629.8	0.00	0.00	0.00
14,400.0	90.00	180.00	10,762.0	-3,672.3	-856.7	3,729.4	0.00	0.00	0.00
14,500.0	90.00	180.00	10,762.0	-3,772.3	-856.7	3,829.1	0.00	0.00	0.00
14,600.0	90.00	180.00	10,762.0	-3,872.3	-856.7	3,928.8	0.00	0.00	0.00
14,700.0	90.00	180.00	10,762.0	-3,972.3	-856.7	4,028.5	0.00	0.00	0.00
14,800.0	90.00	180.00	10,762.0	-4,072.3	-856.7	4,128.1	0.00	0.00	0.00
14,900.0	90.00	180.00	10,762.0	-4,172.3	-856.8	4,227.8	0.00	0.00	0.00
15,000.0	90.00	180.00	10,762.0	-4,272.3	-856.8	4,327.5	0.00	0.00	0.00
15,100.0	90.00	180.00	10,762.0	-4,372.3	-856.8	4,427.2	0.00	0.00	0.00
15,200.0	90.00	180.00	10,762.0	-4,472.3	-856.8	4,526.8	0.00	0.00	0.00
15,300.0	90.00	180.00	10,762.0	-4,572.3	-856.8	4,626.5	0.00	0.00	0.00
15,400.0	90.00	180.00	10,762.0	-4,672.3	-856.8	4,726.2	0.00	0.00	0.00
15,500.0	90.00	180.00	10,762.0	-4,772.3	-856.8	4,825.9	0.00	0.00	0.00
15,600.0	90.00	180.00	10,762.0	-4,872.3	-856.8	4,925.5	0.00	0.00	0.00
15,700.0	90.00	180.00	10,762.0	-4,972.3	-856.8	5,025.2	0.00	0.00	0.00
15,800.0	90.00	180.00	10,762.0	-5,072.3	-856.8	5,124.9	0.00	0.00	0.00
15,900.0	90.00	180.00	10,762.0	-5,172.3	-856.8	5,224.6	0.00	0.00	0.00
16,000.0	90.00	180.00	10,762.0	-5,272.3	-856.8	5,324.2	0.00	0.00	0.00
16,100.0	90.00	180.00	10,762.0	-5,372.3	-856.8	5,423.9	0.00	0.00	0.00
16,200.0	90.00	180.00	10,762.0	-5,472.3	-856.8	5,523.6	0.00	0.00	0.00
16,300.0	90.00	180.00	10,762.0	-5,572.3	-856.8	5,623.2	0.00	0.00	0.00
16,400.0	90.00	180.00	10,762.0	-5,672.3	-856.8	5,722.9	0.00	0.00	0.00
16,500.0	90.00	180.00	10,762.0	-5,772.3	-856.8	5,822.6	0.00	0.00	0.00
16,600.0	90.00	180.00	10,762.0	-5,872.3	-856.8	5,922.3	0.00	0.00	0.00
16,700.0	90.00	180.00	10,762.0	-5,972.3	-856.8	6,021.9	0.00	0.00	0.00
16,800.0	90.00	180.00	10,762.0	-6,072.3	-856.8	6,121.6	0.00	0.00	0.00
16,900.0	90.00	180.00	10,762.0	-6,172.3	-856.8	6,221.3	0.00	0.00	0.00
17,000.0	90.00	180.00	10,762.0	-6,272.3	-856.8	6,321.0	0.00	0.00	0.00
17,100.0	90.00	180.00	10,762.0	-6,372.3	-856.8	6,420.6	0.00	0.00	0.00
17,200.0	90.00	180.00	10,762.0	-6,472.3	-856.8	6,520.3	0.00	0.00	0.00
17,300.0	90.00	180.00	10,762.0	-6,572.3	-856.8	6,620.0	0.00	0.00	0.00
17,400.0	90.00	180.00	10,762.0	-6,672.3	-856.8	6,719.7	0.00	0.00	0.00
17,500.0	90.00	180.00	10,762.0	-6,772.3	-856.9	6,819.3	0.00	0.00	0.00
17,600.0	90.00	180.00	10,762.0	-6,872.3	-856.9	6,919.0	0.00	0.00	0.00
17,700.0	90.00	180.00	10,762.0	-6,972.3	-856.9	7,018.7	0.00	0.00	0.00
17,800.0	90.00	180.00	10,762.0	-7,072.3	-856.9	7,118.4	0.00	0.00	0.00
17,900.0	90.00	180.00	10,762.0	-7,172.3	-856.9	7,218.0	0.00	0.00	0.00
18,000.0	90.00	180.00	10,762.0	-7,272.3	-856.9	7,317.7	0.00	0.00	0.00
18,100.0	90.00	180.00	10,762.0	-7,372.3	-856.9	7,417.4	0.00	0.00	0.00
18,200.0	90.00	180.00	10,762.0	-7,472.3	-856.9	7,517.1	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	Compass	Local Co-ordinate Reference:	Well Jefferies 5301 43-12B
Company:	Oasis	TVD Reference:	WELL @ 2118.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2118.0ft (Original Well Elev)
Site:	153N-101W-13/24	North Reference:	True
Well:	Jefferies 5301 43-12B	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
18,300.0	90.00	180.00	10,762.0	-7,572.3	-856.9	7,616.7	0.00	0.00	0.00
18,400.0	90.00	180.00	10,762.0	-7,672.3	-856.9	7,716.4	0.00	0.00	0.00
18,500.0	90.00	180.00	10,762.0	-7,772.3	-856.9	7,816.1	0.00	0.00	0.00
18,600.0	90.00	180.00	10,762.0	-7,872.3	-856.9	7,915.8	0.00	0.00	0.00
18,700.0	90.00	180.00	10,762.0	-7,972.3	-856.9	8,015.4	0.00	0.00	0.00
18,800.0	90.00	180.00	10,762.0	-8,072.3	-856.9	8,115.1	0.00	0.00	0.00
18,900.0	90.00	180.00	10,762.0	-8,172.3	-856.9	8,214.8	0.00	0.00	0.00
19,000.0	90.00	180.00	10,762.0	-8,272.3	-856.9	8,314.5	0.00	0.00	0.00
19,100.0	90.00	180.00	10,762.0	-8,372.3	-856.9	8,414.1	0.00	0.00	0.00
19,200.0	90.00	180.00	10,762.0	-8,472.3	-856.9	8,513.8	0.00	0.00	0.00
19,300.0	90.00	180.00	10,762.0	-8,572.3	-856.9	8,613.5	0.00	0.00	0.00
19,400.0	90.00	180.00	10,762.0	-8,672.3	-856.9	8,713.1	0.00	0.00	0.00
19,500.0	90.00	180.00	10,762.0	-8,772.3	-856.9	8,812.8	0.00	0.00	0.00
19,600.0	90.00	180.00	10,762.0	-8,872.3	-856.9	8,912.5	0.00	0.00	0.00
19,700.0	90.00	180.00	10,762.0	-8,972.3	-856.9	9,012.2	0.00	0.00	0.00
19,800.0	90.00	180.00	10,762.0	-9,072.3	-856.9	9,111.8	0.00	0.00	0.00
19,900.0	90.00	180.00	10,762.0	-9,172.3	-856.9	9,211.5	0.00	0.00	0.00
20,000.0	90.00	180.00	10,762.0	-9,272.3	-856.9	9,311.2	0.00	0.00	0.00
20,100.0	90.00	180.00	10,762.0	-9,372.3	-857.0	9,410.9	0.00	0.00	0.00
20,200.0	90.00	180.00	10,762.0	-9,472.3	-857.0	9,510.5	0.00	0.00	0.00
20,300.0	90.00	180.00	10,762.0	-9,572.3	-857.0	9,610.2	0.00	0.00	0.00
20,400.0	90.00	180.00	10,762.0	-9,672.3	-857.0	9,709.9	0.00	0.00	0.00
20,500.0	90.00	180.00	10,762.0	-9,772.3	-857.0	9,809.6	0.00	0.00	0.00
20,600.0	90.00	180.00	10,762.0	-9,872.3	-857.0	9,909.2	0.00	0.00	0.00
20,700.0	90.00	180.00	10,762.0	-9,972.3	-857.0	10,008.9	0.00	0.00	0.00
20,800.0	90.00	180.00	10,762.0	-10,072.3	-857.0	10,108.6	0.00	0.00	0.00
20,900.0	90.00	180.00	10,762.0	-10,172.3	-857.0	10,208.3	0.00	0.00	0.00
21,000.0	90.00	180.00	10,762.0	-10,272.3	-857.0	10,307.9	0.00	0.00	0.00
21,100.0	90.00	180.00	10,762.0	-10,372.3	-857.0	10,407.6	0.00	0.00	0.00
21,200.0	90.00	180.00	10,762.0	-10,472.3	-857.0	10,507.3	0.00	0.00	0.00
21,300.0	90.00	180.00	10,762.0	-10,572.3	-857.0	10,607.0	0.00	0.00	0.00
21,313.7	90.00	180.00	10,762.0	-10,586.0	-857.0	10,620.6	0.00	0.00	0.00
TD at 21313.7									

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (ft)	+N/-S (ft)	+E/-W (ft)	Northing (m)	Easting (m)	Latitude	Longitude
- hit/miss target									
- Shape									
Jefferies 5301 43-12B P	0.00	0.00	10,762.0	-10,586.0	-857.0	121,862.97	367,676.20	48° 3' 13.577 N	103° 37' 2.975 W
- plan hits target center									
- Point									

Casing Points					
Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)	
2,170.0	2,170.0	9 5/8"	9.625	13.500	
11,085.0	10,762.0	7"	7.000	8.750	

Oasis Petroleum

Planning Report

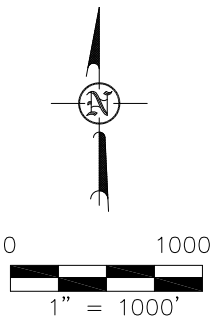
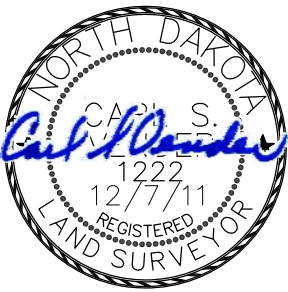
Database:	Compass	Local Co-ordinate Reference:	Well Jefferies 5301 43-12B
Company:	Oasis	TVD Reference:	WELL @ 2118.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2118.0ft (Original Well Elev)
Site:	153N-101W-13/24	North Reference:	True
Well:	Jefferies 5301 43-12B	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Formations						
Measured Depth (ft)	Vertical Depth (ft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,968.0	1,968.0	Pierre				
4,634.0	4,634.0	Greenhorn				
5,030.0	5,030.0	Mowry				
5,461.0	5,461.0	Dakota				
6,377.0	6,377.0	Rierdon				
6,896.0	6,896.0	Dunham Salt				
6,963.0	6,963.0	Dunham Salt Base				
6,968.0	6,968.0	Spearfish				
7,212.0	7,212.0	Pine Salt				
7,337.0	7,337.0	Pine Salt Base				
7,365.0	7,365.0	Opeche Salt				
7,444.0	7,444.0	Opeche Salt Base				
7,625.0	7,625.0	Broom Creek (Top of Minnelusa Gp.)				
7,668.0	7,668.0	Amsden				
7,844.0	7,844.0	Tyler				
8,031.0	8,031.0	Otter (Base of Minnelusa Gp.)				
8,380.0	8,380.0	Kibbey				
8,527.0	8,527.0	Charles Salt				
9,150.0	9,150.0	UB				
9,227.0	9,227.0	Base Last Salt				
9,275.0	9,275.0	Ratcliffe				
9,451.0	9,451.0	Mission Canyon				
10,025.0	10,025.0	Lodgepole				
10,873.7	10,740.0	False Bakken				
10,934.9	10,755.0	Upper Bakken				
11,013.6	10,762.0	Middle Bakken				
11,013.6	10,762.0	Middle Bakken Sand Target				

Plan Annotations					
Measured Depth (ft)	Vertical Depth (ft)	Local Coordinates			
		+N/-S (ft)	+E/-W (ft)	Comment	
10,321.3	10,321.3	0.0	0.0	KOP-Start Build 13.00	
11,013.6	10,762.0	-392.2	-201.1	EOC-Start 53.7 hold at 11013.6 MD	
11,067.3	10,762.0	-440.0	-225.6	Start DLS 1.00 TFO -90.00	
13,781.6	10,762.0	-3,053.9	-856.7	Start 7532.1 hold at 13781.6 MD	
21,313.7	10,762.0	-10,586.0	-857.0	TD at 21313.7	

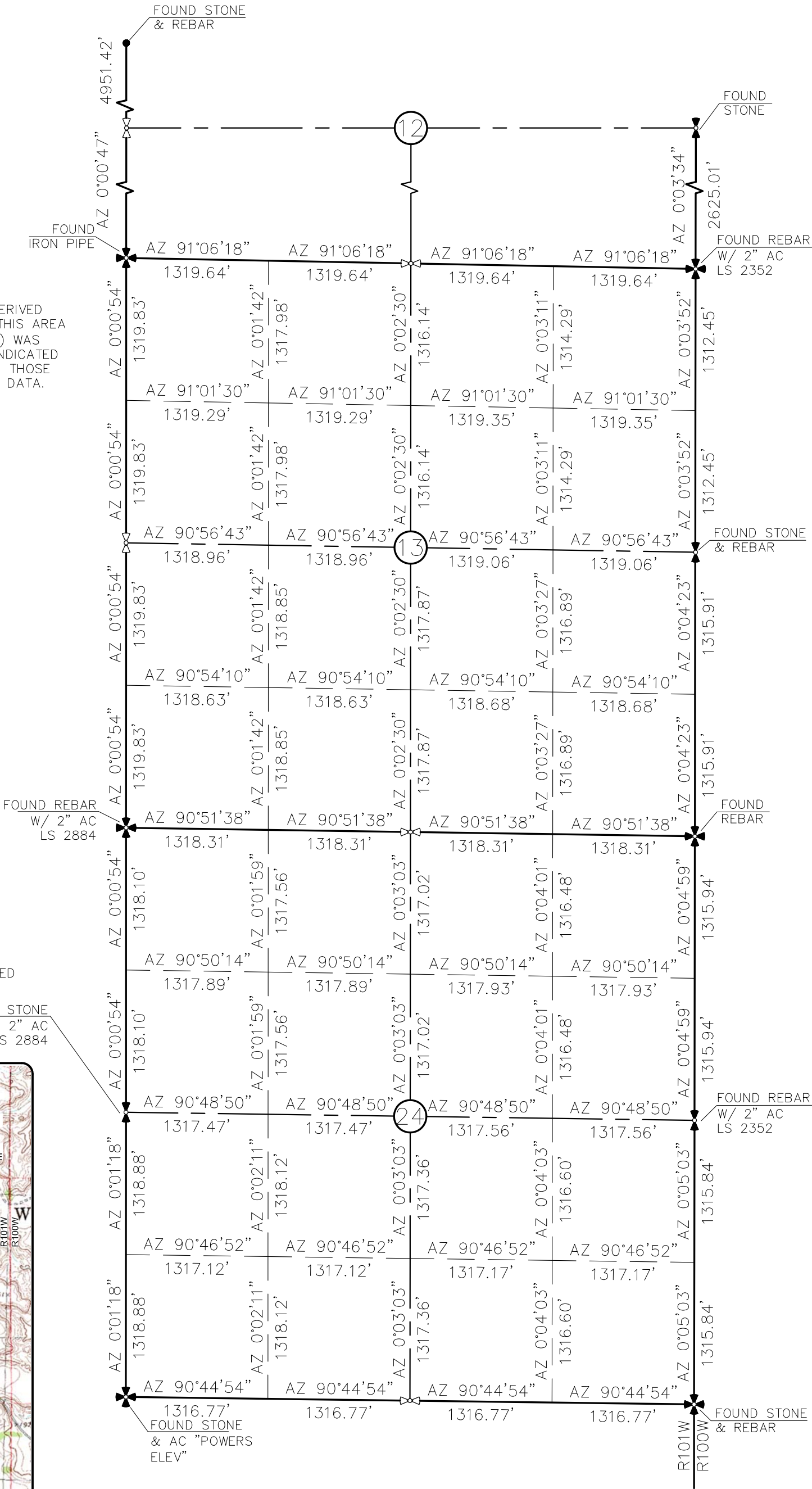
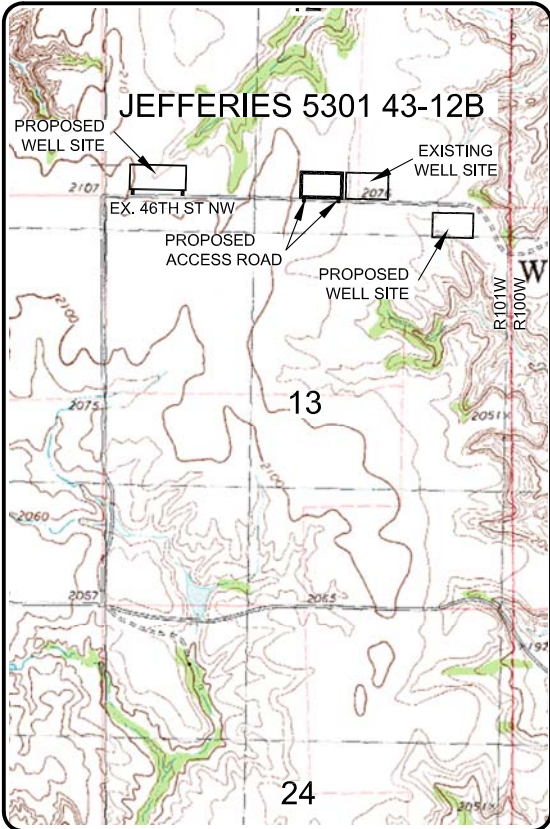
SECTION BREAKDOWN
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"
250 FEET FROM SOUTH LINE AND 2410 FEET FROM EAST LINE
SECTIONS 12, 13, & 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

ALL BEARINGS ARE BASED ON G.P.S. DERIVED BEARINGS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1900. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA.



- ⊕ — MONUMENT — RECOVERED
⊗ — MONUMENT — NOT RECOVERED

VICINITY MAP



© 2011, INTERSTATE ENGINEERING, INC.

2/8

SHEET NO.



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengi.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
SECTION BREAKDOWN
SECTIONS 12, 13, & 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S11-09-361
Checked By: C.S.V. Date: DEC 2011

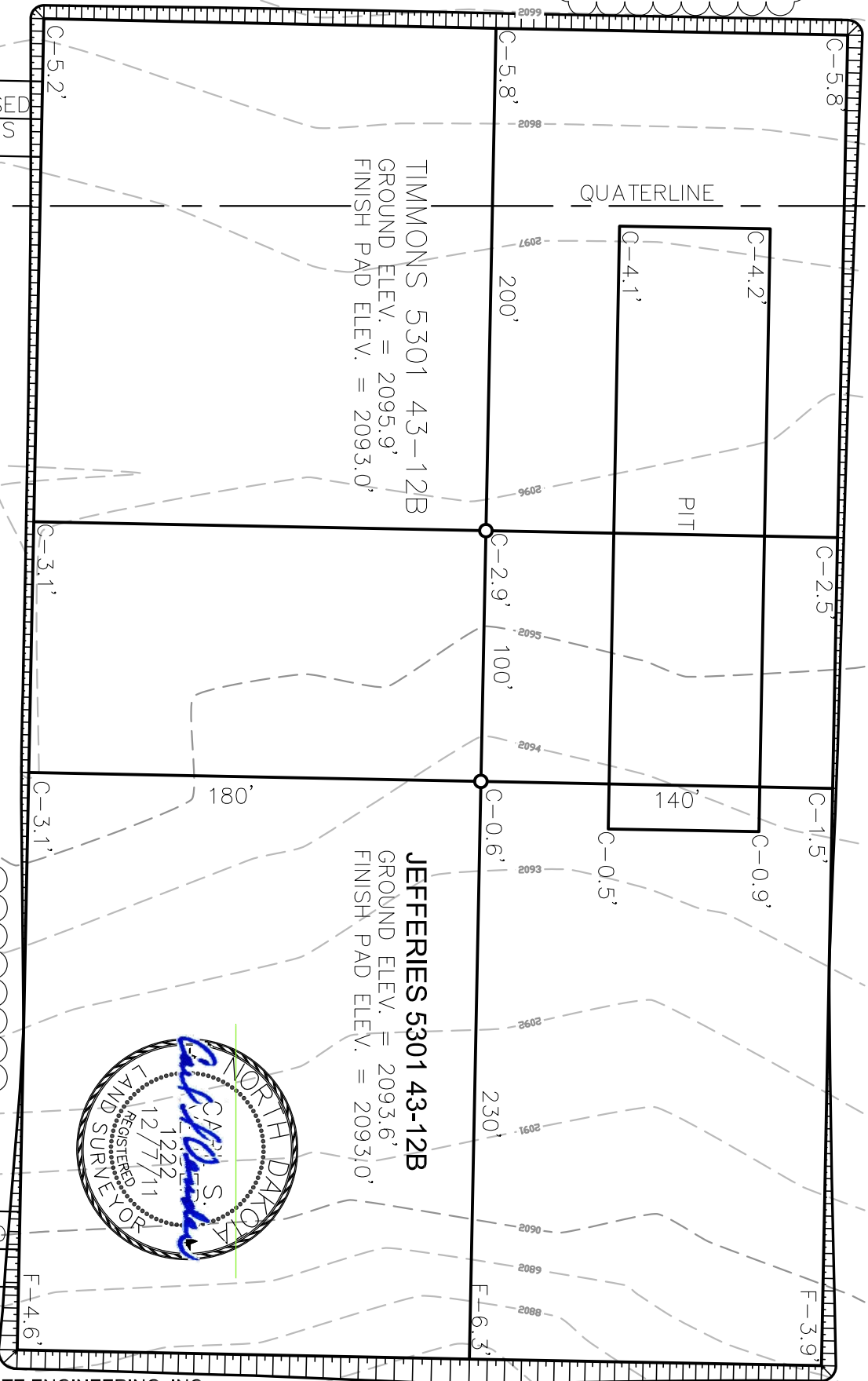
Revision No.	Date	By	Description

C:\2011\S11-09-361\Oasis Petroleum 2 Wells Sec 12-13 & 24 T153N R101W\CAD\JEFFERIES WELL.dwg - 12/7/2011 1:22 PM josh schmieler

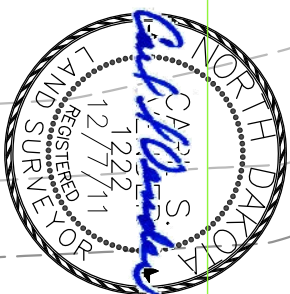
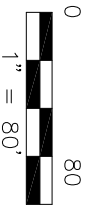
PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"

250 FEET FROM SOUTH LINE AND 2410 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.



© 2011, INTERSTATE ENGINEERING, INC.

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"
250 FEET FROM SOUTH LINE AND 2410 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

JEFFERIES 5301 43-12B

AZ 181°06',
69.9' — BRAY 5301 43-12H

EX. 33' R/W — AZ 181°06', 70°2'

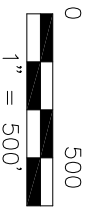
LEX. 33' R/W

TESS 5301 14-13B

FOUND REBAR
W/ 2" AC
LS 2352

EX. 33' R/W

EX. 33' R/W



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.



4/8



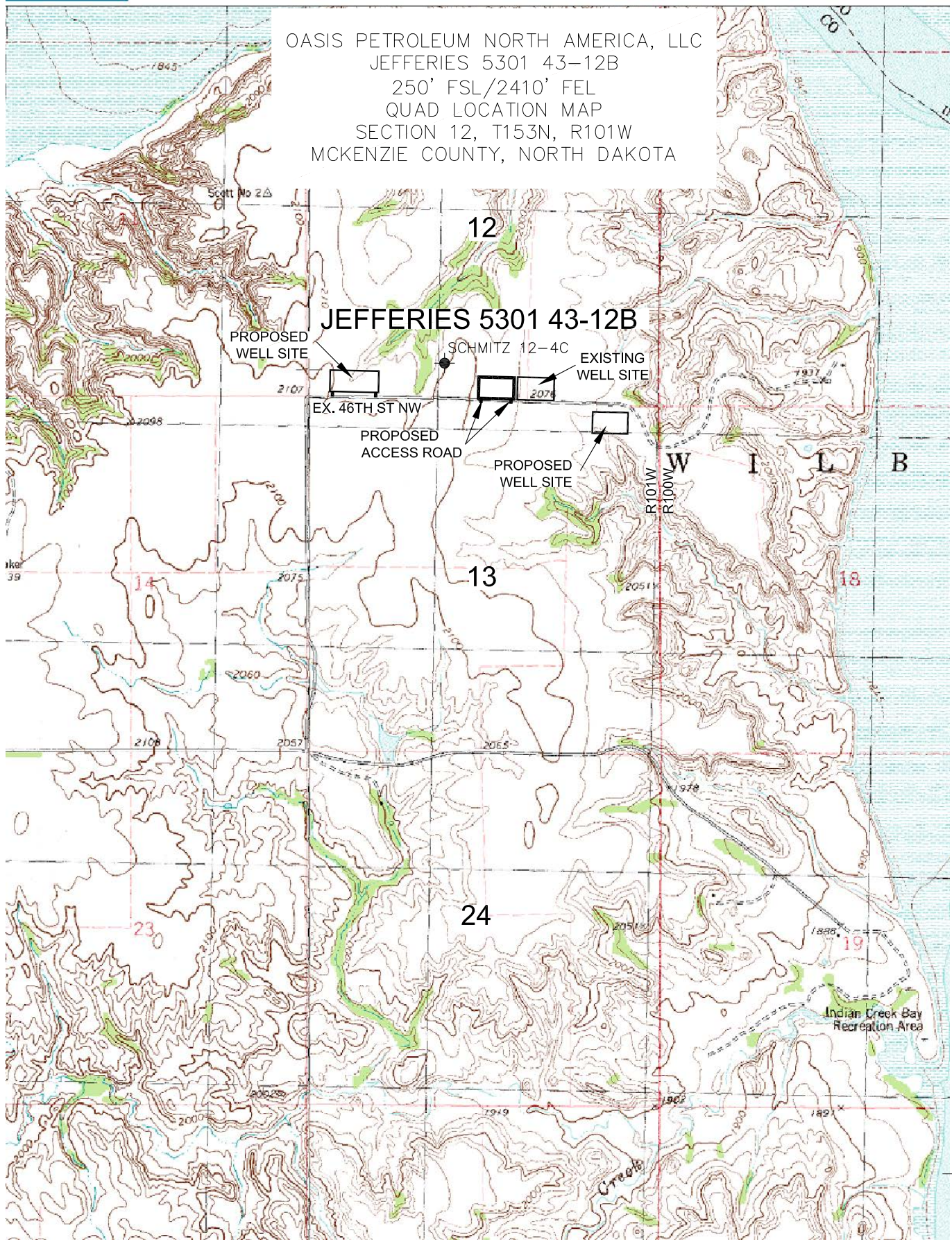
Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengi.com

OASIS PETROLEUM NORTH AMERICA, LLC
ACCESS APPROACH
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: <u>J.J.S.</u>	Project No.: <u>S11-09-361</u>
Checked By: <u>C.S.V.</u>	Date: <u>DEC 2011</u>

[illegible]

OASIS PETROLEUM NORTH AMERICA, LLC
JEFFERIES 5301 43-12B
250' FSL/2410' FEL
QUAD LOCATION MAP
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA



© 2011, INTERSTATE ENGINEERING, INC.

5/8

SHEET NO.



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.ieng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 12, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S11-09-361
Checked By: C.S.V. Date: DEC 2011

Revision No.	Date	By	Description

WELL LOCATION SITE QUANTITIES

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500 HOUSTON, TX 77002
"JEFFERIES 5301 43-12B"

250 FEET FROM SOUTH LINE AND 2410 FEET FROM EAST LINE
SECTION 12, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

WELL SITE ELEVATION 2093.6
WELL PAD ELEVATION 2093.0

EXCAVATION 12,909
PLUS PIT 3,150
16,059

EMBANKMENT 4,705
PLUS SHRINKAGE (30%) 1,412
6,117

STOCKPILE PIT 3,150

STOCKPILE TOP SOIL (6") 3,284

STOCKPILE FROM PAD 3,508

DISTURBED AREA FROM PAD 4.07 ACRES

NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
CUT END SLOPES AT 1:1
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION

2410' FEL
250' FSL

SURFACE DAMAGE SETTLEMENT AND RELEASE

In consideration for the sum of _____ Dollars
(\$ _____) paid by Oasis Petroleum North America LLC
("Oasis") to the undersigned surface owners, Larry P. Heen, a married man dealing in his sole & separate property
("Owners," and together with Oasis, the "Parties") for themselves and their heirs, successors,
administrators and assigns, hereby acknowledge the receipt and sufficiency of said payment as a full
and complete settlement for and as a release of all claims for loss, damage or injury to the Subject
Lands (as defined herein) arising out of the Operations (as defined herein) of the Jefferies 5301 43-12B &
Timmons 5301 43-12B the "Well(s)" located on the approximately (6) six acre tract
of land identified on the plat attached hereto as Exhibit "A" (the "Subject Lands") and which is situated on
the following described real property located in McKenzie County, State of North Dakota, to wit:
Township 153 North, Range 101 West, 5th P.M.
Section 12: SE4SW4, SW4SE4

This pad shall accommodate the drilling of the Jefferies 5301 43-12B well
and the Timmons 5301 43-12B well on the same location. The undersigned is
fully aware that the cuttings generated from the drilling of the above described wells will be buried on
site on the above described location.

The Parties agree that the settlement and release described herein does not include any claims by any
third party against the Owners for personal injury or property damage arising directly out of Oasis's
Operations, and Oasis agrees to indemnify, defend and hold harmless Owners against all liabilities
arising from such claim (except as such claim arises from the gross negligence or willful misconduct of
the Owners).

In further consideration of the payments specified herein, Oasis is hereby specifically granted the right to
construct, install and operate, replace or remove pads, pits, pumps, compressors, tanks, roads,
pipelines, equipment or other facilities on the above described tract of land necessary for its drilling,
completion, operation and/or plugging and abandonment of the Well(s) (the "Operations"), and to the
extent such facilities are maintained by Oasis for use on the Subject Lands, this agreement shall permit
Oasis's use of such facilities for the Operations on the Subject Lands.

Should commercial production be established from the Well(s), Oasis agrees to pay Owners an annual
amount of: _____ per year beginning one year after the completion of the Wells and to be paid
annually until the Wells is plugged and abandoned.

The Parties expressly agree and acknowledge that the payments described herein to be made by Oasis
to the Owners constitute full satisfaction of the requirements of Chapter 38.11.1 of the North Dakota
Century Code and, once in effect, the amended Chapter 38.11.1 of the North Dakota Century Code
enacted by House Bill 1241. The Parties further expressly agree and acknowledge that the \$
_____ payment set forth above constitutes full and adequate consideration for damage and disruption required
under Section 38.11.1-04 of the North Dakota Century Code, and that the _____ payment set forth
above constitutes full and adequate consideration for loss of production payments under Section
38.11.1-08.1 of the North Dakota Century Code.

Oasis shall keep the Site free of noxious weeds, and shall take reasonable steps to control erosion and
washouts on the Site. Oasis shall restore the Site to a condition as near to the original condition of the
Site as is reasonably possible, including the re-contouring, replacing of topsoil and re-seeding of the Site
(such actions, the "Restoration").

The surface owners grant Oasis access to the Wells in the location(s) shown on the plats attached
hereto as Exhibit "A".

Upon written request and the granting of a full release by the Owners of further Restoration by Oasis
with respect to the affected area described in this paragraph, Oasis shall leave in place any road built by
it in its Operations for the benefit of the Owners after abandoning its Operations, and shall have no
further maintenance obligations with respect to any such road.

This agreement shall apply to the Parties and their respective successors, assigns, parent and
subsidiary companies, affiliates and related companies, trusts and partnerships, as well as their
contractors, subcontractors, officers, directors, agents and employees.

This agreement may be executed in multiple counterparts, each of which shall be an original, but all of
which shall constitute one instrument.

[Signature Page Follows.]

DATED this 13th day of December

2011

SURFACE OWNERS

Larry P. Heen
Larry P. Heen, a married man dealing in his sole &
separate property

Address: 14033 45th Street NW

Williston, ND 58801

Phone: 701-572-6991

STATE OF North Dakota

} SS.

ACKNOWLEDGMENT INDIVIDUAL

COUNTY OF McKenzie

BE IT REMEMBERED, That on this 13th day of December, 2011 before me, a Notary Public, in and for said County and State, personally appeared Larry P. Heen, a married man dealing in his sole & separate property, to me known to be the identical person described in and who executed the within and foregoing instrument and acknowledged to me that he executed the same as his free and voluntary act and deed for the uses and purposes therein set forth.

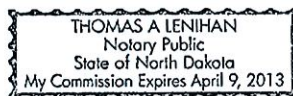
IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal, the day and year first above written.

My commission expires: April 9, 2013

Thomas A. Lenihan

Notary Public

NOTARY STAMP



STATE OF _____)

ACKNOWLEDGMENT CORPORATION

COUNTY OF _____)

Before me the undersigned, a Notary Public, in and for said County and State, on this _____ day of _____, 2011, personally appeared _____, to me known to be the identical person who subscribed the name of the maker thereof to the foregoing instrument as its _____ and acknowledged to me that _____ executed the same as _____ free and voluntary act and deed and as the free and voluntary act and deed of such corporation, for the uses and purposes therein set forth.

Given under my hand and seal of office the day and year last above written.

My commission expires: _____

Notary Public

NOTARY STAMP